GENERAL BIOLOGY 102
VIRUS

Virus:
Virus is a small, simple infectious agent.
It can be defined as a non-cellular, infectious entity made up of genetic material and protein that can invade and reproduce only within the living cells of bacteria, plants and animals. The nucleic acid of viruses, which serves as genetic material, is **either DNA or RNA**(a small piece of genetic information ,DNA or RNA) inside a protective protein coating called **capsid**. Viral infections are any illness you get from virus and this virus uses the cells of living organisms to reproduce or make copies of themselves in a process called **replication**. Since human cells have the machinery needed to make more cells and virus do not, they have to break into cells of living organisms to be able to **replicate** and have higher load which makes its carrier sick. The Low risk viruses go away on their own, while the high risk viruses cause life-threatening or chronic illness. Viruses usually enter the body of its host through the mucous membranes(inner linings) of organs such as eyes, nose, mouth, stomach, penis, vagina and anus. Some viruses also get in through a break or a cut on the skin or from a bite from a mosquito or tick and other virus transmitting vectors. Viruses can cause several illness such as respiratory illnesses, sexually transmitted infections (STIs), skin conditions and many other kinds of illnesses. Some commonly known viral infections include: Common cold, The flu (influenza), COVID-19, Respiratory syncytial virus infection, Chickenpox, Measles, HIV/AIDS, Human Papillomavirus Virus(Warts on the body, genital etc), Herpes simplex (Herpes Simplex Virus), Poliomyelitis, Rabies, Mpox, Zika virus infection, Hepatitis etc.

Characteristics of Virus.
Virus is a sub-microscopic particle that lives by infecting the cells of a living organism. When compared with prokaryotic these are smaller in size and range from 20-300 nm. Individual virus is known as a **Virion**. Viruses do not have specialised machinery such as cell membranes, nucleus, mitochondria, endoplasmic reticulum, and many other cell organelles required for growth, reproduction, energy production, maintaining homeostasis etc. Viruses can replicate only by infecting a host cell. Viruses cannot be classified either a living organism or non-living organism. This is due to the fact that they possess certain defining characteristic features of living organisms and non-living entities. Viruses are significantly smaller than bacteria. They exist outside of the living cell as active particles surrounded by a protein coat.
Virus can replicate in plant, animal as well as bacteria cells etc. It enters and attaches itself to a specific host cell and injects its genetic material and reproduces by using the host genetic material and finally the host cell splits open, releasing the new viruses.

CLASSIFICATIONT OF VIRUS
Viruses are classified based on their different criteria;

1.CLASSIFICATION BASED ON THE PRESENCE OF NUCLEIC ACID
(A) DNA virus
The virus, having DNA as its genetic material. There are two different types of DNA virus
i. Single-stranded (ss) DNA virus are: e.g. Picornaviruses, Parvovirus, etc.
ii. Double-stranded (ds) DNA virus are: e.g. Adenovirus, Herpes virus, etc.
(B) RNA virus
The virus, having RNA as its genetic material. There are two different types of RNA virus

i. Double-stranded (ds) RNA virus are: e.g. Reovirus, etc.
ii. Single-stranded (ss) RNA virus are . It is further classified into two Positive sense RNA (+RNA) and negative sense RNA (-RNA). Poliovirus, Hepatitis A, Rabies virus, Influenza virus are examples of single-stranded RNA virus.

2. CLASSIFICATION BASED ON THE REPLICATION PROPERTIES AND SITE OF REPLICATION

Here, viruses invade into the host cell, where it replicates and assembly within the cell organelles.

(A).Replication within the cytoplasm of the host cell.
E.g. All RNA viruses except the Influenza virus.

(B). Replication within the nucleus and the cytoplasm of the host cell.
E.g. Influenza virus, Poxvirus, etc.

(C). Replication within the nucleus of the host cell.
All DNA viruses except Pox virus.

(D). Replication of the virus through the double-stranded DNA intermediate.
E.g. All DNA viruses, Retrovirus and some tumour causing RNA virus.

(E). Replication of the virus through a single-stranded RNA intermediate.
E.g. All RNA viruses except Reovirus and tumour-causing RNA viruses.

3.CLASSIFICATION BASED ON THE HOST RANGE

Based on the type of host, there are four different types of viruses:

(A). Animal viruses

These viruses infect by invading the cells of animals, including humans. Prominent examples of animal viruses include the influenza virus, mumps virus, rabies virus, poliovirus, Herpes virus, etc.

(B). Plant viruses

These viruses infect plants by invading the plant cells. Replication of plant viruses is obligate and does not happen without a host. Well-known examples of plant virus include the potato virus, tobacco mosaic virus (TMV), beet yellow virus, and turnip yellow virus, cauliflower mosaic virus, etc.

4.CLASSIFICATION BASED ON THE MODE OF TRANSMISSION

Airborne infections - This is the transmission of the virus through the air into the respiratory tract. E.g. Swine flu, and Rhinovirus.

Fecal oral route -This is the transmission of the virus through the contaminated water or food. E.g. Hepatitis A virus, Poliovirus, Rotavirus.

Sexually transmitted diseases- Transmission of the virus through sexual contacts with the infected person. E.g. Retrovirus, human papillomavirus, etc.

Transfusion-transmitted infections- Transmission of the virus through blood transfusion. E.g. Hepatitis B virus, Human Immunodeficiency Virus, etc.

Zoonoses -Transmission of the virus through the biting of infected animals, birds, and insects to human. E.g. Rabies virus, Alpha virus, Flavivirus, Ebola virus(emerging zoonotic disease) etc

TYPES OF VIRUS

Viruses are categorized into family and genus based on related features, **such as size, shape and the type of genetic material they carry**. Some common types of viruses include:

Influenza viruses (Orthomyxoviridae): The **Orthomyxoviridae** family of viruses includes influenza A and B, which cause the flu. Strains of influenza A also causes avian flu (bird flu) and swine flu (H1N1).

Human herpesvirus (Herpesviridae):
Herpesviridae is a large family of viruses. It have two variants 1 and 2. They cause several types of illnesses, like oral and genital herpes, chickenpox, shingles, Epstein-Barr and cytomegalovirus (CMV).
Coronaviruses:
These are a large family of respiratory viruses that includes Severe Acute Respiratory Syndrome COV-1(2002-2004) and Severe Acute Respiratory Syndrome COV-2, Middle east respiratory syndrome (MERS), Named due to their crown-like spikes on their surface. They have symptoms such as fever, cough, sore-throat, runny nose, and can progress to more serious ailments such as pneumonia or bronchitis.
Human papillomavirus (HPV):
Human papillomaviruses are part of the Papillomaviridae family of viruses. They are small, non-enveloped DNA virus that infects the skin or mucosal cells. This causes wart and some types of cancer.
Enteroviruses:
Enterovirus is a genus (one level smaller than the group called a family) of viruses that infect your intestinal tract. Types of enteroviruses cause polio virus disease, hand, foot and mouth disease.
Flaviviruses:
Viruses in this genus are often spread by **mosquitoes**. They cause illnesses like Zika virus infection(by **Aedes mosquitoes),** West Nile (by **Culex mosquitoes),** dengue fever (by **Aedes mosquitoes)** and yellow fever (by **Aedes mosquitoes)**.
Orthopoxviruses:
Viruses in the genus Orthopoxvirus causes flu-like symptoms and blistering rashes. examples are **Mpox, Cowpox .**
Hepatitis viruses:
Though they don't all belong to the same family or genus, hepatitis viruses all infect your liver leading to inflammation but they differ in modes of transmission, severity of the illness and prevention methods. There are **Hepatitis A,B,C,D,E**. **Oncoviruses:**
Oncoviruses are viruses that can cause cancer. Viruses that have been linked to specific cancers include: Human Pappilloma Virus, Human herpes virus 8 (HHV-8).
TYPES OF VIRAL INFECTION
Respiratory infections: Respiratory infections affect the nose, throat, lungs and airways. Some respiratory viruses can cause bronchitis, Sinusitis, Ear infections, pneumonia. Examples include Common cold, The flu(Influenza), Covid'19 etc.
Digestive systems infection: Viral infections in your digestive system can affect the stomach, and intestines or the liver. They can cause stomach flu(gastroenteritis).Examples: Norovirus, Rotavirus ,they can also cause chronic liver disease.egs. Hepatitis virus.
Viral hemorrhagic fever: These viral infection can affect how blood clots and can weaken the blood vessels, causing potentially life threatening bleedings. examples: Ebola(mainly fruit bats), yellow fever, Lassa fever.
Sexually transmitted infections: These are diseases mainly spread by sexual contact(anal, oral, vaginal). Examples: **HIV, Human Papilloma Virus, Hepatitis B, Herpes simplex virus**.
Exanthematous infections: These viral infections cause rashes that can appear as bumps, or rashes on the skin or spots of blood under the skin. They can also cause respiratory or other symptoms. Examples: Measles, Chicken pox, Mpox and other related infections.

Neurological infections: These viruses attack the cells of the nervous system (brain and spinal cord).It can cause paralysis, swelling (inflammation) of the brain or brain covering. Examples: **Polio, Rabies** etc.

Congenital infections: These are viral infections that occur from birth(present from birth). The pregnant woman passes it to her foetus while it is developing or during child birth. It can cause vision or hearing impairment, delays in development, and neurological conditions. Examples: Zika virus, Rubella(German measeles).

DISEASES CAUSED BY VIRUS

Common cold:

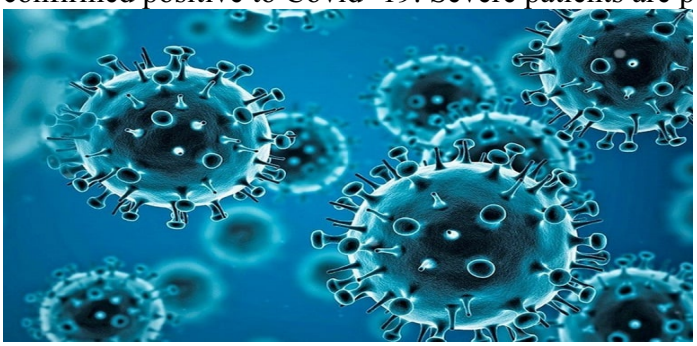
This is a viral infection that affects the upper respiratory tract, it can be caused by a variety of viruses but the most common cause is the rhinoviruses and it is transmitted from person to person or surface to surface. It presents gradual nasal symptoms such as stuffy nose, sneezing, loss of smell or taste, congestion, watery nasal secretions, sore throat, cough, fatigue, chills e.t.c.

The flu (influenza):

This is a respiratory ailment, an infection of the nose, throat and lungs. The causative organism of flu is influenza virus and It presents sudden symptoms such as fever, muscle aches, chills, sore throat, headache, shortness of breath, Dry persistent cough etc. Also dehydration, seizures, in children. Certain group of people are at higher risk of having flu complications than the others - Young children below one year, pregnant women, adults older than 65 years old, those taking care of the sick especially in the hospital, generally ,those with weakened immune system.

COVID-19:

Cooronavirus disease(Covid-19) is an infectious disease and are a subfamily of viruses. Covid-19 is caused by **SARS-CoV-2** (Severe Acute Respiratory Syndrome Corona Virus 2).They are crown like appearance. It spreads between people in close contact presenting common symptoms such as fever, chills, sore throat, persistent cough, shortness of breath, appetite loss, loss of taste and smell, difficulty sleeping etc. People who have preexisting medical conditions or those with compromised immune system are at higher risk of mortality when they have Covid-19. Examples are those with chronic heart, lungs or liver problems, HIV, diabetes, cancer or dementia patients etc. There are numerous measures towards management of Covid-19 infections, There is vaccine for Covid-19 which provide strong protection against serious illness, hospitalisation and death, Crowded places should be avoided, properly fitted mask should be worn, hands should be cleaned regularly with alcohol swab(or hand sanitizer with at least 60% alchohol), regular washing of hands with soap and clean running water as well as self isolation if confirmed positive to Covid -19. Severe patients are put under ventilators to prevent death.



Covid -19 virus

Respiratory syncytial virus :

This is a common respiratory virus that infects the nose, throat, lungs and breathing passages. usually causes mild, cold-like symptoms. RSV can cause serious illness in mostly infants, older

adults and people with compromised immune systems. It spreads through contact with respiratory droplets(sneezing, coughing or kissing) from an infected person or touching surfaces contaminated with the virus and then touching your eyes, nose, mouth. And has symptoms such as congested or runny nose, dry cough, sore throat, sneezing etc RSV is a vaccine preventable disease.

Chickenpox:

This is a highly contagious viral infection causing an itchy rash with small fluid-filled blisters on the skin. It spreads through. This disease mostly affects those who haven't had the disease or who haven't been vaccinated against it. The causative organism of Chicken pox is **Varicella - zoster** virus and it is transmitted from person to person through bodily fluids(coughing and sneezing, secretions from eyes, nose, mouth) and bodily contact and presents symptoms such as low grade fever, headache, an itchy blister-like skin rash, scabs after the blister breaks.

Measles (Rubeola):

This is an extremely contagious airborne disease causative organism of measles is an RNA virus called morbillivirus. It presents symptoms that usually appear 7-14 days after exposure, and it includes a fever (possibly up to 40°C), cough, runny nose, sneezing, watery eyes, body aches, white spots in the mouth and a red rash appearing about 3-5 days after the onset of the symptoms. This usually appears on the face, spreads to the chest and back. It is transmitted from person to person when an infected person breathes, coughs, sneezes, or talks and droplets from these body parts are inhaled by person who doesn't have the virus. Vaccines offer effective protection against measles.

HIV/AIDS:

The causative organism of AIDS is a retrovirus **Human immunodeficiency virus(HIV)** and it is transmitted from person to person through sexual contact (unprotected sex, anal sex, oral sex), mother to child during pregnancy and delivery, breast milk, blood transfusion with an infected blood, exchange of body fluids using unsterilized or contaminated needles. This viral disease attacks the immune system by destroying the CD 4 Cells (CD4 T Lymphocytes) of the immune system thereby exposing the infected person to other deadly infections and diseases such as tuberculosis, other sexually transmitted infections and cancer as a result of the immune system being compromised by the HIV infection. (It cannot be transmitted or spread through saliva, sweat, tears, sharing towels and beddings, insect vectors, toilet seats .etc. But can be present in semen, breast milk, blood, vaginal secretions etc. They also suffer from stigmatization from the society as a result of HIV infection, this can lead to depression or even death. Though there is no cure for HIV/AIDS, the antiretroviral drugs are administered to drastically reduce the viral load on the patient.

Human Papillomavirus Virus/genital warts:

This is transmitted from person to person through intimate skin contact. It infects mostly the skin and epithelial cells. This is the most common sexually transmitted infection which causes warts on different parts of the body depending on the variety of infection. It transmitted through intimate skin to skin contact. Especially by direct sexual contact such as vaginal, anal, and oral sex. Some types/strains of HPV may not cause any health issues while some can lead to cold sores, genital warts on surrounding skin, inside the mouth or in the throat. even cancers of the cervix, anus and throat.(and penile).The strains that causes various cancer varies.

People infected with HPV are advised to seek medical assistance instantly if you notice any unusual bumps or lesions on the penis, scrotum or anus.

This is a viral infection that causes warts in various parts of the body. It spreads by sexual contact, it's usually a non curable chronic infection which can last for several years or lifelong. Some types of Human Papilloma virus can progress to cervical cancer, anal, mouth and throat cancers e.t.c. Getting vaccinated and receiving regular Pap smears can prevent cervical cancer.

Herpes Infection:

This is a common viral infection that causes painful blisters or ulcers. It is caused by **herpes simplex virus** and it spreads by skin to skin contact, though not curable but it can be treated. There are two types of HSV: Type 1(HSV-1) and Type 2(HSV-2).

TYPE 1(Oral herpes or cold sores)- This is spread by oral contact ,and it causes infection in or around the mouth. Most adults are re-infected by this type of HSV

TYPE 2-(Genital herpes)- This is spread by sexual contact and it causes genital herpes. This infection causes painful blisters or ulcers that can recur over time, medications can reduce the symptoms but can't cure the infections(It is stigmatizing, with impact on sexual relationship), though have mild or no symptoms. Though the recurrent symptoms of both oral and genital herpes may be distressing .

Oral herpes.

Poliomyelitis (Polio):

This is a disease caused by **polio virus**. It causes mild or no symptoms in most people, but it can cause paralysis or death. It is highly infectious viral disease that affects mostly children under the age of 5. There are about three variations of polio virus 1,2 and 3(2 and 3 have been eradicated).This virus is transmitted from person to person through the fecal -oral-route or less frequently by consuming contaminated food or water. This multiplies in the intestine from where it can invade the nervous system and cause paralysis. It's initial symptoms are fever, fatigue, headache, fatigue, vomiting, stiffness of the neck and pain in the limbs. There is no cure currently for polio but can be prevented with polio vaccine given multiple times and can protect the child for a lifetime. There are oral and inactivated polio(intramuscular) vaccines. Those at risk are Live or travel to places where polio is prevalent, poor sanitation areas, children under 5 years of age.

Rabies:

This is a zoonotic, viral disease affecting the central nervous system. Rabies spreads to people and animals via saliva, usually through bites, scratches, or direct contact with mucosa (such as eyes, mouth, open wounds) and it can be fatal. Dogs are responsible for the virus transmission of rabies in human. This infects mammals, dogs, livestock, wildlife. Causative organism of rabies is the rabies virus and other animals that can transmit rabies virus are bats, raccoons, foxes, jackals, cats etc.

Mpox (Formerly called Monkey pox):

This is a viral disease caused by the monkey pox virus(genus Orthopxvirus),it can be transmitted to humans through physical contact, kissing, or sex (vaginal, oral and anal) as well as from respiratory droplets,from an infectious persons, with animal when hunting ,skinning or cooking it, or contaminated materials such as bed sheets, clothes, needles, pregnant mothers may also pass the virus to their unborn babies. It presents symptoms such as severe skin rashes accompanied by fever, headache, muscle aches, back pain, swollen lymph nodes, low energy. Laboratory confirmation is by PCR(testing the skin lesion). Treatment is by supportive care, as well as vaccine help prevent the infection for people at risk.

Zika Virus:

This is a viral disease mostly transmitted through the bite of an infected aedes mosquito, mostly found in the tropics and subtropics. It can also be transmitted from mother to foetus during pregnancy which can lead to certain birth defects, during sexual contact, blood transfusion as well as organ transplants. It's mostly asymptomatic(sometimes present rashes, fever, headache, muscle and joint pain) The main management is on relieving the symptoms and this includes rest, oral rehydration, fever and pain medications such as NSAID's-Non steroidal anti-inflamatory drugs.

Hepatitis:

This refers to an **inflammatory condition of the liver** as a result of viral infection, autoimmune hepatitis(when the body makes antibodies against the liver tissues) as well as hepatitis that occur as a result of medications, drugs(abuse), toxins(exposure), and alcohol which can lead to liver damage as well as inflammation. The five main viral classifications of hepatitis are
Hepatitis A, B, C, D, E, F . Hepatitis A and E causes acute infection, while B, C, and D causes both acute and chronic infections.
The common route of transmission of these viruses are
Hepatitis A-Exposure to Hepatitis A Virus in food or water.
Hepatitis B- contact with Hepatitis B Virus in body fluids, such as blood, vaginal secretions or semen.
Hepatitis C- contact with Hepatitis C Virus in the body fluids(blood, vaginal secretions or semen).
Hepatitis D- Contact with blood Hepatitis D Virus.
Hepatitis E-exposure to Hepatitis E Virus in food or water.

Ecological Adaptations

Adaptation is the trait that enables an organism survive and reproduce in the environment it finds itself. Such trait or Adaption makes the organism fitted to their environment.

An ecological adaptation is any morphological, physiological, or behavioral attribute or trait that an organism has, that enables it survive and reproduce in its habitat.

There are three main types of ecological adaptation based on how organisms express their genetic traits. These are structural, behavioral, and physiological adaptation.

1. Structural adaptation has to do with the traits of how an organism's body is, such as color, skin, and shape of the body.
2. behavioral adaptation has to do with what the organism does or how it acts
3. physiological adaptation has to with mechanisms found in organisms that allow them to undertake specific metabolic activities to live in their natural habitat

Plant Adaptations and Ecological Classification of Plants

Plants are adapted to different environmental conditions

A. Adaptation to Light

Plants are classified into two based on their adaptation to light

1. Heliophytes:
Theses are Plants adapted to high light intensity or bright light. These plants have high photosynthetic, metabolic and respiratory rates E.g. Sun plants.

2. Sciophytes:
These are Plants adapted to low light intensity. These plants have low photosynthetic, metabolic and respiratory rates. They grow well under shade E.g. ferns and several herbs growing on the forest floor under the canopy of trees and shrubs.

B. Adaptations to water

Plants adapted to growing in water, water-logged soil, wetlands or on substrata with an abundance of water are Aquatic Plants (or Hydrophytes).

Plants living in aquatic environment **face several challenges in the form of low oxygen content, lack of soil and nutrients and less sunlight.**

These aquatic plants are divided into the following five categories, depending on whether they come in contact with water, air and soil.

1. **Free-floating hydrophytes:** These plants float freely on the water. They are in constant contact with water and air, but not with soil. Examples: Eichhornia and Pistia. Lemna, Wolffia, Trapa, Azolla, Salvinia,
2. **Rooted floating hydrophytes:** The roots of these plants are buried deep in the ground, while their leaves and flowers float on the water. Soil, water, and air all come into contact with these plants. Victoria, *Potomogeton, Marsilea,* Nelumbo and Nymphaea are examples.

3. **Rooted submerged hydrophytes:** These plants are completely immersed in water, have roots in the soil and do not interact with air. Examples: Hydrilla and Vallisneria.

4. **Submerged floating hydrophytes:** These plants are completely immersed in water and do not interact with soil and air. Examples: Utricularia and Ceratophyllum.

5. **Amphibious hydrophytes:** These plants are both aquatic and terrestrial. They grow in shallow water. Examples: Sagittaria, Scirpus, Typha and Ranunculus. They are also called emergent hydrophyte

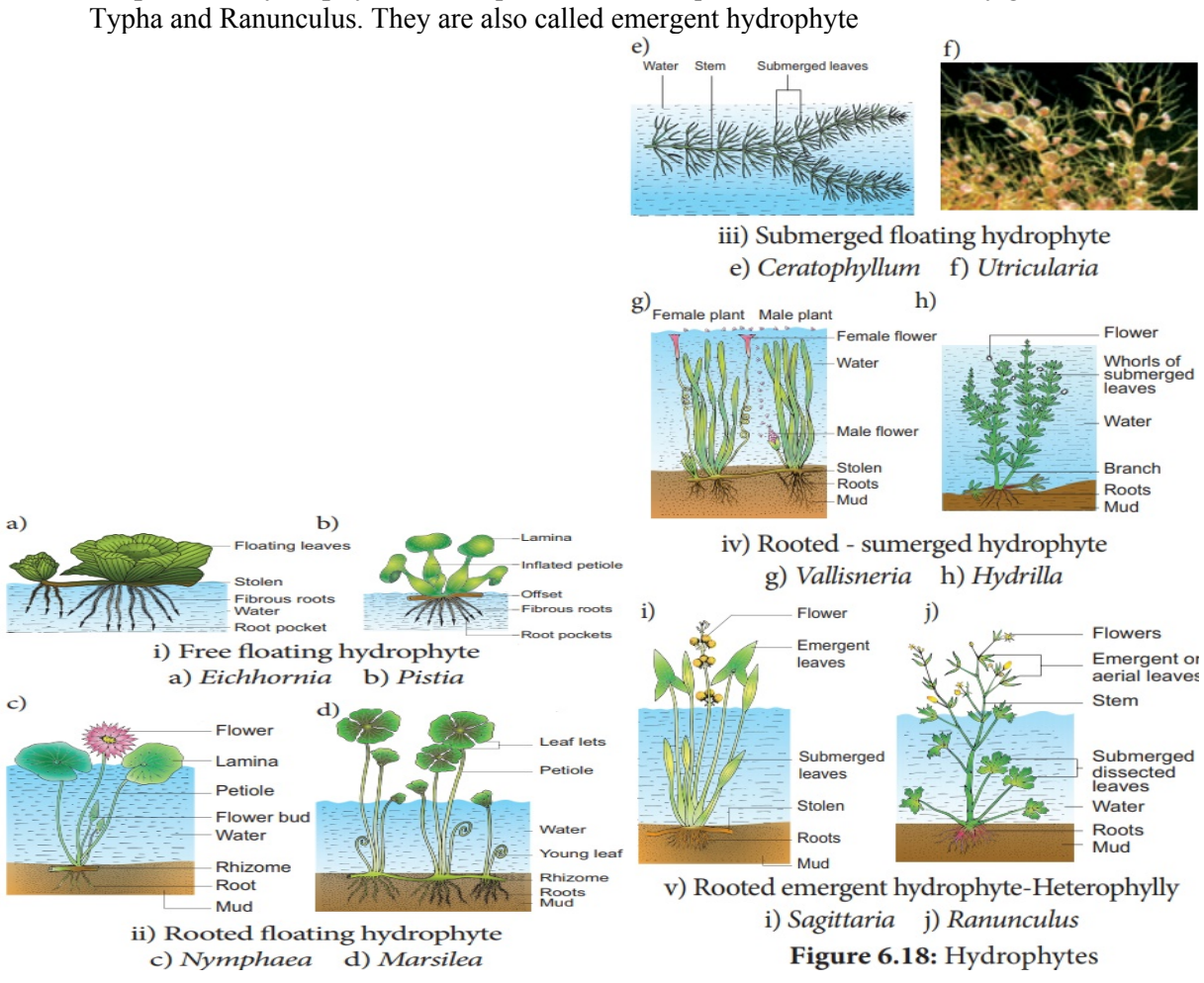


Figure 6.18: Hydrophytes

C. Adaptations to dry conditions

Plants adapted to growing in dry (or xeric conditions) or where the availability of water is negligible, are known as xerophytes.

There are different types of these plants

1. Ephemeral or Drought Escapers:
These plants complete their life cycle in a very short period only during rains. They live only for a period of 4-6 weeks, e.g., Tribulus terrestris, Solanum xanthocarpum.

2. Annuals or Drought Evaders:
The plants live for about 3 or 4 months, a few weeks longer than the rains. The size remains small, leaves have thick waxy, hairy coating with or without prickles to reduce transpiration, e.g., Echinops echinatus, Solanum surattense.

3. Succulents or Drought Resisting:
These xerophytes store water in their roots, stems, or leaves for drought periods. Succulents are stored in stems (Opuntia, Cacti), leaves (Aloe, Agave), and roots (Asparagus sp.).

4. Nonsucculents or Drought Enduring:
These are true xerophytes that actually endure drought conditions. They suffer from both external and internal dryness, e.g., Acacia, Ziziphus, Calotropis, etc.

5. Drought Avoiding Plants:
The life cycle of these plants is very short and these ephemerals complete their life cycle before summer comes, e.g., Argemone.

D. Adaptations to salty conditions

Plants adapted to growing in saline habitats are called halophytes.

The saline habitats have high concentrations of salts in soil or water. They occur in tidal marshes and coastal dunes, mangroves, along ocean edges and saline soils.

Examples of plants adapted to saline habitat are Rhizophora, Avicennia and Sundarban
The important adaptations of halophytes are as follows.

E. Adaptations of Plants to normal condition

Land plants adapted to grow under average conditions of temperature and moisture are called Mesophytes e.g Hibiscus

F. Adaptations of Plants to cold in Polar Regions

Polar Regions include one of the coldest regions on earth. And the vegetation found here is quite sparse.

Sedges, dwarf shrubs, grasses and certain mosses and lichens are some plant species found in these places.

The special features of these plants include short stature (not more than 12 inches), hairy parts, darker colour etc.

Only plants with shallow roots grow in the tundra because the deep layers of permanently frozen ice prevent the roots from breaking through the permafrost

Sometimes, trees hold on to dead leaves for insulation. However, larger plants such as trees as uncommon as the roots cannot penetrate deeply because of the permafrost.

G. Adaptations of Plants against Herbivory

Herbivory is the consumption of plant matter by any organism.

Since plants are the primary producers in an ecosystem, nearly all consumers depend on them for sustenance.

Therefore, to prevent herbivory, plants developed thorns, spines and chemicals.

Thorns are the most common form of deterrent.

Other plants might use chemicals to leave an unpleasant taste in the mouths of the herbivores. But some of these chemicals are very toxic and might result in the death if consumed.

Animal Adaptations

Like plants, animals also adapt to different environmental conditions to survive and flourish.

Animals have also developed a number of techniques in their bodies to meet the challenges of the environment in which they live.

Animals have adaptations to Particular feeding habits like carnivory and herbivory, Protection from predators, Camouflage for easy predation, Structural and behavioural adaptation to attract mate, Physiological and behavioural adaptation to environmental variations and stress conditions, e.g., migration, hibernation, aestivation, etc.

Some forms of animal adaptations are

1. Migration

It is a movement of an animal group to other places for food, climate, and other reasons.

Migration is of three types – daily, seasonal, and periodic.

The distance may be short or long. Long-distance migration is observed on Arctic Terns. They nest close to the North Pole in summer, fly south to Antarctica in autumn, and return North Pole again in each spring. Short-distance migration is observed in whales. Migration occurs in locusts when their number increases beyond the feeding capacity of the homeland.

2. Camouflage

Some animals have the capacity to blend with the surroundings or background. It is called camouflage.

It is a common adaptation in animals to remain unnoticed for protection or aggression, e.g., many insects, reptiles, and mammals.

It is difficult to distinguish leaf-like grasshoppers or Praying Mantis from the surrounding foliage. Similarly, stick insects, leaf insects and dead leaf butterflies can not be noticed unless and until they show movement.
Desert animals like lizards, camels, and lions are sand coloured and hardly visible against their background unless they move. The chameleon can change its colour according to its background. Camouflage is protective of animals that are preyed upon by others. It is advantageous in predation for predators like Preying Mantis as they remain unnoticed till the prey comes within their striking range.

3. Mimicry

The phenomenon of producing similarities with nonliving objects by which a living animal or plant species may conceal itself in the surroundings or obtain protection from enemies is known as mimicry.

Mimicry may be achieved either by altering the physical appearance or by changing behaviour. The living form which undergoes mimicry is known as mimetic and the object with which similarity is established is known as model.

Classification of Mimicry:
Mimicry may be categorized into three types namely, Protective mimicry, Aggressive mimicry, and Conscious mimicry.

1. Protective Mimicry:

To get protection from the predator animals, the mimicry shown by the animals is known as protective mimicry. This may again be of two types concealing mimicry and warning mimicry.
(a) Concealing mimicry:
Sometimes the animals try to conceal themselves in the surroundings so as to evade getting noticed by the enemies. In this case, the mimetic organism develops a similarity with an inanimate object either in physical appearance or in colour and because of this, it becomes difficult to locate the mimetic animal.

Examples:
Cryptolithode (a white-coloured crab) takes the shape of pebbles in the seashore and therefore they are difficult to be recognized.
A **type of butterfly named Papilio machaon** produces a pupa that when sticks to a tree branch appears green. Again when it attaches to the tree trunk or stone it becomes grey in colour.
The structure of the body and colour of the **butterfly Kallima paralekta** becomes such that it appears to be a dry leaf.
The **larva of the geometric moth, Selenia tetralunaria** when remains attached to the branch of the tree appears to be a twig.
The body shape and colour of the **insect Phyllium** appear like a fungus-infected green leaf.
The body structure of a stick insect is like that of a dry twig of a plant.
Arisaema is known as the Snake plant. In this plant, the exterior part of the inflorescence looks like a greenish-violet colour with the appearance of a cobra snake.

(b) Warning mimicry:
With the help of this mimicry, an animal may get protection from an enemy. In this case mimetic animal is usually non-poisonous, and this animal may take the shape of the model organism which is harmful or poisonous. In this case, both the body shape and colour of the organism may be altered. As a result of this, the enemies take the mimetic animal as harmful or poisonous.

Warning mimicry is of two types: i. Batesian and ii. Mullerian mimicry

Examples:
i. **Batesian mimicry** is when an unprotected and edible animal mimic another animal that is protected and unpalatable, the mimicry is termed as **Batesian mimicry**.

The **mimicry of butterfly Pieris** mimics tasteless Helicornius and gets protection from the predator animal.

ii. Mullerian mimicry is when two or more unpalatable species of animal mimic each other.

The **Ctenuchid moth** shows Mullerian mimicry by mimicking a type of wasp. In this case mimetic animals and models both get privilege.
Besides Batesian and Mullerian mimicry, there are certain examples of other warning mimicry.
Some non-poisonous coral snakes of the family Colubridae take the colour of the snakes of the poisonous Elaphidae family.
Some spiders live together with ants and take the colour ants.
Sesia crabroniformis, a type of moth mimics the dreaded wasp, Vespa crabro.

2. Aggressive mimicry:

This is the mimicry of the predator animals for capturing the prey.

The mimicry in this case helps the mimetic animal in attack instead of defence. The mimicry of this type is termed as aggressive mimicry.

Aggressive mimicry may be of two types such as i. Concealling mimicry and ii. Alluring mimicry.

i. **Concealing mimicry**: this is where the predator animal remains in ‘such a disguise that the prey cannot recognize it in the environment. As a result, the prey may easily be caught by the predator animal.
e.g., **some spiders** take the colour and shape of the flower when it remains within the flower. This helps the spider to conceal itself in the environment.

ii. Alluring mimicry: this is where the predator animal remains in such a disguise that the prey becomes attracted to it.
The **American horned toad Ceratophrys** sits quietly and rotates one finger over the head. Because of this the prey animals become attracted and go towards the toad. **The deep sea angler fish, Lophius** for capturing prey may exhibit alluring mimicry.

3. Conscious mimicry:

Some animals pretend as dead when they realize the possibility of any danger. Such mimicry is known as conscious mimicry. **American opossum, Didelphis**, stays like a dead animal when it senses any danger.

4. Colouration

In the living world, there is the existence of various colours. Like mimicry, animals may achieve adaptation through colouration also. However, in many cases, the animals may get protection through colouration.

Colouration may be divisible into two types namely concealing coloration and warning colouration.

i. **Concealing colouration**: With the help of such colouration the animals may conceal themselves within the surroundings in such a way that it becomes difficult to find them. If mimicry is associated with colouration, it gets special adaptation in nature.

ii. **Warning colouration**: The colouration, in this case, protects the animal in such a manner that the predator animal gets frightened to attack the prey. Batesian and Mullerian mimicry exhibit this type of mimicry. Dark spots on the wings of some moths confuse the predators and they get protection from this.

5. Echolocation

Bats are nocturnal flying mammals that do not employ eyesight for the location of their path, food, place of rest, etc.
They produce high-frequency sound which produces echoes after striking various objects on the principle of sonar.
Echoes are analyzed by the bats to know their path. Echolocation is also used by other animals, e.g., whales.

6. Adaptation to Excessive Cold (Cold Hardening)

Animals have also developed adaptations to extreme cold conditions.

Animals of colder areas **possess thick fur and sub-cutaneous fat**, they help in the conservation of heat.

During excessive cold, they may undergo hibernation. However, sea animals can not undergo **hibernation**.

Sessile animals cannot migrate. Certain sessile animals, such as barnacles and mollusks which live in very cold intertidal zones or northern shores, several insects and spiders resist the effect of cold by a process, known as **cold hardening**. Such animals are called **freeze-tolerant animals**. They possess ice-nucleating proteins, which induce ice formation in the extracellular spaces at very low sub-zero temperatures.

Some animals can tolerate environmental temperatures below 0°C by **accumulating glycerol or antifreeze proteins that lower the freezing point of their body fluids**. Such animals are called **freeze-avoiding animals.** The **fishes in the Antarctica region** remain active in cold seawater due to the presence of antifreeze compounds.

7. Adaptation to Water Scarcity

The **animals, which live in dry or xeric habitats**, are called **xerocoles**.

The xerocoles are of two types: i. drought evaders and ii. drought resistants.

i. **Drought evaders**: The drought-evading xerocoles adopt either a short annual life cycle or undergo aestivation or some other dormant state during the dry season.

The eggs and pupae of many insects (ants, bees, butterflies) and eggs of some invertebrates may remain dormant for several months during the dry season. In dry habitats, the birds breed and make nests during the rainy season, when there is abundant food.

ii. Drought resistants: These xerocoles can carry out their normal function throughout the year. They are able to resist aridity and heat through morphophysiological adaptations or by modifying their feeding activity patterns.

Some animals avoid heat by adopting nocturnal habits and remaining underground during the daytime.

Kangaroo and desert rats seldom drink water. They possess thick coats to minimize evaporative desiccation. 90% of their water requirement is met from metabolic water while 10% is obtained from food. Loss of water is minimized by producing nearly solid urine and feces.

Camel has a number of adjustments to desert conditions economical in water consumption, tolerance to fluctuations of temperature, and no sweating till body temperature rises to 55°-66°C. The animal produces dry feces and concentrated urine. During periods of non-availability of water the animal stores urea and does not produce urine. When water is available, a camel can rehydrate itself quickly by drinking a large quantity of water, some 80 liters in 10 minutes. Birds and reptiles in the desert retain water as their metabolic wastes are excreted in the form of an insoluble white compound called uric acid.

Cold-blooded animals entirely lack sweat glands as they rely on the external environment to regulate body temperature

8. Behavioural Adaptations in Desert Lizards

Desert lizards lack the physiological ability that mammals have to deal with high temperatures.
They keep their body temperature fairly constant by behavioural means.

They enjoy the sun and absorb heat when their body temperature drops below the comfort zone, but move into shade when the surrounding temperature starts increasing.

Some species are capable of burrowing into the soil to hide and escape from too much heat. Therefore, not only desert lizards but almost all reptiles are called cold-blooded animals, as they cannot maintain a constant inner temperature. So, behavioral adaptation helps them to do so.

Other Sources of Water
Natural sources of water, such as lakes and river are almost non-existent or are seasonal. So, animals derive water from desert plants such as cactus. Some insects also tap fluids such as saps and nectars from various parts of the plants.

Specialized Appendages
Animals like the jackrabbit have very large ears that have a network of blood vessels. When these animals rest in the shade, their enormous ears dissipate the excess heat from their bodies.

LECTURE NOTE ON BIO 102 (GENERAL BIOLOGY 2) SECOND SEMESTER 2024

BY Dr N.V. Elosiuba

FUNGI

Up until the mid-20th century many Scientists classified fungi as plants. Until recently the biology of fungi was poorly understood. Based on fossil evidence, bfungi appeared in the Precambrian era, about 450 million years ago. Fungi, like most plants are sessile and seem to be rooted to a spot. They possess a stem like structure similar to plants, as well as having a root-like fungal mycelium in the soil. The field of Mycology, which is the scientific study of fungi, brought a lot of progress in field of fungal biology.

The word fungus comes from the Latin word for mushrooms. The Kingdom Fungi includes a large group of living organisms collectively referred to as Eumycota (True fungi). About 100, 000 species of fungi have been identified by scientists which are only a fraction of about 1 million species probably present on Earth. Organisms like mushrooms, yeasts, mould of all types are all fungi. *Penicillium notatum* used in the production of Penicillin drug is also a member of the Kingdom Fungi.

Fungi were once considered as plants but, molecular studies have thrown more light and it was discovered that fungi have more of animal characteristics than plants hence they were carved into their own separate kingdom. Fungi play very important roles in the ecosystem. They are decomposers and play great roles in the cycling of nutrients, by breaking down organic materials into simple molecules. They often interact with other organisms to form mutualistic associations. They could also be harmful, causing diseases in plants, animals and humans

BASIC CHARACTERISTICS OF Fungi

Fungi are eukaryotic organisms with a complex cellular organization. Their nucleus is membrane bound. The DNA in the nucleus is wrapped

around with histone proteins (which package DNA into structural units called nucleosomes). Few fungal cells have structures that can be compared to bacterial plasmids (loop of DNA) however there is no horizontal transfer of genetic information as is often the case in plasmids.

Fungi contain mitochondria, endoplasmic reticulum and golgi apparatus. They do not have chloroplasts and chlorophyll; however they have other cellular pigments giving them multiple colours ranging from red, green to black. The pigments in fungi are associated with their cell wall and function protectively against U.V. radiations. Some of these pigments are poisonous. Fungal cells have thick cell walls made of chitin and glucans which give structural strength to the cell walls. Chitin is similarly found in the exoskeleton of insects.

Fungi have plasma membrane similar to other eukaryotes except that the structure is stabilized by ergosterol a steroid molecule replacing the cholesterol found in animals cell membranes. Fungi are non-motile, it is only the gametes of the primitive phylum Chytridiomycota that produces flagella. Most fungi are multicellular, while some are unicellular example the yeast cells. They display two distinctive morphological stages;

- The vegetative stage
- The reproductive stage

They reproduce by means of spores. They exhibit alternation of generation (they alternate between two different life stages or generations in their life cycle) a haploid stage (gametophyte) a diploid stage (sporophyte). They develop from the spores, they have no embryonic stage. Reproduction can be by sexual or asexual means.

Fungi have a filamentous structure except in yeast cells (Yeast cells are single celled). These filamentous/thread-like structures are known as hyphae, which come together to form a mesh-like structure called mycelium. Mycelium is a network of fungal threads or hyphae. It often grows underground. The fruiting bodies of fungi such as mushrooms sprout from mycelium.

CLASSIFICATION OF FUNGI:

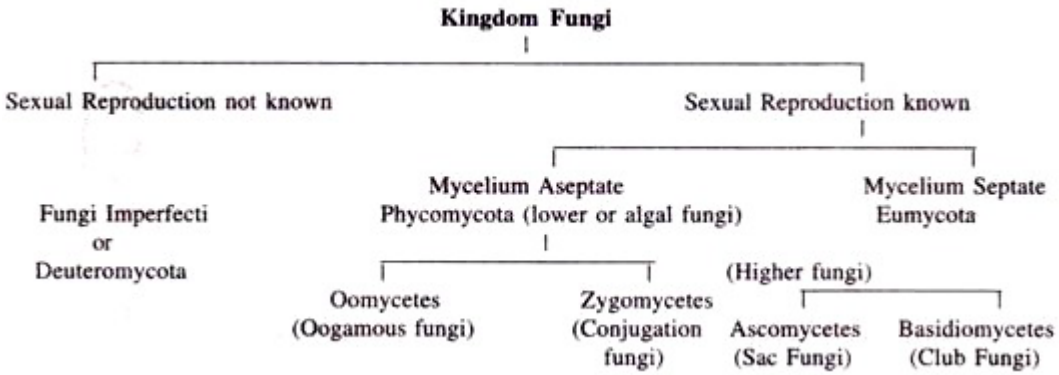
Based on mode of nutrition, fungi can be classed into 3.

Saprophytic mode of nutrition: These saprotrophs feed on dead and decaying organic matter. Examples include *Rhizopus stolonifer* – the black mould of bread. *Penicillium chrysogenum* – used in the production of penicillin antibiotic. Aspergillus – a spore forming mould fungus.

Parasitic mode of nutrition: these live on animals and plants causing harm to them. Examples include *Taphrina deformans* – causing peach leaf curl disease (phyto-pathogenic fungi). Those that live on animals are called zoo-pathogenic fungi. Examples include ringworm infections, epidermophytosis and *Tinea vescicolor, Candida spp*. Those that live on insects are called entomopathogenic fungi.

Symbiotic mode of nutrition: Interdependent relationship whereby both organisms are mutually beneficial e.g. Lichens (relationship between algae and fungi) and Mycorrhiza (relationship between fungi and plants).

Fungi can also be classified based on their spore formation and reproductive structures.



1. Deuteromycota – The fungi imperfecti

This is an artificial class of fungi created to include all those fungi whose sexual stage is either absent or not known. Some are unicellular like yeast cells, but most are multicellular with septate mycelium. They reproduce asexually by spore formation (Conidia formation). They live mainly on land. Most food spoilages are caused by this group of fungi and are also responsible for most human fungal infections. Examples include red-rot of sugar cane caused by *Colletotrichum falcatum. Helminthosporium oryzae*, the leaf spot of rice, early blight of potato and tomatoes etc. They occur as saprophytes on a wide range of substrates but a large number of them are parasites of plants and animals.

2. Oomycota - (Oogamous fungi)

Mycelium is coenocytic (multinucleate and aseptate). Hyphal wall contain cellulose and in some cases chitin. Asexual reproduction involves spore formation contained in sacs or sporangia. In aquatic conditions sporangia produce zoospores, but in terrestrial condition sporangia behave as spores known as conidia hence the name conidiosporangia. Sexual reproduction is by gametangial contact in which the male sex organ or antheridium passes its product into the female sex organ oogonium or archegonium through a fertilization tube. The product of sexual reproduction here is **oospore. Examples** include blight diseases, white rust disease, damping off, downy mildew, etc.

3. Zygomycota – (The conjugation fungi)

Their mycelium is coenocytic. Hyphal wall contain chitin or fungus cellulose. There is absence of motile cells (zoospores and plasmogametes). Mitospores are non-motile and are called sporangiospores. These are terrestrial fungi which are mostly saprotrophic and rarely parasitic. Sexual reproduction occurs through gametangial copulation or conjugation, hence they are called conjugation fungi. Sexual reproduction gives rise to to **zygospore** (diploid spore). Examples include squirting fungus (*Pilobolus crystallins*), *Rhizopus stolonifer* – black bread mould, *Mucor caninus*

– dung mould, both are saprotrophic.

4. Ascomycota – (The sac fungi)

This is a class of diverse fungi having over 30000 species. They are saprotrophs, decomposers, coprophilous (living on dung) and parasitic. They have mycelium with septate hyphae, with the exception in yeasts. Cell wall contains chitin and fungus cellulose. No motile structures in their life cycle. In most ascomycetes asexual reproduction are by formation of conidia borne on conidiophores, which maybe branched or unbranched. We find them growing on bread, citrus fruit, and old leather etc. Examples are *Penicillium* and *Aspergillus*. They also include pigmented moulds (brown, green, blue, pink), powdery mildew, yeasts, morels and truffles. Sexual reproduction can take place in this group also.

5. Basidiomycota – (the club fungi).

These are the most advanced and most commonly seen fungi as their fructifications are large and conspicuous. The class contains about 25,000 species. They are the best wood decomposers. They can decompose both lignin and cellulose with the help of secreted enzymes. They have no motile structures or cells. They have two types of mycelia, 1^0 and 2^0 mycelium. Sexual reproduction is by Plasmogamy (fusion of protoplasts without fusion of nuclei. Sex organs are not involved. Examples include: Rusts – *Puccinia graminic tritici* (black rust of wheat), *Puccinia glumarun* (yellow rust of wheat). Smuts: *Ustilago maydis* – (smut of corn). *Ustilago triciti* – (loose smut of wheat). Mushrooms: with umbrella-like basidiocarp – *Agaricus campestris, Voluariella volvacea* – (paddy straw mushroom). Toadstool: these are non-edible, often poisonous with white spores *Amanita acesarea* (Caesaer' mushroom). *Amanita phalloides* (death cap), Bracket fungi or shelf fungi also known as **polypores.**

BIO 102

VIRUS

- **CHARACTERISTICS OF VIRUS**
- **METHODS OF IDENTIFICATION OF VIRUS**
- **CLASSIFICATION OF VIRUS**
- **DISEASES TRANSMITTED BY VIRUS**

1. CHARACTERISTICS OF VIRUSES

i. **Virus posses acellular nature**: Viruses are not made up of cells or cell components instead rather virus exists as a virion when it's outside of a host cell. The virion is the infectious form of the virus (consisting of genetic material DNA or RNA) enclosed within a protective protein coat called capsid. Viruses are considered acellular because they do not have a cellular structure. This means they lack the typical components of a cell, such as organelles, ribosomes, a plasma membrane etc.

ii. **Genetic Composition**: Each virus contains either DNA or RNA as its genetic material, but never both. This nucleic acid of virus (DNA and RNA) can be either single-stranded or double-stranded. Eg. ssDNA, dsDNA, ssRNA, dsRNA. This type and structure of the nucleic acid can influence the virus's replication strategy, genetic stability and interaction with host cells.

iii. **Dependence on Host Cells**: Viruses are obligate intracellular parasites that rely entirely on a host cell to reproduce and perform metabolic functions. Therefore they cannot reproduce or carry out metabolic activities without infecting a host cell. Viruses are extremely small, usually ranging from 20 to 300 nanometers, and are much smaller than most pathogens.

iv. **Protective Protein Coat**: The viral genetic material is surrounded by a protective protein coat called the capsid. The capsid is the outer protein shell of a virus. It surrounds and protects the viral genetic material (DNA or RNA) from damage.

v. **Presence of an Envelope**: Some virus possess a lipid envelope derived from the host cell membrane that surrounds the capsid. A viral envelope is a lipid membrane that surrounds the capsid of certain viruses. It is typically derived from the host cell's membrane during viral replication or budding.

vi. **Host Specificity**: Viruses exhibit specificity for their hosts, usually infecting only particular species or cell types based on receptor compatibility. Viruses can only infect certain hosts or specific types of cells because they need to bind to particular receptors on the surface of those cells. This virus has proteins that fit only certain receptors on host cells. Examples: HIV (Human Immunodeficiency Virus): Infects mostly human immune cells that have the CD4 receptor and a co-receptor (CCR5 or CXCR4). Influenza Virus infects respiratory tract cells because it binds to sialic acid receptors on those cells. Rabies Virus: Infects nerve cells because it binds to receptors found on neuronal cells.

vii. **Lack of Metabolic Activity**: Viruses do not perform independent metabolism; they neither respire nor grow on their own.

viii. **High Mutation Rate:** Viruses replicate very quickly, often producing millions of copies in a short time, especially among RNA viruses, allowing viruses to evolve

quickly and adapt to changing environments. The more the replication, the higher the chances of mutation

2. METHODS OF IDENTIFICATION OF VIRUSES

Identifying viruses is a crucial step in diagnosing viral infections, conducting epidemiological surveillance, and developing targeted treatment strategies. Viruses can be identified by detecting the whole virus, its components, or the host's immune response. The main methods are:

1. **Virus Isolation and Culture**: This involves growing viruses in cell cultures, eggs, or animals and observing cytopathic effects. It's the gold standard but time-consuming (e.g., Herpesvirus, Influenza).
2. **Electron Microscopy (EM):** Electron microscopy is the visualization of virus particles to determine size and shape, useful for rapid identification of large or unknown viruses (e.g., Poxvirus, Adenovirus).
3. **Detection of Viral Antigens**: This method involve identifying viral proteins in samples using assays like immunofluorescence and ELISA for early diagnosis (e.g., Influenza, Respiratory Syncytial Virus-RSV).
4. **Detection of Viral Nucleic Acids**: Detection of Viral Nucleic Acids using molecular techniques such as PCR and next-generation sequencing to detect viral DNA or RNA with high sensitivity (e.g., HIV, SARS-CoV-2
5. **Serological Methods**: Serological Methods detects antibodies produced by the host in response to infection to diagnose current or past infections (e.g., Measles, Rubella).
6. **Rapid Diagnostic Tests (RDTs):** These are point-of-care tools that provide quick results, typically within minutes. These immunochromatographic assays detect either viral antigens or antibodies and are highly useful in field conditions and emergency settings. Commonly used for HIV, hepatitis, and COVID-19, malaria etc

3. CLASSIFICATION OF VIRUSES

Viruses are classified based on several criteria including their b

1. CLASSIFICATION OF VIRUSES BASED ON THE TYPE OF THEIR GENETIC MATERIAL AND THEIR METHOD OF REPLICATION

This classification is called the Baltimore classification system. It categorizes viruses according to the type of their genetic material and their method of replication. This system, proposed by David Baltimore, divides viruses into seven groups based on whether their genome consists of DNA or RNA, whether it is single-stranded or double-stranded, and how the virus synthesizes messenger RNA (mRNA) during infection. The are classified as:

Group I: Double-Stranded DNA Viruses (dsDNA)
These viruses have a double-stranded DNA genome. They typically replicate inside the host cell nucleus using the host's DNA-dependent RNA polymerase to produce viral mRNA. Examples include adenoviruses and herpesviruses.

Group II: Single-Stranded DNA Viruses (ssDNA)
Viruses in this group carry single-stranded DNA genomes. Upon entering the host cell, their

ssDNA is converted into double-stranded DNA before transcription into mRNA occurs. Parvoviruses are typical members of this group.

Group III: Double-Stranded RNA Viruses (dsRNA)
These viruses contain double-stranded RNA genomes. They carry their own RNA-dependent RNA polymerase to transcribe the negative strand of their RNA into positive-sense mRNA. Reoviruses are examples.

Group IV: Single-Stranded Positive-Sense RNA Viruses (+ssRNA)
These viruses have single-stranded RNA genomes that act directly as mRNA. This means their RNA can be immediately translated by the host's ribosomes to produce viral proteins. Examples include poliovirus and hepatitis C virus.

Group V: Single-Stranded Negative-Sense RNA Viruses (-ssRNA)
Viruses in this group contain single-stranded RNA genomes complementary to mRNA. They must carry an RNA-dependent RNA polymerase enzyme within their virions to transcribe their genome into positive-sense mRNA after infection. Influenza virus and rabies virus are notable members.

Group VI: Single-Stranded RNA Viruses with Reverse Transcriptase (ssRNA-RT)
These are retroviruses that contain single-stranded RNA genomes but replicate through a DNA intermediate. They use reverse transcriptase to convert their RNA into DNA, which integrates into the host genome. HIV is a prime example.

Group VII: Double-Stranded DNA Viruses with Reverse Transcriptase (dsDNA-RT)
Viruses in this group have double-stranded DNA genomes but replicate via an RNA intermediate. They use reverse transcription during their replication cycle. Hepatitis B virus belongs to this category.

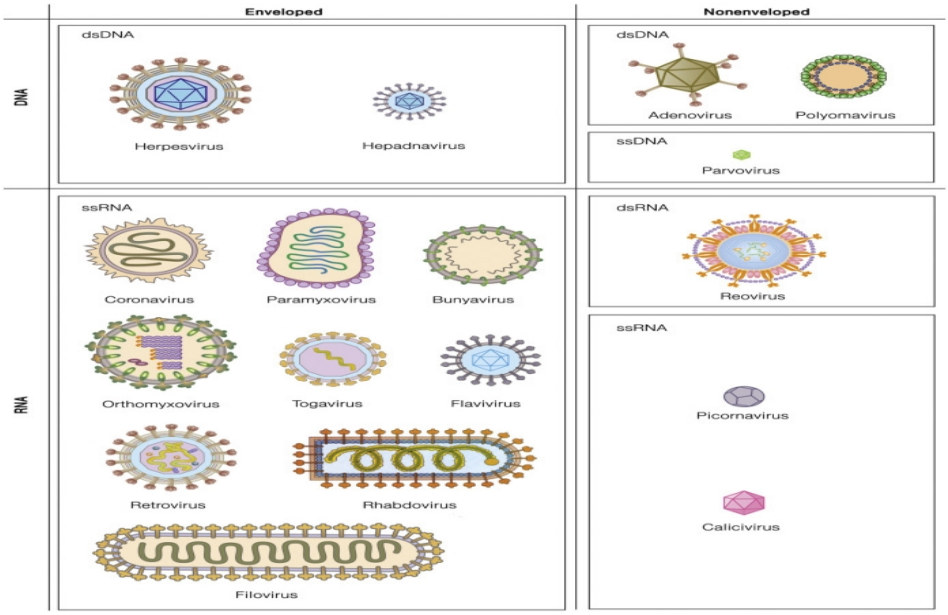


Fig. 1. Classification of Virus

2. CLASSIFICATION BASED ON SHAPE AND STRUCTURE

Viruses, though diverse in nature, can also be classified according to their shape and structural characteristics. Viruses exhibit different capsid shapes based on their structural organization.

Icosahedral viruses: Icosahedral viruses have capsid arranged in a symmetrical 20-sided shape (e.g., Adenovirus).

Helical viruses: The capsid proteins arranged in spiral or helical form (e.g., Tobacco mosaic virus).

Complex viruses: Complex viruses possess more intricate architectures that do not fit into either the icosahedral or helical categories (e.g., Poxviruses with a brick-like shape).

3. BASED ON PRESENCE OR ABSENCE OF A LIPID ENVELOPE

This is a fatty outer layer that surrounds some viruses. It is derived from host cell membrane. This characteristic affects how viruses are transmitted, how they enter host cells, and how they are protected in the external environment.

i. **Enveloped viruses**: These are viruses surrounded by a lipid membrane derived from host cells (e.g., Influenza virus (responsible for flu), Herpesvirus(causes cold sores,genital herpe).

ii.**Non-enveloped (naked) viruses**: These are the viruses that lack the lipid envelope. Non-enveloped viruses are often transmitted via the fecal-oral route or through contact with contaminated surfaces. (e.g., Adenovirus, Poliovirus-causative agent of polio)

4. VIRUS CLASSIFICATION BASED ON HOST SPECIFICITY

Viruses are incredibly diverse in terms of their hosts, and one of the most fundamental ways to classify viruses is based on the type of organisms they infect. Host specificity refers to the specific type of organism or cell that a virus is capable of infecting. This characteristic is crucial for understanding viral transmission, pathogenicity, and the development of treatments and vaccines. Viruses can be broadly classified into three categories based on their host specificity:

i. Animal Viruses

Animal viruses are those viruses that infect animals and humans. These viruses can cause a wide range of diseases, from mild illnesses to severe and life-threatening conditions. Animal viruses are highly adapted to their specific host cells, often targeting particular tissues or organs. For example, the Human Immunodeficiency Virus (HIV) targets immune cells such as CD4 cells, leading to acquired immunodeficiency syndrome (AIDS), while Influenza viruses affect the respiratory tract. These viruses may be transmitted through direct contact, airborne particles, or bodily fluids, depending on the nature of the virus.

ii. **Plant Viruses**

Plant viruses specifically infect plants, causing diseases that can severely affect crop yields and agricultural productivity. These viruses do not infect animals or humans but instead are specialized to infect plant cells. Plant viruses are usually transmitted by insects (such as aphids), through contaminated tools, or by infected seeds. One well-known example of a plant virus is the Tobacco Mosaic Virus (TMV), which infects tobacco plants and other species, causing characteristic mottling and distortion of leaves. Plant viruses often lack specialized receptors for cell entry; instead, they rely on mechanical transmission, such as through wounds in plant tissues or vector organisms. Plant viruses can result in significant agricultural losses and are a major concern in the food industry. Control measures often involve genetic resistance in plants, crop rotation, and vector control to minimize transmission.

iii. **Bacteriophages**

Bacteriophages are viruses that specifically infect bacteria. These viruses are highly specific to their bacterial hosts and can be used to target particular bacterial strains. Bacteriophages have an important ecological role in controlling bacterial populations, and they are considered natural regulators of bacterial growth. For example, the T4 bacteriophage infects *Escherichia coli*.

iv. Viruses that infect fungi are called mycoviruses (fungus virus). Mycoviruses are distinct from other viruses due to their unique interactions with fungal hosts.

METHODS OF IDENTIFICATION OF VIRUSES

Viruses can be identified through various methods that detect the virus itself, its components, or the immune response it triggers in the host. These identification techniques are essential for diagnosing viral infections, guiding treatment, and conducting epidemiological studies. The main approaches include virus isolation, electron microscopy, antigen detection, nucleic acid detection, and serological methods.

One of the primary methods is virus isolation, which involves growing the virus in susceptible systems such as cell cultures, embryonated eggs, or laboratory animals. During this process, the virus replicates within the host cells, often causing visible changes known as cytopathic effects (CPE). Virus isolation remains the gold standard for identifying many viruses because it confirms the presence of infectious virus particles. This method is commonly employed for viruses such as herpesvirus, adenovirus, and influenza.

Another direct method is electron microscopy (EM), which allows visualization of virus particles in clinical samples. By using a focused electron beam, EM provides detailed images of the viral morphology, enabling rapid preliminary diagnosis. This technique is particularly useful for detecting large or previously unknown viruses, with examples including poxvirus and herpesvirus.

Detection of viral components is also widely practiced, especially through the identification of viral antigens. Techniques such as the

Immunofluorescence assay (IFA), enzyme-linked immunosorbent assay (ELISA), and rapid antigen tests target viral proteins in patient specimens. These methods facilitate early diagnosis of infections and are commonly used for viruses like influenza, respiratory syncytial virus (RSV), and rotavirus.

Advancements in molecular biology have led to highly sensitive and specific techniques for detecting viral nucleic acids. Methods including polymerase chain reaction (PCR), reverse transcriptase PCR (RT-PCR), nucleic acid hybridization, and next-generation sequencing (NGS) allow direct identification of viral DNA or RNA from clinical samples. These molecular approaches are invaluable for diagnosing infections caused by viruses such as human immunodeficiency virus (HIV), hepatitis B virus (HBV), and SARS-CoV-2.

Serological methods play an important role in viral identification by detecting antibodies produced by the host in response to infection or by identifying viral antigens. Common serological tests include ELISA for IgM and IgG antibodies, neutralization tests, hemagglutination inhibition (HI) assays, and complement fixation tests (CFT). These techniques are useful for diagnosing both current and past infections as well as assessing immune status. Viruses frequently identified through serology include measles, rubella, and various hepatitis viruses.

SOME ROUTES OF VIRAL DISEASE TRANSMISSION

Viral diseases can spread through various routes depending on the virus and its characteristics. Understanding these transmission routes is key to developing strategies for prevention and control. Below are the main routes of viral disease transmission:

1. Airborne Transmission

- Route: Through the air via respiratory droplets when an infected person coughs, sneezes, or talks. Some viruses can remain in the air for a longer time, especially in enclosed spaces.
- Examples:
    - Influenza (Flu)
    - COVID-19 (SARS-CoV-2)
    - Measles
    - Chickenpox (Varicella)

2. Direct Contact Transmission

- Route: This occurs when a person comes into direct physical contact with an infected person or their bodily fluids, such as saliva, blood, or mucous membranes.
- Examples:
    - HIV/AIDS
    - Herpes Simplex Virus (HSV)
    - Ebola Virus Disease
    - Hepatitis B and C
    - Human Papillomavirus (HPV)

3. Fecal-Oral Transmission

- Route: Transmission occurs when a person ingests food or water that has been contaminated by the feces of an infected person. This is common in areas with poor sanitation.
- Examples:
    - Hepatitis A
    - Rotavirus
    - Polio (Poliomyelitis)
    - Norovirus

4. Vector-Borne Transmission

- Route: Involves vectors, usually insects (mosquitoes, ticks, etc.), that carry the virus from an infected person or animal to a healthy person. These vectors typically bite and transmit the virus.
- Examples:
    - Dengue Fever (via Aedes mosquitoes)
    - Zika Virus (via Aedes mosquitoes)
    - Yellow Fever (via Aedes and Haemagogus mosquitoes)
    - West Nile Virus (via mosquitoes)
    - Rabies (via bites from infected animals)

7. Sexual Transmission

- Route: Involves the transmission of viruses through sexual contact, either through vaginal, anal, or oral sex. The virus may be transmitted through bodily fluids such as semen, vaginal fluids, or blood.
- Examples:
    - HIV/AIDS
    - Human Papillomavirus (HPV)
    - Herpes Simplex Virus (HSV)

8. Dermatologic Viral Infections

Some viruses primarily affect the skin, causing rashes, lesions, and other dermatological issues. Examples include:

- Chickenpox (Varicella), caused by the Varicella-zoster virus, which leads to an itchy rash.
- Herpes Simplex Virus (HSV), responsible for cold sores and genital herpes.
- Molluscum Contagiosum, which causes small skin bumps, particularly in children.

NNAMDI AZIKIWE UNIVERSITY

FACULTY OF BIOSCIENCES

BIO 102 - INTRODUCTORY BIOLOGY II

INTODUCTION

Living things stay alive by doing certain activities that keep them going. These activities help them control what they do every day. One of these activities is called excretion. Inside the cells of living things, a process called metabolism happens. This process produces both helpful and harmful substances. If harmful substances build up, they can hurt the living thing. So, living things get rid of these waste products from their bodies, and this is called excretion. Different living things get rid of waste in different ways. This lecture will look at how plants get rid of waste and how it's different from animals.

Chemical reactions occur in the cells of living organisms all the time to carry out the life processes. The sum of these reactions is called metabolism. Metabolism produces useful products as well as toxic (poisonous) by-products. These toxic substances have to be removed as they are harmful if allowed to accumulate. The removal of metabolic waste products from the body of an organism is known as excretion.

The major excretory products are carbon dioxide, excess water, and nitrogenous compounds like ammonia, urea, uric acid, etc. Carbon dioxide and water are produced in the process of tissue respiration. Nitrogenous compounds are formed from the breakdown of proteins and amino acids. Water and salts in excess of the body's needs are also excreted.

We acquire most of the water with our food and drink and some by metabolism, e.g., the water produced during cellular respiration. Other excretory products include chemicals from medicines, toxic substances, and circulating hormones that have already served their purpose. We will learn how metabolic wastes get eliminated.

Excretion in animals

Many unicellular organisms like Amoeba throw out their wastes by diffusion from their body surface. Protozoan's have no organs for excretion. As they live in an aquatic habitat, their wastes are eliminated by diffusion through the plasma membrane.

Simple multicellular organisms like Hydra throw out solid waste matter through their mouth. Higher multicellular organisms have well-defined specialized excretory organs. These organs could be simple tubular structures as in flatworms and leech.

The excretory organs of insects (e.g., grasshopper, cockroach and housefly) are also tubular. They remove nitrogenous wastes from the body fluid and help in maintaining the water balance in the body.

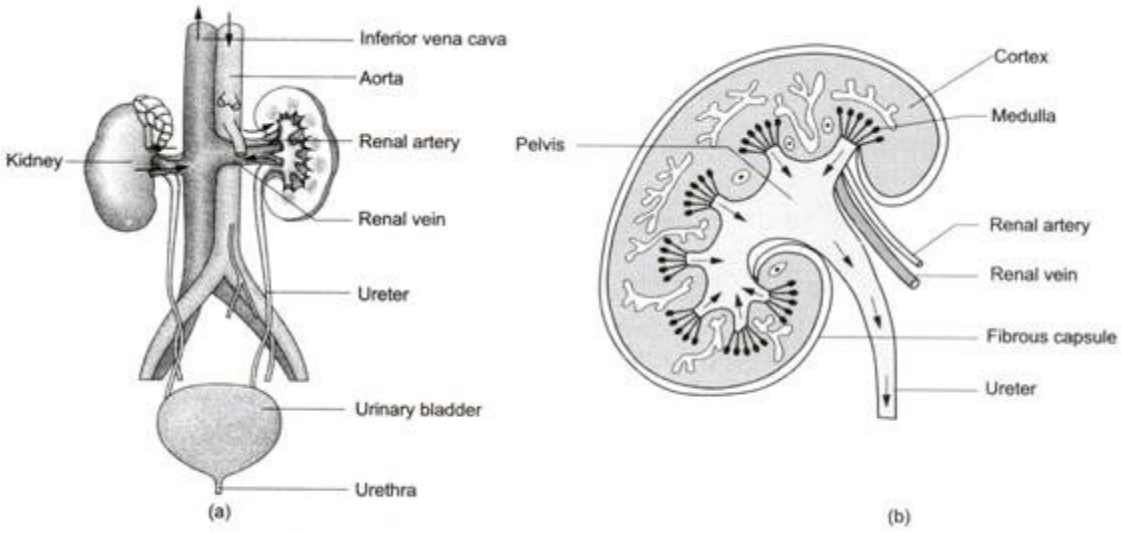
In vertebrates, the main organs of excretion and maintenance of water balance are the kidneys.

Excretion in human beings

Although the kidneys are the main organs of excretion, the skin, lungs and liver also help in excretion.

Skin: Our skin has sweat glands, through which we excrete small amounts of water, urea and salts.
Liver: The liver excretes bile, which contains bile pigments. These are produced by the breakdown of old RBCs in the liver. As hemoglobin breaks down, its iron is retained, while the pigment (haem) is excreted with the bile. The liver also excretes cholesterol.
Lungs: The lungs help in getting rid of carbon dioxide, formed as a result of cellular respiration, through exhalation.
Kidney: Our excretory system consists of kidneys, blood vessels that join them, ureters, urinary bladder and urethra. They help produce and excrete urine.
There are two bean-shaped kidneys that lie in the abdominal cavity, one on either side of the vertebral column. The kidneys are reddish brown. Each of them is about 10 cm long and weighs about 150 g. Although they weigh less, they receive a lot of blood for filtration.

A volume of blood nearly equivalent to that in the whole body passes through the kidneys every four or five minutes. The kidneys produce urine to filter out the waste products, like urea and uric acid, from the blood.



(a) Excretory organs of man (b) Internal structure of a kidney

Urine leaves each kidney through a tube called ureters. The ureters from both the kidneys are corrected to the urinary bladder that collects and stores urine. Ureters carry urine from the kidneys into the urinary bladder. The urethra is a canal that carries urine from the bladder and expels it outside the body.

Excretion in Plants

When plants get rid of harmful and waste substances from their bodies, it's called excretion. Unlike animals, plants don't have a fancy system just for excretion. They don't have special organs dedicated to it. So, excretion in plants isn't as complicated as in animals.

Different Types of Excretion in Plants

Transpiration: Transpiration is when a plant loses water vapor from its surface, mainly from its leaves. This loss of water helps to move water up through the plant's xylem. Plants can lose quite a bit of water through transpiration.

Guttation: Guttation is when a plant loses water in the form of droplets from its hydathodes. Droplets of water come out from the edges of leaves of certain plants. This happens when there's strong pressure in the roots but low transpiration.

Minerals, acids, sugars, and other substances can be found in these droplets. Plants like garden nasturtium, grasses, tomatoes, potatoes, and colocasia can exhibit guttation.

Bleeding: Bleeding is when sap flows out from injured parts of a plant. This happens because of pressure in the roots, causing sap to flow out of cuts made in the stem, especially when the soil is moist.

Respiration: Plants use carbon dioxide, water, sunlight, and chlorophyll to make food in a process called photosynthesis. Oxygen is produced as a result. This oxygen is released as waste through a process called diffusion.

Excretion is carried out in plants in the following ways:

- Plants get rid of their waste gases like oxygen, carbon dioxide, and water vapor through tiny openings called stomata on their leaves and lenticels on their stems.
- Some waste collects in leaves and bark. When leaves and bark fall off, the waste goes away too.
- Other waste turns into harmless solids and stays in the plant. Things like raphides, tannins, resins, gum, rubber, and essential oils are examples of this.
- For instance, the oil from oranges, eucalyptus, and jasmine, latex from rubber trees, papaya trees, and gums from acacia, are all forms of stored waste products. Sometimes, these products even get released into the soil.

Plant Excretory Product

- Plants mainly get rid of carbon dioxide, extra water from breathing, and nitrogen compounds from breaking down proteins. They release two gases: oxygen when they make food and carbon dioxide when they breathe out.
- When animals breathe, they release a gas called carbon dioxide. Plants use this carbon dioxide, along with sunlight, to create their food in a process known as photosynthesis. As a result, plants release oxygen into the air, which is crucial for all living beings.
- Plants absorb water from the soil through their roots. Sometimes, they take in more water than they need. To get rid of this excess water, plants release it through small pores on their leaves called stomata. This process is called transpiration.
- In some cases, the water or soil where plants grow may contain too much salt. When plants absorb this excess salt, they must eliminate it from their systems. They do so by depositing the salts in their tissues, forming crystals.
- As animals and plants go about their daily activities, they produce waste substances. One type of waste is called nitrogenous waste, which includes compounds like urea. These wastes are expelled from the body through processes like diffusion.
- During their metabolic processes, plants produce organic acids. Some of these acids, like oxalic acid, can accumulate in plants and cause harm if not properly managed.
- Plants also produce chemicals called tannins, which act as a defense mechanism against pests. These tannins give certain plants a bitter taste and help protect them from being eaten by animals.
- Certain plants, like rubber trees, produce a milky white fluid called latex. This latex serves to seal wounds in the plant and provides protection against insects and diseases.

- Another protective substance produced by plants is resin, which is a thick liquid with a strong aroma. Resins help shield plants from damage caused by insects and fungi.
- Plants also generate gums, which are sticky substances that help seal wounds and protect against infections. These gums play a crucial role in maintaining the health of the plant.

Plant Organs involved in Excretion

- **Old Leaves:** When leaves get old, they store waste and then fall off the tree.
- **Old Xylem:** Waste like resins and tannins builds up in the old xylem, making it stop working.
- **Bark:** Dead cells in bark collect impurities like tannins, which are used for making dyes and inks.
- **Central Vacuole:** Waste mostly stays in the central vacuole of a plant's cell, but it doesn't affect the cell's work because of a special membrane.
- **Root Excretion:** Plants push some waste out through their roots.
- **Detoxification:** Harmful acids are turned into harmless crystals by plants.
- **Salt Glands:** Plants also get rid of extra salt through special glands.

REPRODUCTION IN ANIMALS AND PLANTS

Reproduction in plants is a fundamental biological process that enables the continuation of plant species. Plants can reproduce through both asexual and sexual means, each having distinct mechanisms and advantages.

Asexual Reproduction in Plants
Asexual reproduction in plants involves a single parent and results in offspring that are genetically identical to the parent. This method is common in many plants and occurs through various mechanisms:

Vegetative Propagation:
Example: Strawberry (runners), Potato (tubers)
Plants reproduce from vegetative parts such as roots, stems, and leaves. For instance, strawberry plants produce runners (stolons) that develop into new plants. Potatoes produce tubers, which are underground stems that can sprout into new plants. Bryophyllum and Begonia have leaf buds which develop into new plants.

Cuttings:
Example: Rose, Sugarcane
A piece of the plant, usually a stem or leaf, is cut and planted in soil. This piece develops roots and grows into a new plant. Many horticultural plants are propagated this way.

Grafting:
Example: Apple, Grapes
A shoot (scion) from one plant is attached to the rootstock of another plant. The two parts grow together and form a new plant. This method is often used in fruit tree cultivation.

Layering:
Example: Blackberry, Jasmine

A stem is bent to the ground and covered with soil. It develops roots at the buried portion and eventually becomes an independent plant.

Bulbs, Corms, and Rhizomes:
Example: Tulip (bulbs), Gladiolus (corms), Ginger (rhizomes)
These are underground storage organs that give rise to new plants. Bulbs (like in tulips) contain a shoot surrounded by fleshy leaves. Corms are swollen stem bases, and rhizomes are horizontal underground stems.

Spore formation:
Example: Rhizopus, fern
Observed in fungi like bread mold. Sporangium has spores which develop to a new plant when it finds moisture.

Budding
Example: yeast.
Formation of an outgrowth called a bud.

Fragmentation
Example: spirogyra.
It involves breaking into fragments, each fragment develops into a new organism.

Sexual Reproduction in Plants
Sexual reproduction in plants involves the fusion of male and female gametes, leading to offspring with genetic variation. This process occurs through the formation of flowers, pollination, and seed development.

Flower Structure
Flowers are the reproductive structures of angiosperms (flowering plants). A typical flower consists of the following parts:

Sepals: Protect the developing flower bud.
Petals: Attract pollinators with their color and scent.

Stamens: The male reproductive organs, consisting of anthers (produce pollen) and filaments.
Pistils (Carpels): The female reproductive organs, including the stigma (receives pollen), style (connects stigma to ovary), and ovary (contains ovules).
Pollination
Pollination is the transfer of pollen from the anther to the stigma. It can occur through various agents:

Wind Pollination:
Example: Grasses, Pine Trees
Wind carries pollen from one plant to another. Wind-pollinated plants typically have small, inconspicuous flowers and produce large amounts of pollen.

Insect Pollination:
Example: Bees pollinate sunflowers and apple trees.
Insects, such as bees, butterflies, and beetles, transfer pollen as they move from flower to flower. These plants usually have brightly colored flowers and produce nectar to attract pollinators.

Bird Pollination:
Example: Hummingbirds pollinate trumpet vine.
Birds transfer pollen while feeding on nectar. Bird-pollinated flowers are often red or orange and have tubular shapes.

Water Pollination:
Example: Vallisneria
Water can carry pollen to female flowers in aquatic plants.
Fertilization and Seed Development
After pollination, fertilization occurs when the pollen tube grows from the stigma down the style to the ovary, allowing the male gametes to reach and fuse with the female gametes (ovules). This results in the formation of seeds, which contain the embryonic plant and stored food. The ovary develops into a fruit, which protects the seeds and aids in their dispersal.

Seed Dispersal

Seeds are dispersed from the parent plant to reduce competition and increase the chances of successful germination. Common dispersal mechanisms include:

Wind:
Example: Dandelion, Maple
Seeds have structures like wings or fluff that allow them to be carried by the wind.

Water:
Example: Coconut
Seeds can float and be carried by water to new locations.

Animals:
Example: Berries, Acorns
Seeds are enclosed in fruits that are eaten by animals. The seeds pass through the digestive system and are excreted at a different location. Some seeds have hooks or barbs that attach to animal fur.

Explosive Mechanisms:
Example: Touch-me-not (Impatiens)
Some plants have pods that burst open, propelling seeds away from the parent plant.

Examples of Plant Reproduction
Self-Pollinating Plants:

- Example: Pea, Tomato
- These plants have flowers that can fertilize themselves. They often have mechanisms to ensure that pollen from the same flower can reach the stigma.

Cross-Pollinating Plants:

- Example: Apple, Pumpkin
- These plants require pollen from a different individual of the same species. Cross-pollination promotes genetic diversity.

Monoecious Plants:

- Example: Corn, Squash

- Monoecious plants have both male and female flowers on the same plant, allowing for self- or cross-pollination.

Dioecious Plants:

- Example: Holly, Kiwi
- Dioecious plants have separate male and female plants. Cross-pollination is necessary for fertilization.

Advantages of Asexual Reproduction:

- Rapid Population Increase: Asexual reproduction allows for quick and efficient population growth.
- Energy Efficiency: No need for pollinators or energy-intensive processes like flower production.
- Genetic Consistency: Offspring are clones of the parent, ensuring stability in stable environments.

Advantages of Sexual Reproduction:

- Genetic Diversity: Promotes variation, which enhances adaptability and resilience to environmental changes.
- Evolutionary Potential: Allows for the combination of beneficial traits, driving evolution and adaptation.

Disadvantages of Asexual Reproduction:

- Lack of Genetic Diversity: Makes populations more vulnerable to diseases and environmental changes.
- Mutation Accumulation: Harmful mutations can accumulate over generations.

Disadvantages of Sexual Reproduction:

- Energy and Resource Intensive: Requires more energy for producing flowers, attracting pollinators, and developing seeds.
- Dependence on Pollinators: Success depends on the presence and behavior of pollinators, which can be affected by environmental changes.

REPRODUCTION IN ANIMALS

Reproduction is a fundamental biological process by which new individual organisms, known as offspring, are produced from their parents. The process can be broadly categorized into two main types: asexual and sexual reproduction.

Asexual Reproduction

Asexual reproduction involves a single parent and results in offspring that are genetically identical to the parent. This type of reproduction is common in lower organisms, including many plants, bacteria, and some animals. Common methods of asexual reproduction include:

1. **Binary Fission**:
    - **Example**: *Amoeba, paramecium and plasmodium*
    - In binary fission, a single organism divides into two equal parts, each becoming a new organism. It is a common method of reproduction in prokaryotes such as bacteria.
2. **Budding**:
    - **Example**: *Hydra*,
    - In budding, a new organism grows out of the body of the parent organism and eventually detaches to live independently.
3. **Fragmentation**:
    - **Example**: *Planaria*
    - Fragmentation occurs when a parent organism breaks into fragments, each of which can develop into a new organism.
4. **Parthenogenesis**:
    - **Example**: *Aphids*, *Komodo Dragon*
    - Parthenogenesis is the development of an embryo from an unfertilized egg cell. It occurs in some insects, reptiles, and amphibians.

Sexual Reproduction

Sexual reproduction involves the combination of genetic material from two parents, resulting in offspring with genetic variation. This process typically

involves the production of gametes (sperm and egg cells) through meiosis, followed by fertilization. Sexual reproduction is common in most plants, animals, and fungi.

Key Processes in Sexual Reproduction

1. **Gamete Production**:
    - **Example**: Humans produce sperm and egg cells.
    - Gametes are produced through meiosis, a type of cell division that reduces the chromosome number by half, resulting in four haploid cells.
2. **Fertilization**:
    - **Example**: In frogs, external fertilization occurs where eggs and sperm are released into the water.
    - Fertilization is the fusion of male and female gametes, resulting in a diploid zygote.
3. **Development**:
    - **Example**: In mammals, the zygote undergoes several stages of development, including cleavage, blastula formation, gastrulation, and organogenesis.
    - After fertilization, the zygote undergoes mitotic divisions and differentiation to form a multicellular organism.

Examples of Sexual Reproduction in Animals

1. **Mammals**:
    - **Example**: Humans
    - Mammals have internal fertilization. The male deposits sperm inside the female's reproductive tract, where fertilization occurs. The embryo develops inside the female's uterus and is nourished through the placenta.
2. **Birds**:
    - **Example**: Chickens
    - Birds also have internal fertilization. Eggs are fertilized within the female, and then laid in a nest where they develop outside the

mother's body. The egg contains all the necessary nutrients for the developing embryo.

3. **Amphibians**:
    - **Example**: Frogs
    - Many amphibians, like frogs, have external fertilization. Females lay eggs in water, and males release sperm over the eggs to fertilize them. The larvae, known as tadpoles, undergo metamorphosis to become adult frogs.
4. **Fish**:
    - **Example**: Salmon
    - Fish typically exhibit external fertilization. Female fish lay eggs in water, and males release sperm to fertilize them. Some species, like sharks, have internal fertilization and give birth to live young.
5. **Insects**:
    - **Example**: Butterflies
    - Insects like butterflies undergo complete metamorphosis, starting from egg, larva (caterpillar), pupa (chrysalis), and finally, adult. Fertilization is internal, and females lay fertilized eggs on host plants.

Advantages of Asexual Reproduction:

- **Speed**: Asexual reproduction can occur rapidly, allowing for quick population growth.
- **Energy Efficiency**: It requires less energy since it does not involve finding a mate or producing gametes.
- **Consistency**: Produces genetically identical offspring, which can be advantageous in stable environments.

Advantages of Sexual Reproduction:

- **Genetic Diversity**: Sexual reproduction generates genetic variation, which enhances the adaptability and survival of a species in changing environments.
- **Evolutionary Potential**: Greater genetic diversity allows for more rapid evolution and adaptation to new challenges, such as diseases or environmental changes.

Disadvantages of Asexual Reproduction:

- **Lack of Genetic Diversity**: Offspring are genetically identical to the parent, making them vulnerable to diseases and environmental changes.
- **Mutation Accumulation**: Harmful mutations can accumulate over generations.

Disadvantages of Sexual Reproduction:

- **Energy and Time Consuming**: Finding a mate and producing gametes requires significant energy and time.
- **Risk of Predation**: The act of mating and the vulnerable stages of offspring development can expose individuals to predators.

Growth and Development in Plants and Animals

Growth and development are essential aspects of life for both plants and animals, ensuring that they mature, reproduce, and adapt to their environments. Despite their differences, both groups undergo complex processes to reach maturity. Let's explore these processes in detail.

Growth and Development in Plants

Plant growth and development involve a series of stages from seed germination to maturity and reproduction.

Stages of Plant Growth and Development.

1. **Seed Germination**:
    - **Example**: A bean seed
    - Germination begins when a seed absorbs water, causing it to swell and break its coat. The embryo inside starts to grow, and the radicle (embryonic root) emerges first, followed by the shoot (plumule).
2. **Seedling Growth**:
        i. The seedling develops its first true leaves and begins photosynthesis, producing its own food. This stage is critical as the young plant is highly vulnerable to environmental factors.
3. **Vegetative Growth**:
    - The plant undergoes rapid growth, developing roots, stems, and leaves. This stage focuses on increasing the plant's size and energy production through photosynthesis.
4. **Reproductive Development**:
    - The plant matures and begins to produce flowers. Flowers are the reproductive organs where pollination and fertilization occur, leading to the formation of seeds and fruit.
5. **Senescence**:
    - The final stage of development where the plant's metabolic activities decline. Leaves may fall, and the plant prepares for dormancy or death.

Factors Influencing Plant Growth

- **Light**: Essential for photosynthesis. Plants require different light intensities and durations.
- **Water**: Vital for nutrient transport and photosynthesis. Too much or too little water can harm the plant.
- **Nutrients**: Elements like nitrogen, phosphorus, and potassium are crucial for growth. Deficiencies or excesses can cause problems.
- **Temperature**: Affects enzyme activity and growth rates. Different plants have optimal temperature ranges.
- **Hormones**: Chemicals like auxins, gibberellins, cytokinins, and ethylene regulate growth and development.

Examples of Plant Growth

- **Annual Plants**: *Tomatoes* grow, flower, produce seeds, and die within a year.
- **Perennial Plants**: *Oak trees* grow for several years, produce seeds multiple times, and can live for decades or even centuries.

Growth and Development in Animals

Animal growth and development involve a series of stages from conception to adulthood, often characterized by more complex behaviors and physiological changes compared to plants.

Stages of Animal Growth and Development

1. **Fertilization**:
    - **Example**: Human conception
    - The fusion of sperm and egg cells results in a zygote, which contains genetic material from both parents.
2. **Embryonic Development**:
    - The zygote undergoes rapid cell division (cleavage) to form a blastocyst. This stage involves differentiation where cells start to become specialized for various functions.
3. **Fetal Development**:
    - The embryo develops into a fetus, with organs and systems forming and maturing. This stage involves significant growth and complexity.
4. **Birth or Hatching**:

    - The young animal is born or hatches from an egg, entering the world outside the womb or egg.
5. **Juvenile Growth**:
    - The animal grows rapidly, developing its muscles, bones, and other tissues. This stage often includes learning important survival skills.
6. **Adulthood**:
    - The animal reaches sexual maturity and can reproduce. Growth may slow down, but development continues as the animal maintains and repairs its body.
7. **Aging and Senescence**:
    - The final stage where the animal's physiological functions decline. This stage varies greatly among species, with some animals showing signs of aging early while others live long, healthy lives.

Factors Influencing Animal Growth

- **Nutrition**: Essential for energy, growth, and repair. Balanced diets provide the necessary proteins, carbohydrates, fats, vitamins, and minerals.
- **Genetics**: Determines growth potential, development rates, and overall health.
- **Environment**: Factors like temperature, humidity, and habitat quality affect growth and development.
- **Hormones**: Regulate growth, metabolism, and development. For example, growth hormone influences overall body size, while sex hormones trigger reproductive development.
- **Social Interactions**: In some species, social structures and interactions can influence growth and behavior. For instance, dominance hierarchies in social animals can impact access to food and mating opportunities.

Examples of Animal Growth

- **Humans**: Human development follows a predictable pattern from infancy, through childhood, adolescence, and adulthood. Each stage involves physical, cognitive, and emotional growth.
- **Insects**: *Butterflies* undergo metamorphosis, transforming from larvae (caterpillars) to pupae (chrysalises) and finally to adults (butterflies).

- **Amphibians**: *Frogs* start as eggs, hatch into tadpoles, and undergo metamorphosis to become adult frogs.

Similarities:

- Both plants and animals go through distinct stages of development from immature forms to fully mature adults.
- Both are influenced by genetic, environmental, and hormonal factors.
- Both need adequate nutrition for proper growth and development.

Differences:

- Plants rely heavily on light for photosynthesis, while animals require a balanced diet of various nutrients.
- Animals often exhibit more complex behaviors and mobility compared to plants.
- Plant growth can continue indefinitely in some species (e.g., trees), while animals typically reach a fixed size at maturity.

GENERAL BIOLOGY 2 (BIO 102)

DIFFERENCES AND SIMILARITIES BETWEEN PLANTS AND ANIMALS

There are many differences between plants and animals. When we say the plant, you think of green things which can prepare their own food. Similarly, when you hear animal, you think of organisms having a well-developed body with a proper organ system. Thus, they are easy to differentiate. However, there is a lot of difference between plants and animals which can be observed at a physical as well as the cellular level. Plants and animals fall under the category multicellular, eukaryotes. They share the same ecosystems, surroundings and their dependability on each other.

Definition of Plants

Plants fall under the multicellular, photosynthetic eukaryotes category. Further, they belong to the kingdom Plantae. There are different kinds of plants found in different areas and climates. Moreover, they are the primary producers on the Earth's ecosystem.

We call plants autotrophs because they are capable of producing their own food through photosynthesis. Generally, plants consist of a proper root system and the shoot system. The root system is one found below the ground.

Whereas, the shoot system contains parts above the ground like flowers, fruits, leaves, buds and more. There are different types of plants. They range from angiosperms, gymnosperms, ferns, conifers and mosses to liverworts, hornworts, green algae and more.

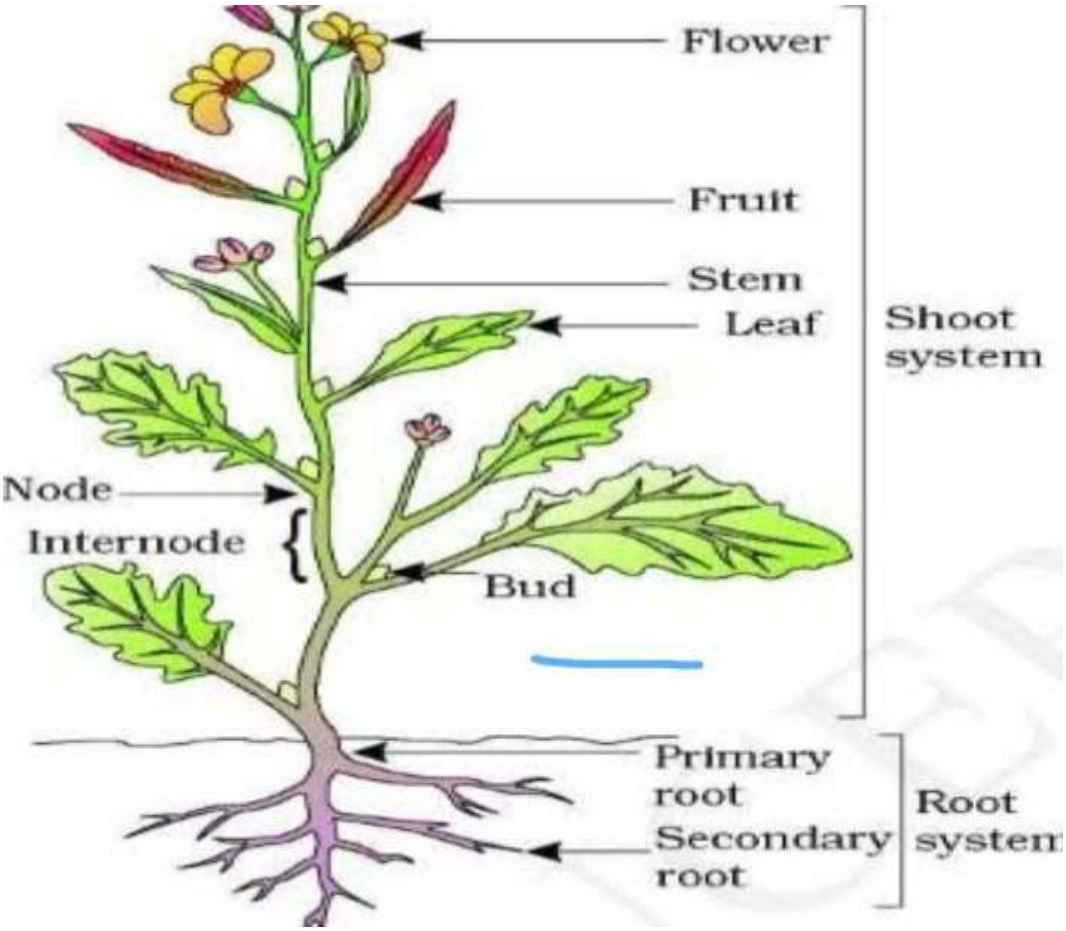


Fig 1. A diagrammatic representation of a typical plant

Definition of Animals

Belonging to the kingdom of Animalia, animals also vary from one place to another. Further, they fall under the category of multicellular, eukaryotes. Moreover, they are found in air, water and land.

Animals also have sub-divisions. For instance, they may be vertebrates or invertebrates, oviparous or viviparous, and even cold-blooded or warm-blooded. Generally, the most common groups of animals are classified as mammals, amphibians, reptiles, fish, birds and insects.

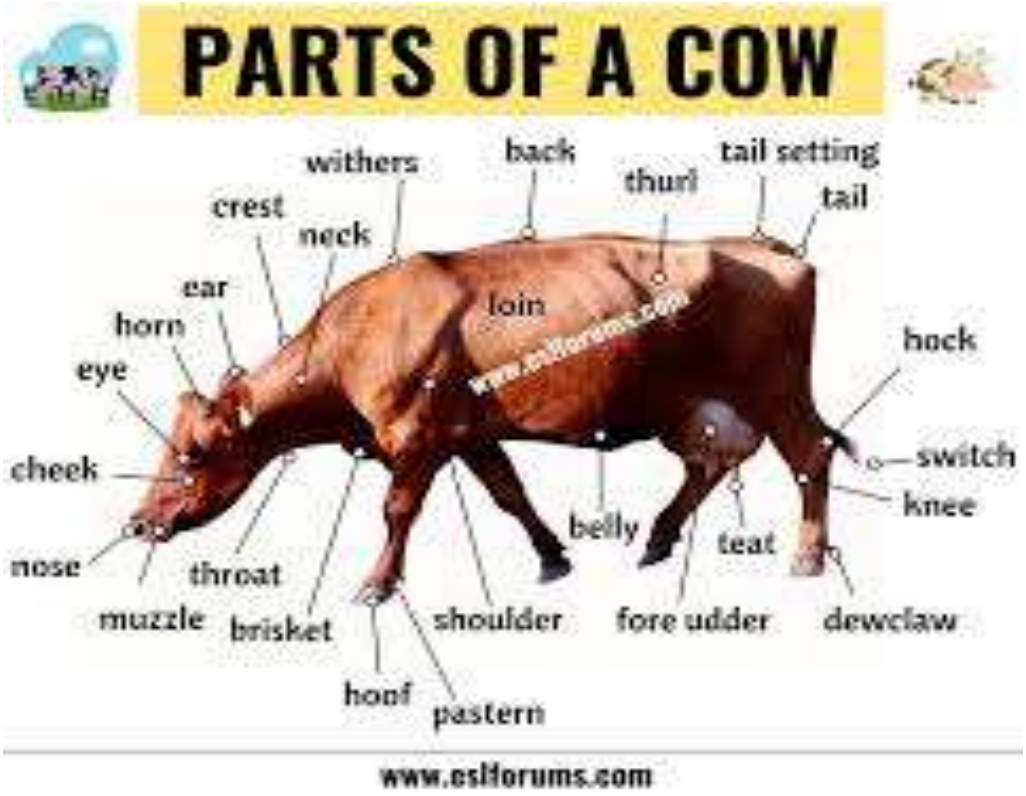


Fig 2. A diagrammatic representation of a typical Animal

Similarities Between Plants and Animals

Plants and animals are living things. They feed, respire, excrete, grow, move, reproduce and are sensitive to their environment. Animals and plants need food for energy but they feed in different ways. Animals eat plants and other animals, but plants make their own food.

Important Differences Between Plants and Animals

S/N	COMPARISON	PLANTS	ANIMALS
1	**Movement**	Cannot move as they are rooted in the ground. Exceptions- Volvox and Chlamydomonas.	Can move freely from one place to the other. Exception- Sponges and Corals.
2	**Digestive System**	Absent	Present
3	**Food Storage**	Store food in the form of starch	Store food in the form of glycogen
4	**Structure of Cell**	Contains cell wall, chloroplast, plasmodesmata, plastids and other organelles	Do not have cell walls, have other organelles like tight junction and cilia.
5	**Respiration**	Take in Carbon Dioxide and release Oxygen	Take in oxygen and release carbon dioxide
6	**Respiration through**	Occurs through stomata	Occurs through lungs, gills, skin and more.
7	**Growth**	The meristematic system in the tip of roots and stems supports growth	Organs and organ system support the growth
8	**Reproduction Method**	Asexual reproduction by budding, vegetative methods, wind, spores, insects.	Animals reproduce sexually whereas some lower animals like algae have asexual reproduction
9	**Response**	Show response through touch and light	The proper nervous system allows responding quickly

10	**Sensitive**	Less sensitive	Highly sensitive

Interdependency of Plants and Animals

1. Animals breathe oxygen and exhale carbon dioxide. Plants take in carbon dioxide and release oxygen back into the air.

2. Animals need plants for food and shelter.

3. Plants need animals for seed dispersal and pollination.

4. Some animals eat plants. Some animals eat other animals. When animals die and decompose, the plants use the nutrients that were deposited back into the soil from the decomposing organism.

5. Forests and wetlands filter water so it is clean, and animals need clean water to drink.

Dr. K.C Irikannu

BIO 102: GENERAL BIOLOGY II
TOPIC: MODES OF NUTRITION IN PLANTS, AND ANIMALS
Lecturer: Mr. Nwankwo Samuel Ikwunne

Definition of Nutrition:
It is the process by which living organisms obtain, and utilize nutrients to support various body functions.

Nutrients are chemical substances needed by the body to sustain basic functions.
It includes **macronutrients** like carbohydrates, proteins, fats and oil, and **micronutrients** like vitamins and minerals.

Modes of Nutrition:
There two modes of nutrition:

1. Autotrophic mode of nutrition
2. Heterotrophic mode of nutrition

Autotrophic Mode Of Nutrition
In autotrophic mode, organisms produce their own food. Examples are plants, algae and bacteria (including cyanobacteria). Organisms that undergo autotrophic mode of nutrition are called **autotrophs** (producers).

This mode of nutrition is divided into two:

1. Photosynthetic nutrition
2. Chemosynthetic nutrition

Photosynthetic Nutrition
In this type of autotrophic nutrition, organisms (green plants) are capable of manufacturing their own organic food from simple inorganic substances (carbon (IV) oxide, water) in the presence of sunlight, and chlorophyll (green colouring matter) through a process called **photosynthesis**. Organisms that undergo photosynthetic nutrition are called **photoautotrophs.**

$$6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2$$

From the simple sugars formed, other food substances such as proteins, oils, and vitamins are manufactured.

Chemosynthetic Nutrition
In this type of autotrophic nutrition, organisms manufacture their own organic food from simple inorganic substances (carbon (IV) oxide, water, and hydrogen sulphide) using the energy released during chemical reaction in the absence of sunlight, and chlorophyll (green colouring matter) through a process called **chemosynthesis**. Organisms that undergo chemosynthetic nutrition are called **chemoautotrophs.**

For example, the sulphur bacteria in the soil can oxidize hydrogen sulphide around it to produce energy.

$$2H_2S + O_2 \xrightarrow{\text{Sulphur bacteria}} S + 2H_2O + \text{chemical energy}$$

This chemical energy is then used to combine hydrogen sulphide and carbon (IV) oxide in the soil to form sugar, water, and sulphur.

$$12H_2S + 6CO_2 \xrightarrow{\text{Chemical energy}} C_6H_{12}O_6 + 6H_2O + 12S$$

Also nitrifying bacteria (Nitrosomonas and Nitrobacter) in the soil oxidize ammonia to nitrites and nitrates respectively. The chemical energy released is used in combining water and carbon (IV) oxide in the soil to form sugar. The bacteria also use the nitrates in the soil with the sugar produced to form protein.

Heterotrophic Mode Of Nutrition
This is the type of nutrition carried out by organisms that are not capable of manufacturing their own food from simple inorganic substances. It is a mode of nutrition in which organisms eat other organisms to survive. Organisms that undergo heterotrophic mode of nutrition are called **heterotrophs** (consumers). Examples are all animals, fungi, bacteria, and few parasitic flowering plants.

Heterotrophs are classified into three based on the form of food item they consume:
1. Macrophagus – they take large food items which need the aid if teeth to break down before entering the digestive gut. E.g. mammals, rodents etc.
2. Microphagus – they take tiny food particles into their gut without being broken down with teeth. E.g. Paramecium, mosquito larvae etc.

3. Fluid feeders – they feed on fluids. E.g. adult mosquitoes, aphids butterflies etc.

There four main types of heterotrophic mode of nutrition namely:
1. Holozoic
2. Saprophytic
3. Symbiotic
4. Parasitic

1. Holozoic Nutrition
This is a type of heterotrophic mode of nutrition, which involves the intake of complex organic food materials into the body using the process of ingestion. This involves five steps: ingestion, digestion, absorption, assimilation, and excretion (egestion). Holozoic animals because of the need to constantly search for their food and catch them have developed sensory, nervous and muscular structures to enable them achieve such feat. Also, they developed efficient digestive systems to digest ingested food materials into simple materials for absorption into the body. Nearly all animals are holozoic in nutrition.

Holozoic nutrition is further classified into four:

Herbivores: they are animals that eat only plants to gain energy. E.g. cows, goat etc. they are known as primary consumers
Carnivores: they are animals that eat the flesh of other animals. E.g. lions, tigers etc. they are known predators.
Omnivores: they are animals that eat both plants and animals e.g. humans
Scavengers: they are animals that that feed on dead animals e.g. vulture

2. Saprophytic Nutrition
This is a type of nutrition in which organisms (mainly non-green plants) feed on dead and decaying organic matter. Enzymes are secreted from cells to the dead organic matter and the digestion of the dead organic matter takes place outside the cells (extracellular digestion) of the saprophytic plant. The cells then absorb the digested and soluble food substances.
Organisms, which feed saprophytically, are known as **saprophytes**. Most fungi (*Mucor*, mushroom, and yeast), and bacteria are examples.

3. Symbiotic Nutrition
This is a type of nutrition in which two organisms of different species live together to take advantage of each other. This type of association is called

symbiosis. In some cases the two organisms become dependent on each other that each cannot live alone.
For example, the association between an alga and a fungus in lichens is mutualistic. The fungus provides moisture (water), mineral salts, and support for the alga, which the alga uses in manufacturing food by photosynthesis for the fungus and itself. Both fungus and alga cannot live independently.
Another example is the symbiotic feeding between herbivorous animals and cellulose-digesting bacteria that live in the caecum and colon of herbivores. The bacteria are capable of digesting the cellulose of leaves and converting it to sugar that is absorbed by the herbivore. The bacteria in turn feed on the abundant nutrients of the herbivore as well as obtaining shelter in the animal's caecum.
This type of feeding is also exhibited in hermit crab and sea anemone.

4. Parasitic Nutrition
This is a type of nutrition in which one organism obtains food from another organism and during the feeding process, the organism is harmed and does not benefit. The organism that benefits is called **parasite** while the organism being harmed is called the **host**. The association between a parasite and a host is called **parasitism.**
There are two types of parasites namely:
1. **Ectoparasites**: these parasites live on the body surfaces of their hosts. E.g. mites, lice (singular: louse), ticks, bedbugs, fleas etc.
2. **Endoparasites**: these parasites live inside the bodies of their hosts. E.g. Plasmodium, hookworm, round worm, tapeworm, liver fluke.

Parasitic Nutrition In Plants
Dodder plant (*Cassytha filiformis*) is a parasitic flowering plant. It has small-scale leaves, which are not green. The stem is thin and ropelike and coils round the stem of a host plant. Dodder develops haustoria (suckers) which are sent to the vascular bundles of the host from where it obtains food and water it requires for its metabolism.

Parasitic Nutrition In Animals
Tapeworm (Taenia solium) it is a flat, ribbonlike or tapelike parasite. The head or scolex has four muscular suckers on the sides. Hooks surround the rostellum. The body consists of a series of proglottides. The long and tape-like body provides a large surface area for absorption of digested food all over the body by diffusion. The presence of four suckers supported by hooks helps the tapeworm to fasten itself onto the wall of the small intestine.

The host suffers an abdominal disorder and malnutrition and if tapeworm embryos migrate to the brain they damage the brain and may cause epilepsy.

Carnivorous Mode Of Nutrition In Plants

A few green plants supplement their nutrients with few organic foods by trapping insects and other small animals. The trapped insects are digested and the amino acids are absorbed. Often, these carnivorous plants grow in soils poor in nitrogen and therefore, animal proteins are needed to supplement the nitrogenous compounds requirement of the plant. Sundews, Venus's flytrap, Bladderwort, Pitcher plant, Butterwort, Water flytrap are examples.

MODE OF RESPIRATION IN ANIMALS AND PLANTS WITH EXAMPLES

Learning objectives

By the end of this unit, each student will be able to:

1. define respiration and the various types of respiration
2. state the various modes of respiration seen in animals and plants
3. list organisms that respire through the various modes

General Overview

Organisms carry out important life processes like synthesis of proteins, fats, carbohydrates and other metabolic reactions. Energy is required for these activities. One might therefore aptly ask: where does the energy come from, how is it stored and how is it made available to the cells of these organisms?

During the process of photosynthesis, light energy is converted to chemical energy and stored in the bonds of complex organic molecules - in the form of carbohydrates, such as starch and glucose. The weakening or breaking of the carbon to carbon bonds of such compounds releases a considerable amount of energy for utilization by the plant. However, the total amount of energy contained within a compound for example, glucose, is not released all at once, but slowly, in a stepwise series of reactions controlled by enzymes.

Generally series of reactions in the cell leading to the synthesis or breakdown of organic compound is referred to as a metabolic pathway. In plants, there are two chemical processes occurring at every particular point in time. These are the building up (synthesis process) called anabolism e.g. photosynthesis and a breakdown process called catabolism e.g. respiration. In animals, respiration is catabolic.

Carbohydrates, produced in photosynthesis serve as ready source of stored energy and are in most cases preferentially utilized as respiratory substrates. The energy is trapped in high energy compounds; the most important among them is the ATP; NADH and NADPH are equally important because of their ability to transfer electrons. The chains of oxidation-reductions reactions involved are collectively referred to as **respiration.**

Respiration as an oxidation process is a chain of chemical reactions involving the degradation of complex organic compound, from which carbon dioxide, water and energy are produced.

Types of Respiration

Respiration is of two types:

1. Aerobic Respiration

Aerobic respiration is a series of enzyme-controlled reactions that release the energy stored up in carbohydrates and lipids during photosynthesis and make it available to living organisms.

In most animals, the tissue oxidation occurs in presence of O_2 with end products CO_2 and H_2O. It requires exchange of O_2 and CO_2 between the organisms and the surroundings. Aerobic respiration takes place inside mitochondria. The pathways involved include Glycolysis, Krebs cycle and Electron transport system as sketchily represented in Fig. 1 below:

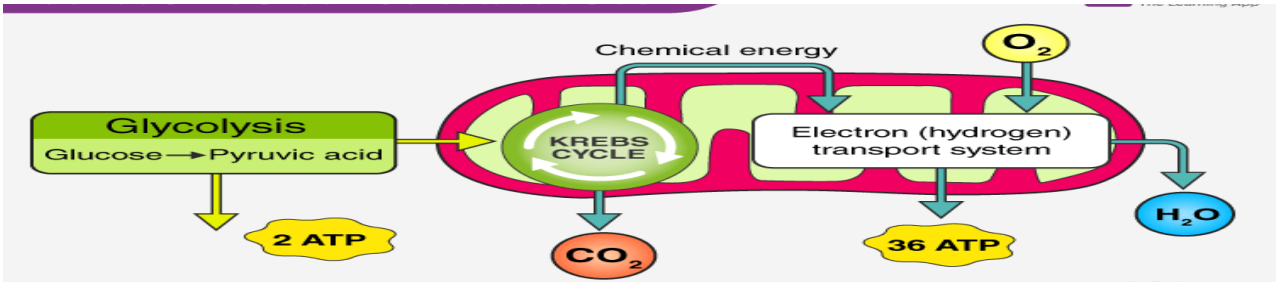


Fig.1. Pathways of Aerobic Respiration in Organisms

Aerobic respiration may be represented by the general equation: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

Anaerobic Respiration

In lower organisms like bacteria, yeast, and so on, the nutrients are oxidized without using O_2 and lactic acid or ethyl alcohol is the end product. It also occurs in some multicellular animals and some tissues (skeletal muscle and RBC) of higher animals. Anaerobic respiration occurs outside mitochondria.

During glycolysis, glucose molecules (six-carbon molecules) are split into two pyruvates (three-carbon molecules) during a sequence of enzyme-controlled reactions. This is the same reaction as occurs in aerobic respiration. Without oxygen, pyruvate is converted to lactic acid in animals or ethanol in plants and yeast. It produces only about 10% of the energy released in the complete oxidation of glucose. Anaerobic respiration in humans takes place when muscle undergoes extreme contraction as in vigorous exercise. When oxygen is limited the oxidation of NADH to NAD^+ by the electron transport chain is insufficient to maintain glycolysis. Under these conditions NAD^+ is regenerated by the reduction of pyruvate to lactate.

In yeast, pyruvate is converted to ethanal and then to ethanol. The latter stage oxidizes NADH to NAD^+, allowing glycolysis to continue.

Differences between Aerobic and Anaerobic Respiration

	Aerobic respiration	Anaerobic respiration
Usage of Oxygen	It uses molecular oxygen	It does not use molecular oxygen.
Release of CO_2	Always release CO_2.	May or may not release CO_2.
Production of water	It produces water.	It does not produce water
Quantity of energy produced	It produce much more energy (whole energy present in glucose)	It produces less energy (only 5% of that available in glucose).
End product	It yields inorganic end products only.	It yields organic end products with or without inorganic product.

Respiration process can be divided into two phases:

1. **External Respiration:** In this process, animals take in oxygen and release carbon dioxide. It is also known as breathing. This is mainly two phases.

a) Inspiration: It is the process by means of which oxygen is taken to the lungs.

b) Expiration: During expiration, carbon dioxide is expelled out from the respiratory organs into the environment.

2. **Internal Respiration or Cellular Respiration:** In this process, oxygen is used in chemical reactions within the cells. These reactions release energy from food substances and produce carbon dioxide and water as waste products. Sugar + $O_2 \longrightarrow CO_2 + H_2O$ + Energy.

Mode of Respiration in Animals

The mode of external respiration varies greatly from organism to organism. **Organisms adopt different modes of respiration depending on their types, complexity, size and the habitat in which they are found.** Respiratory structures are also tailored to the organsism's need for oxygen. Thus, below are respiratory structures and mechanisms used by various organisms:

1. Diffusion through Cell/Plasma Membrane/**Body Surface**

In unicellular animals, such as amoeba, exchange of gases takes place through the cell surface. They absorb oxygen from the surrounding air or water and give out carbon dioxide by diffusion.

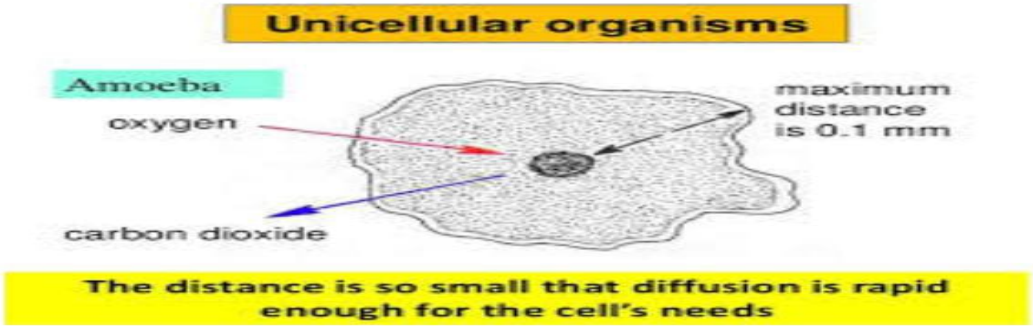


Fig.2: Diffusion in Amoeba

Organism like tapeworms, earthworms, and leeches also use their skin for the exchange of gases.



Fig. 3: Respiration in Earthworm

The skin of earthworms is very thin and moistened. Many blood cells are spread on this skin. These blood cells are known as capillaries. The exchange of gases occurs at capillaries. Oxygen needed for respiration is dissolved in water. On the principle that molecular gases diffuse from a region of higher concentration to region of lower concentration, dissolved oxygen which is of higher concentration in water than in the cell will diffuse into the cell. After gaseous exchanges, carbon(iv)oxide which is now higher in concentration in the cell than in water will diffuse out of the cell into the water. They die of suffocation if their skin is dried up.

2. Through the Trachea

Trachea is a common respiratory organ of invertebrates. This respiratory organ is a hallmark of insects like cockroaches, grasshopper e.t.c. It is made up of a system of branching tubes that deliver oxygen to, and remove carbon dioxide from the tissues. The pores to the outside called spiracles are typically paired structures, two in the thorax and eight in the abdomen. Periodic opening and closing of the spiracles prevents water loss by evaporation, a serious threat to insects that live in dry environments.

Oxygen for respiration enters the abdomen and thoracic segment from surrounding atmosphere. The expansion of the abdomen causes the volume of the abdomen to increase thereby drawing air by diffusion into the body through the spiracles, tracheae, air sacs and trachioles. When the atmospheric oxygen reaches the fluid of the trachioles, it dissolves. The dissolved oxygen then diffuses into the tissues where cellular or tissue respiration occurs. When the thoracic segments and abdomen contract, carbon (iv)oxide and water vapour diffuse from spiracles into the atmosphere.

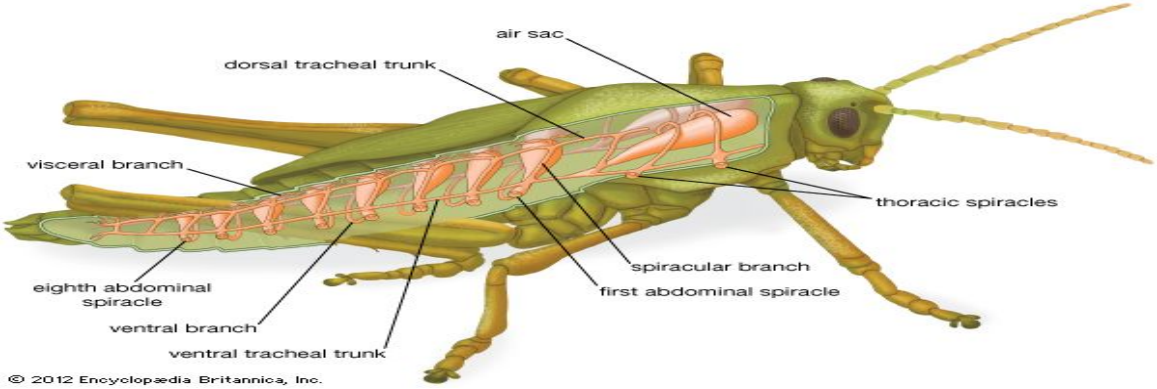


Fig 4: Tracheal System in Grasshopper

Although tracheal systems are primarily designed for life in air, in some insects, **modifications enable the tracheae to serve for gas exchange under water**. Of special interest are the insects that might be termed **bubble breathers,** which, as in the case of the water beetle ***Dytiscus***, take on a gas supply in the form of an air bubble under their wing surfaces next to the spiracles before they submerge. Tracheal gas exchange continues after the beetle submerges and anchors beneath the surface. As oxygen is consumed from the bubble, the partial pressure of oxygen within the bubble falls below that in the water; consequently oxygen diffuses from the water into the bubble to replace that consumed. The carbon dioxide produced by the insect diffuses through the tracheal system into the bubble and thence into the water. The bubble thus behaves like a **gill**. There is one major limitation to this adaptation: As oxygen is removed from the bubble, the partial pressure of the nitrogen rises, and this gas then diffuses outward into the water. The consequence of outward nitrogen diffusion is that the bubble shrinks and its oxygen content must be replenished by another trip to the surface.

A partial solution to the problem of bubble renewal has been found by small aquatic beetles of the family Elmidae (*e.g., **Elmis***, *Riolus*), which capture **bubbles containing oxygen produced by algae and incorporate this gas into the bubble gill**. Several species of aquatic beetles also augment gas exchange by stirring the surrounding water with their posterior legs.

One extraordinary strategy used by the hemipteran insects *Buenoa* and *Anisops* is an **internal oxygen store that enables them to lurk for minutes without resurfacing** while awaiting food in relatively predator-free but oxygen-poor mid-water zones. The internal oxygen store is in the form of hemoglobin-filled cells that constitute the first line of oxygen delivery to actively metabolizing cells, sparing the small air mass in the tracheal system while the hemoglobin store is being depleted.

Immature insects - nymphs

Many immature insects have special adaptations for an aquatic existence. Thin-walled protrusions of the integument, containing tracheal networks, form a series of gills (tracheal gills) that bring water into close contact with the closed tracheal tubes. The nymphs of mayflies and dragonflies have external tracheal gills attached to their abdominal segments, and certain of the gill plates may move in a way that sets up water currents over the exchange surfaces. Dragonfly nymphs possess a series of tracheal gills enclosed within the rectum. Periodic pumping of the rectal chamber serves to renew

water flow over the gills. Removing the gills or plugging the rectum results in lower oxygen consumption. Considerable gas exchange also occurs across the general body surface in immature aquatic insects.

NB: The insect tracheal system has inherent limitations. Gases diffuse slowly in long narrow tubes, and effective gas transport can occur only if the tubes do not exceed a certain length. It is generally thought that this has imposed a size limit upon insects.

3. through Book lung

The respiratory structures of spiders consist of peculiar "book lungs," leaflike plates over which air circulates through slits on the abdomen. The book lungs contain blood vessels that bring the blood into close contact with the surface exposed to the air and where gas exchange between blood and air occurs. In addition to these structures, there may also be abdominal spiracles and a tracheal system like that of insects.

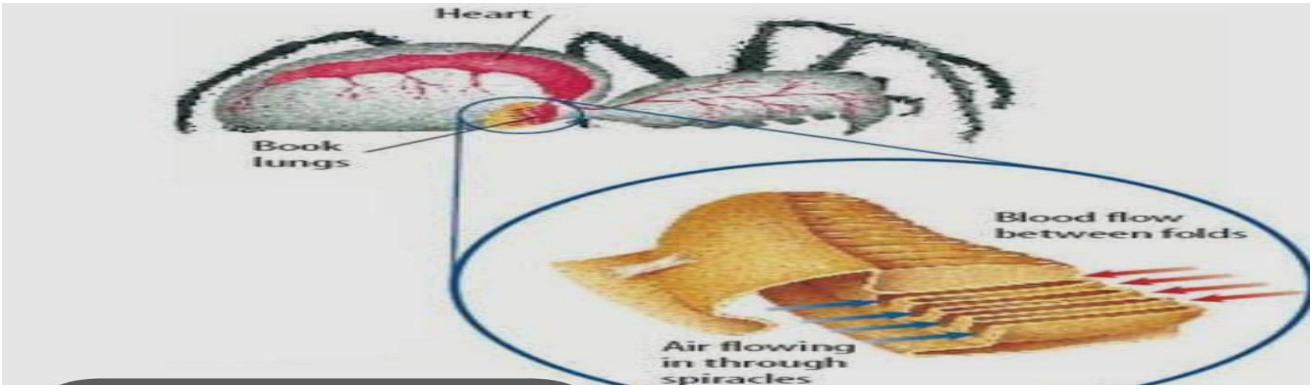


Fig. 5: Book lung of a Spider

Since spiders are air breathers, they are mostly restricted to terrestrial situations, although some of them regularly hunt aquatic creatures at stream or pond edges and may actually travel about on the surface film as easily as on land. The water spider (or diving bell spider), *Argyroneta aquatic* - known for **its underwater silk web**, which resembles a kind of diving bell - is the only species of spider that spends its entire life underwater. Using **fine hairs on its abdomen,** where its respiratory openings lie, the water spider captures tiny bubbles of air at the water's surface, transports them to its silk web, which is anchored to underwater plants or other objects, and ejects them into the interior, thereby inflating the underwater house with air. Research has shown that the inflated web serves as a sort of gill, extracting dissolved oxygen from the water when oxygen concentrations inside the web become sufficiently low to draw oxygen in from the water. As the spider consumes oxygen, nitrogen concentrations in the inflated web rises, causing it to slowly collapse. Hence, the spider must travel to the water's surface for bubble renewal, which it does about once each day. Most of the life cycle of the water spider, including courtship and breeding, prey capture and feeding, and the development of eggs and embryos, occurs below the water surface. Many of these activities take place within the spider's diving bell.

4. Gills of Invertebrates

Gills are evaginations of the body surface. Some open directly to the environment; others, as in fishes, are enclosed in a cavity just as lungs represent invaginations of the body surface. Many invertebrates use gills as a major means of gas exchange; a few, such as the pulmonate land snail, use lungs. Almost any thin-walled extension of the body surface that comes in contact with the environmental medium and across which gas exchange occurs can be viewed as a gill. Gills usually have a large surface area in relation to their mass; pumping devices are often employed to renew the external medium. Although gills are generally used for water breathing and lungs for air breathing, this association is not invariable, as exemplified by the water lungs of sea cucumbers.

The marine polychaete worms use not only the general body surface for gas exchange but also a variety of gill-like structures: segmental flaplike parapodia (in *Nereis*) or elaborate branchial tufts (among the families Terebellidae and Sabellidae). The tufts, used to create both feeding and respiratory currents, offer a large surface area for gas exchange.

In echinoderms (starfish, sea urchins, brittle stars), most of the respiratory exchange occurs across tube feet (a series of suction-cup extensions used for locomotion). However, this exchange is supplemented by extensions of the coelomic, or body-fluid, cavity into thin-walled "gills" or dermal branchiae that bring the coelomic fluid into close contact with seawater. Sea cucumbers (Holothuroidea), soft-bodied, sausage-shaped echinoderms that carry on some respiration through their oral tentacles, which correspond to tube feet, also have an elaborate "respiratory tree" consisting of branched hollow outpouchings off the cloaca (hindgut). Water is pumped in and out of this system by the action of the muscular cloaca, and it is probable that a large fraction of the animals' respiratory gas is exchanged across this system.

The gills of mollusks have a relatively elaborate blood supply, although respiration also occurs across the mantle, or general epidermis. Clams possess gills across which water circulates, impelled by the movements of millions of microscopic whips called cilia. In the few forms studied, the extraction of oxygen from the water has been found to be low, on the order of 2 to 10 percent. The currents produced by cilial movement, which constitute ventilation, are also utilized for bringing in and extracting food. At low tide or during a dry period, clams and mussels close their shells and thus prevent dehydration. Metabolism then shifts from oxygen-consuming (aerobic) pathways to oxygen-free (anaerobic) pathways, which causes acid products to accumulate; when normal conditions are restored, the animals increase their ventilation and oxygen extraction in order to rid themselves of the acid products. In snails, the feeding mechanism is independent of the respiratory surface. A portion of the mantle cavity in the form of a gill or "lung" serves as a gas-exchange site. In air-breathing snails, the "lung" may be protected from drying out through contact with the air by having only a pore in the mantle as an opening to the outside. Cephalopod mollusks, such as squid and octopus, actively ventilate a protected chamber lined with feathery gills that contain small blood vessels (capillaries); their gills are quite effective, extracting 60 to 80 percent of the oxygen passing through the chamber. In oxygen-poor water, the octopus may increase its ventilation 10-fold, indicating a more active control of respiration than appears to be present in other classes of mollusks. Many crustaceans (crabs, shrimps, crayfish) are very dependent on their gills.

Respiratory Organs of Vertebrates

In most vertebrates, the organs of external respiration are thin-walled structures well supplied with blood vessels. Such structures bring blood into close association with the external medium so that the exchange of gases takes place across relatively small distances. There are three major types of respiratory structures in the vertebrates: a) **gills**, b) **integumentary exchange areas**, and c) **lungs**. The gills are totally external in a few forms (as in *Necturus*, a neotenic salamander), but in most they are composed of filamentous leaflets protected by bony plates (as in fish). Some fishes and numerous amphibians also use the body integument, or skin, as a gas-exchange structure. Both gills and lungs are formed from outpouchings of the gut wall during embryogenesis. Such structures have the advantage of a protected internal location, but this requires some sort of pumping mechanism to move the external gas-containing medium in and out.

The quantity of air or water passing through the lungs or gills each minute is known as the **ventilation volume**. The maximum capacity of human lungs is about six litres. During normal quiet respiration, a tidal volume of about 500 millilitres is inspired and expired during every respiratory cycle. The lungs are not collapsed at the close of expiration; a certain volume of gas remains within them. At the close

of the expiratory act, a normal subject may, by additional effort, expel another 1,200 millilitres of gas. Even after the most forceful expiratory effort, however, there remains a residual volume of approximately 1,200 millilitres. By the same token, at the end of a normal inspiration, further effort may succeed in drawing into the lungs an additional 3,000 millilitres.

5. The Gills of Vertebrates

A majority of aquatic animals like fish and prawns breathe through special organs called gills. Gilla are projections of the skin that help in using oxygen dissolved in water. Gills contain blood vessels which help in exchange of gases.

Gill in Amphibians

Tadpole: This is a larval toad or frog that hatch from eggs and can only survive in water. Tadpoles have tiny external gill flaps that extract oxygen from water as it passes over them. Tadpoles open their mouths as they swim and take in water. As the mouth closes, muscles transfer the water to the gills. These consist of thin membranes called lamellae, which take oxygen from the water where it enters the blood stream through the process of diffusion. Tadpoles can also rise to the surface and gulp oxygen from the air. As tadpoles mature the gills are absorbed by the body as other respiratory systems develop. In addition to these structures, frog tadpoles use their large tail fins for respiration; the tail fins contain blood vessels and are important respiratory structures because of their large surface area. As amphibian larvae develop, the gills (and in frogs, the tail fin) degenerate, paired lungs develop, and the metamorphosing larvae begin making excursions to the water surface to take air breaths.

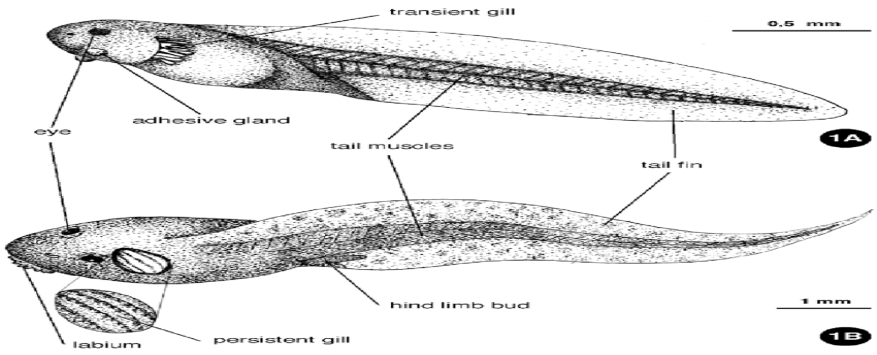


Fig.7 Gill respiration in tadpole

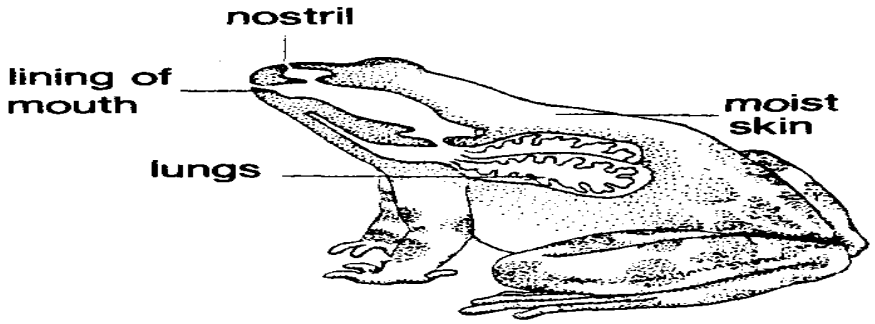
Adult Toad: The adult toad respire in three different ways namely: the mouth, skin and lungs

1. Buccal (Mouth) Gaseous Exchange

The toad is able to utilize its mouth as a respiratory organ because of the following:

- It is very large i.e. has a large surface area.
- It has a thin mucus membrane for easy diffusion
- It is well supplied with blood capillaries

To initiate breathing, the toad closes its mouth, the nostrils are opened and the floor of the buccal cavity is lowered, air is drawn through the nostrils into the buccal cavity. After this, the capillaries and the glottis close and gaseous exchange takes place between the blood and the inhaled air. To get rid of air, the floor of the buccal cavity is raised increasing the air pressure, and hereby forcing the nostrils to open and air in the buccal cavity containing carbon dioxide flows out.



Breathing organs of the frog

2. Skin/Cutaneous Respiration: This is possible because of the large surface area of the skin. The skin is moist because of continuous secretion from the mucus gland, there is adequate supply of blood capillaries and finally, the membrane serving as the skin is thin. Simple diffusion of gases takes place through it both on land and water.

3. Lungs/Pulmonary Respiration in Toad

The trachea of amphibians is not divided into secondary tubes but ends abruptly at the lungs. The relatively simple lungs of frogs are subdivided by incomplete walls (septa), and between the larger septa are secondary septa that surround the air spaces where gas exchange occurs. The diameter of these air spaces (alveoli) in lower vertebrates is larger than in mammals: The alveolus in the frog is about 10 times the diameter of the human alveolus. In order to draw air into its mouth the toad lowers the floor of its mouth, which causes the throat to expand. Then the nostrils open allowing air to enter the enlarged mouth. The nostrils then close and the air in the mouth is forced into the lungs by contraction of the floor of the mouth. To eliminate the carbon dioxide in the lungs the floor of the mouth moves down, drawing the air out of the lungs and into the mouth. Finally the nostrils are opened and the floor of the mouth moved up pushing the air out of the nostrils.

The mechanism of lung inflation in amphibians is the buccal cavity (mouth-throat) pumping mechanism that also functions in air-breathing fishes. To produce inspiration, the floor of the mouth is depressed, causing air to be drawn into the buccal cavity through the nostrils. The nostrils are then closed, and the floor of the mouth is elevated. This creates a positive pressure in the mouth cavity and drives air into the lungs through the open glottis. Expiration is produced by contraction of the muscles of the body wall and the elastic recoil of the lungs, both acting to drive gas out of the lungs through the open glottis. In aquatic amphibians the pressure of water on the body wall can also assist expiration. Many amphibians show rhythmic oscillations of the floor of the mouth between periods of lung inflation; these oscillations are thought to be involved in olfaction by producing a flow of gas over the olfactory epithelial surfaces.

Gills in Fish living in Water

The gills of fishes are supported by a series of gill arches encased within a chamber formed by bony plates (the operculum). A pair of gill filaments projects from each arch; between the dorsal (upper) and ventral (lower) surfaces of the filaments, there is a series of secondary folds, the lamellae, where the gas exchange takes place. Deoxygenated blood from the heart flows in the lamellae in a direction counter to that of the water flow across the exchange surfaces. In a number of fishes the water-to-blood distance across which gases must diffuse is 0.0003 to 0.003 millimeter, or about the same distance as the air-to-blood pathway in the mammalian lung.

The countercurrent flow of blood through the lamellae in relation to external water flow has much to do with the efficiency of gas exchange. Laboratory experiments in which the direction of water flow across fish gills was reversed showed that about 80 percent of the oxygen was extracted in the

normal situation, while only 10 percent was extracted when water flow was reversed. Countercurrent flow is a feature of elasmobranchs (sharks, skates) and cyclostomes (hagfishes, lampreys) as well as bony fishes.

To get oxygen dissolved in water, fish gulp water through the mouth and pump it over the gills. Water passes into the gill chamber through gill slits. In each chamber, the water passes over the filaments. They absorb oxygen from the water and replace it with carbon dioxide formed. The water then passes out through the gill opening and this process is repeated again and again. Among the most primitive of present-day vertebrates are the cyclostomes (lampreys and hagfishes), the gill structures of which are in the form of pouches that connect internally with the pharynx (throat) and open outward through slits, either by a fusion of the excurrent gill ducts into a single tube (in *Myxine*) or individually by separate gill slits (in *Petromyzon*). The gill lamellae of cyclostomes form a ring around the margins of the gill sac, and the series of sacs is supported in a flexible branchial skeleton. The number of paired pouches varies in different forms from six to 14. The pharynx of lampreys divides into an esophagus above and a blind tube below, from which the gill pouches arise. The upper pharynx of hagfishes communicates to the exterior through a nostril, a structure absent in lampreys. When the parasitic lampreys are embedded in the flesh of fish, upon which they live, they maintain a flow of water through the gills by alternate contractions of the gill pouches. When the gill-pouch muscles relax, the pouches expand, and water is sucked in. The water is forced out through the gills by muscular contraction; the branchial musculature apparently prevents reflux of the water into the pharynx while the head of the lamprey is embedded in the flesh of its prey.

A number of vertebrates use externalized gill structures. Some larval fishes have external gills that are lost with the appearance of the adult structures. A curious example of external gills is found in the male lungfish (*Lepidosiren*). At the time the male begins to care for the nest, a mass of vascular filaments (a system of blood vessels) develops as an outgrowth of the pelvic fins. The fish meets its own needs by refilling its lungs with air during periodic excursions to the water surface. When it returns to the nest, its pelvic-gill filaments are perfused with well-oxygenated blood, providing an oxygen supply for the eggs, which are more or less enveloped by the gill filaments.

In the hagfish *Myxine glutinosa*, the major oxygen supply is derived from water drawn in through the nostril that opens into the pharynx. A peculiar respiratory structure, the velum, just behind the nostril opening, dangles from the upper midline of the pharynx, resembling an inverted T. Membranous scrolls attached to this horizontal bar can extend downward and then roll upward like window shades. A combination of velar and gill-pouch contractions directs the flow of water through the gill pouches. Foreign material entering the nostril is expelled from both the mouth and nostril by a violent “sneeze.” This reaction probably protects the respiratory surfaces, since the animals have common respiratory and alimentary ducts. Blood flow in the gills of cyclostomes, as in those of bony fishes, is in a direction counter to that of water flow—an arrangement that increases the efficiency of gas exchange across the respiratory surface.

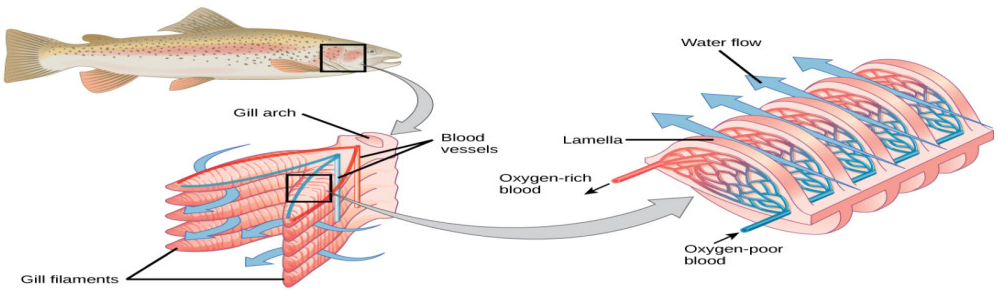


Fig. 8: Gills respiration in fish

Cartilaginous fishes (sharks and rays) and bony fishes employ a double-pumping mechanism to maintain a relatively constant flow of water over the gill exchange surfaces. In sharks and rays a small forward gill slit, the spiracle, also provides a channel for water flow into the gill chamber. Bottom-dwelling forms (*e.g.,* skates) have relatively larger spiracles, and the major portion of the water flow passes through them rather than through the downward-oriented mouth.

The pumping mechanism is not the only method of ventilation; sharks have been observed to keep both mouth and gill flaps open while swimming, ensuring a constant water flow across the gill surfaces. When they slow down or settle to the bottom, the pumping activity is resumed. Tunas and mackerel cannot stop swimming: They have no active respiratory mechanism and are dependent for their gill ventilation on the current that results from their forward motion through the water.

A number of fishes depend in varying degree on aerial respiration. The ability to breathe air enables them to live in places where the oxygen content of water may be low or nil. Two general means of acquiring oxygen are employed. Some fishes stay near the surface of the water, where the oxygen pressure resulting from surface diffusion is highest. Others have developed ancillary respiratory structures in the pharynx or the stomach; the gulping of air at the surface is a means of charging these respiratory surfaces (such as the pharyngeal epithelium in *Electrophorus* or the stomach in *Plecostomus*). The frequency with which these fishes rise to the surface to gulp air corresponds to their current need for oxygen.

The swamp-dwelling *Erythrinus* of Guyana uses both aquatic and aerial respiration, varying them according to the gaseous composition of the water. When the oxygen content is low, respiration through the gills ceases; when the oxygen content of the water is high, the fish relies primarily upon its gills except when the carbon dioxide content is also high—when, again, aerial respiration predominates. In other conditions it uses both modes of respiration. This apparently extends the range of conditions in which *Erythrinus* can survive.

Eels (*Anguilla*) use their skin as a major respiratory surface in addition to their gills. In water, about 15 percent of their oxygen uptake is across the skin, and this rises to around 50 percent when in air. They are capable of making extensive overland migrations during which, in the first few hours, they draw upon oxygen in the swim bladder. Like most fishes, eels when out of water exhibit a reduced heart rate and less oxygen consumption. When they return to water, their heart rate rises, and both oxygen consumption and blood lactic-acid levels rise. Lactic-acid production results from metabolism without oxygen, and such acid products must themselves be metabolized through higher oxygen consumption. Such patterns have been observed in grunions of the California coast that come ashore to breed, and even in flying fishes during their brief aerial excursions.

The lungfishes (Dipnoi) are remnants of the Devonian period and a transitional form between water and air breathers. Like amphibians, they rely on the buccal force pump mechanism to inflate the lung. They are adapted for bimodal respiration so that oxygenated blood leaving the air-exchange organ, either gills or lung, can pass to the afferent branchial circulation and then to the body tissues or can be dispatched to the lungs. The three dipnoan genera differ with respect to their reliance on the gills or lungs. In the Australian lungfish (*Neoceratodus*), the bulk of oxygen uptake and carbon dioxide elimination is by way of well-developed gills; in the African lungfish (*Protopterus*) and the South American lungfish (*Lepidosiren*), the gills are reduced and ventilation depends heavily on the lungs. In the latter two, which rely primarily on lung ventilation, separation of oxygenated and deoxygenated blood is much more complete than in the Australian lungfish.

During long periods of drought, both *Protopterus* and *Lepidosiren* build a subterranean cocoon that opens to the surface via a thin tunnel. They then enter into a state of estivation in which metabolism, respiration, and heart rate fall to low levels. This state of diminished oxygen requirement enables the lungfish to remain viable without food or water for months or years, until the waters return.

The bowfin, *Amia calva*, has both gills and an air bladder that may be used for respiration. It is almost exclusively a water breather at 10° C (50° F), a temperature at which it shows low physical activity. Its air-breathing rate increases with temperature and activity, and, at around 30°C (86°F), it draws about three times as much oxygen from air as from water. As in lungfishes, carbon dioxide elimination is predominantly across the gills. The bowfin's air-breathing frequency varies inversely with the oxygen content of the water; when oxygen tensions in water decline below 40 or 50 millimetres of mercury at 20°C (68°F), air breathing largely replaces water breathing. When an exchange surface (gill or air bladder) is not being utilized as the primary oxygen-exchange site, there is a tendency for blood to bypass it.

The so-called electric eel of South America (*Electrophorus electricus*) inhabits muddy streams that may become severely oxygen deficient. It is an obligatory air breather that depends upon the exchange of oxygen across the membranes of its mouth, expelling the air through its gill slits. Its blood has a high percentage of red corpuscles, is high in hemoglobin, and has an oxygen-absorbing capacity similar to that of mammals. Carbon dioxide elimination is primarily across the skin and, to a lesser extent, through the vestigial gills.

6. Through the Lungs

Amphibians, reptiles, birds and mammals exchange gases through special respiratory organs called lungs. Lungs are air-filled sac-like structures in the chest cavity. They are connected to the outside by a series of tubes and small opening. An important characteristic of lungs is their elasticity. An elastic material is one that tends to return to its initial state after the removal of a deforming force. Elastic tissues behave like springs. As the lungs are inflated, there is an accompanying increase in the energy stored within the elastic tissues of the lungs, just as energy is stored in a stretched rubber band. The conversion of this stored, or potential, energy into kinetic, or active, energy during the deflation process supplies part of the force needed for the expulsion of gases. A portion of the energy put into expansion is thus recovered during deflation. The elastic properties of the lungs have been studied by inflating them with air or liquid and measuring the resulting pressures. Muscular effort supplies the motive power for expanding the lungs, and this is translated into the pressure required to produce lung inflation. It must be great enough to overcome (1) the elasticity of the lung and its surface lining; (2) the frictional resistance of the lungs; (3) the elasticity of the thorax or thoraco-abdominal cavity; (4) frictional resistance in the body-wall structures; (5) resistance inherent in the contracting muscles; and (6) the airway resistance. The laboured breathing of the asthmatic is an example of the added muscular effort necessary to achieve adequate lung inflation when airway resistance is high, owing to narrowing of the tubes of the airways.

Lungs in Reptiles

To survive on land, the reptiles had to develop a skin relatively impermeable to water, so as to prevent desiccation, and hence not well suited for respiration. Thus, while a few specialized reptiles (for example, sea snakes) can acquire nearly half of their oxygen supply through their skin, most reptiles depend almost entirely on the lungs for gas exchange. Reptilian lungs are considerably more complex than those of amphibians, showing much more internal partitioning to provide additional

surface area for gas exchange between lung gas and blood. The most complex reptilian lungs are found in sea turtles such as *Chelonia mydas*, the green turtle. This species can develop a high metabolic rate associated with its prolific swimming ability. Its lungs are suited to providing a high rate of gas exchange, with extensive branching of the airways leading to the numerous gas sacs of the lungs.
The mechanism for lung inflation in reptiles is an aspiration (suction) pump, which is the same in general principle as the lung inflation mechanism in birds and mammals. In most reptiles inspiration is produced by muscular expansion of the rib cage and body wall, creating a sub-atmospheric pressure within the lungs that causes air to flow in. Crocodiles and alligators have a specialized muscle attached to the posterior surface of the liver; the anterior surface of the liver in turn is attached to the posterior surface of the lungs. Contraction of this muscle pulls on the liver and results in expansion of the lungs.

The adoption of a rigid shell by turtles and tortoises necessitated the development of highly specialized skeletal muscles to inflate the lungs. In the tortoise *Testudo graeca*, lung ventilation is achieved by changing the volume of the body cavity. Expiration is brought about by the activity of muscles that draw the shoulder girdle back into the shell, compressing the abdominal viscera. The increased pressure in the body cavity is transmitted to the lungs. Inspiration involves opposite muscular actions that produce an increase in the volume of the body cavity and thus a subatmospheric lung pressure. Because of the rigidity of its shell, the tortoise, unlike other reptiles, cannot use the potential energy of abdominal wall structures to assist in respiration, and hence both expiration and inspiration are active energy-consuming events. In aquatic turtles, however, the pressure of water on the front and rear limbs assist in expiration.

The breathing patterns of most reptiles are not regular, usually consisting of a series of active inspirations and expirations followed by **relatively long pauses.** In aquatic reptiles diving occurs during these pauses, which may last an hour or more in some turtles and aquatic snakes. Even terrestrial reptiles show intermittent periods of breathing and breath-holding. The metabolic rate of most reptiles is one-fifth to one-tenth that of birds or mammals, and constant lung ventilation is unnecessary in most reptiles.

Lungs in Birds
Birds must be capable of high rates of gas exchange because their oxygen consumption at rest is higher than that of all other vertebrates, including mammals, and it increases many times during flight. The gas volume of the bird lung is small compared with that of mammals, but the lung is connected to voluminous air sacs by a series of tubes, making the total volume of the respiratory system about twice that of mammals of comparable size. The trachea divides into primary bronchi, each of which passes through a lung and onward to the paired abdominal air sacs; they also give rise to secondary bronchi supplying the other air sacs. Tertiary bronchi penetrate the lung mass and, from the walls of the tertiary bronchi, rather fine air capillaries arise. These air capillaries have a large surface area; their walls contain blood capillaries connected with the heart. Gas exchange takes place between the air capillaries and blood capillaries, making this surface analogous to the alveolar surface in mammals.

There are several important differences in the mechanism and pattern of lung ventilation in birds compared with other vertebrates with lungs. The lungs of birds do not inflate and deflate but rather retain a constant volume. Also, the lungs are unidirectionally ventilated rather than having a tidal, bidirectional flow, as in other vertebrates with lungs. To achieve this unidirectional flow, the various air sacs are inflated and deflated in a complex sequence, like a series of interconnected bellows. The lungs, which are located midway between air sacs in terms of the flow of gas, are continuously

ventilated in a single direction with freshly inspired air during both inspiration and expiration at the nostrils. Aspiration into the air sacs is produced by expansion of the chest and abdominal cavity. The sternum (breastbone) swings forward and downward, while the ribs and chest wall move laterally. Expiration is caused by compression of the air sacs by skeletal muscle.

As a consequence of the continual, unidirectional airflow, the lungs of birds are more completely ventilated than the lungs of mammals. The flow of gas and blood within the bird lung is carefully arranged to maximize gas exchange, which is far more efficient than in the mammalian lung: Himalayan geese have been observed not only to fly over human climbers struggling to reach the top of Mount Everest, but to honk as they do so. The ventilation of pigeons increases around 20-fold during flight, brought about by more rapid breathing and not by taking in more air at a breath. There is a precise synchrony between breathing and wing motion: the peak of expiration occurs at the downstroke of the wingbeat.

The respiratory system of birds is also used for communication through song. The “voice box” is the syrinx, a membranous structure at the lower end of the trachea. Sound is produced only when air flows outward across the syrinx. In canaries, notes or pulses are synchronous with chest movements; the trills, however, are made with a series of shallow breaths. The song of many small birds is of long duration relative to their breathing frequencies.

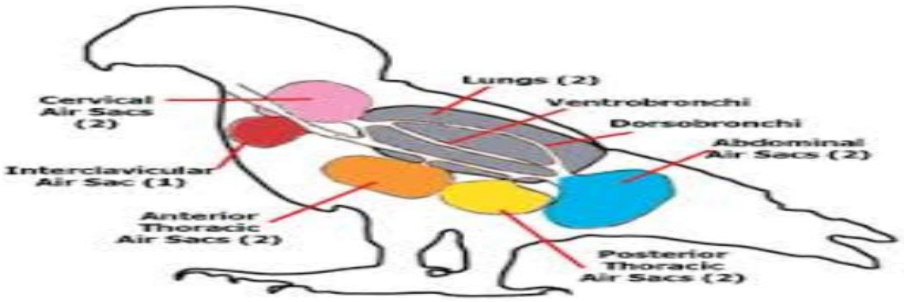


Fig. 9: Anatomy of a bird respiratory organs

Lung by Mammals

An increasing subdivision of the airways and the development of greater surface area at the exchange surfaces appear to be the general evolutionary trend among the higher vertebrates like man.

In the embryo, lungs develop as an outgrowth of the forward portion of the gut. in varying degree among different vertebrate groups.

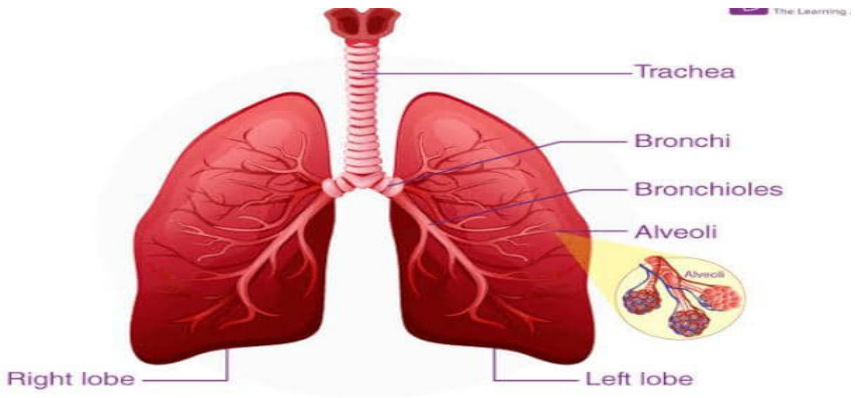


Fig. 10: The human lung

To provide the gas exchange necessary to support the elevated metabolic rate of mammals, mammalian lungs are subdivided internally. The trachea (wind pipe) branches into two bronchi. The presence of cartilagenous rings in the trachea and bronchi prevents them from collapsing when the air pressure in them is low. Each bronchus leads to the lung where it branches into small tubes called bronchioles. The alveoli are tiny balloon shaped air sacs at the end of the bronchioles richly supplied with blood capillaries and are sites where gaseous exchange takes place.

The repetitive subdivisions of the lung airways provide gas to the tiny alveoli (gas sacs) that form the functional gas-exchange surface area of the lungs.

Mechanism of Inspiration or Inhalation in Man

- The intercostal muscle contract
- The rib are moved upward and outward
- The diaphragm becomes flattened
- There is an increase in the volume of thoracic cavity
- Consequently air is drawn from the nostril to the trachea, bronchi, bronchioles, and finally to the alveoli.

Mechanism of Expiration or Exhalation in Man

- The intercostal muscle relax
- The ribs are moved downward and inward
- The diaphragm becomes dome shape
- There is decrease in the volume of thoracic cavity.
- Consequently air is drawn from the alveoli, bronchioles, bronchi, and trachea.
- Consequently air containing waste products like CO_2 and water vapour from inside the alveoli or lungs are forced out through bronchioles, bronchi, trachea, and finally to the exterior through the nos

MODE OF RESPIRATION IN PLANTS

Yes, like animals and humans, plants also breathe. Plants do require oxygen to respire, the process in return gives out carbon dioxide. Unlike humans and animals, plants do not possess any specialized structures for exchange of gases, however, they do possess stomata (found in leaves) and lenticels (found in stems) actively involved in the gaseous exchange. Leaves, stems and plant roots respire at a low pace compared to humans and animals.

We know that plants possess a specific ability to synthesize their own food through photosynthesis. Photosynthesis takes place in only those parts of the plants which have chlorophyll, the green plant parts. Photosynthesis is so evident that at times it seems to mask the respiratory process in plants. Respiration must not be mistaken for photosynthesis. Respiration occurs all through the day, but the photosynthesis process occurs in the daytime, in the presence of sunlight only. Consequently, respiration becomes evident at night time in plants. This is the reason we often hear people warn against sleeping under a tree during nighttime, as it may lead to suffocation due to excess amounts of carbon dioxide liberated by trees following respiration.

The Process of Respiration in Plants

Oxygen, carbon dioxide, and water vapour are released by simple diffusion process in plants. Oxygen diffuses into plants through the stomata and lenticels while CO_2 and water vapour diffuses out of the plant through the same opening. This is facilitated by the difference in the concentration gradient of these gases. The plant takes in oxygen mostly during the night and gives out carbon dioxide and water vapour during the day due to photosynthetic activities of the plant. Oxygen is the by-product of photosynthesis.

Respiration in Roots

Roots - the underground part of the plants, absorbs air from the air gaps/spaces found between the soil particles. Hence, absorbed oxygen through roots is utilized to liberate the energy that in the future, is used to transport salts and minerals from the soil.

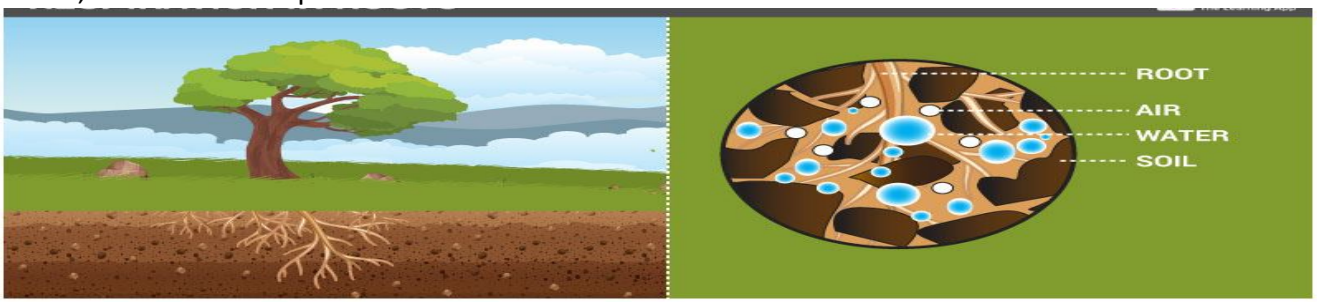


Fig. 12: Respiration by plant roots

Respiration in Stems



Fig. 13: Respiration in Stems

The air in case of stem diffuses into the stomata and moves through different parts of the cell to respire. During this stage, the carbon dioxide liberated is also diffused through the stomata. Lenticels are known to perform gaseous exchange in woody or higher plants.

Respiration In Leaves

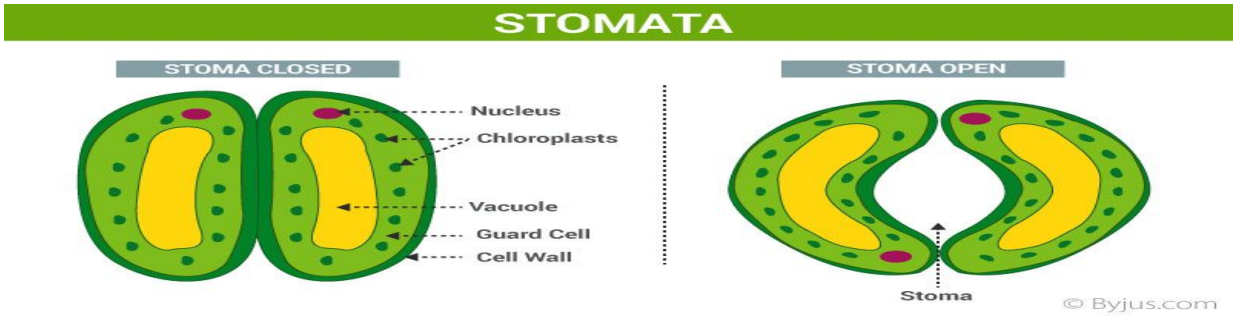


Fig. 14: Respiration by plant stomata

Leaves consist of tiny pores known as stomata. Gaseous exchange occurs through diffusion via stomata. Guard cells regulate each of the stomata. Exchange of gases occurs with the closing and opening of the stoma between the inferior of leaves and the atmosphere.

SUMMARY

Minute life-forms, such as protozoans, exchange oxygen and **carbon dioxide** across their entire surfaces. Multicellular organisms, in which **diffusion** distances are longer, generally resort to other strategies. Aquatic worms, for example, lengthen and flatten their bodies to refresh the external medium at their surfaces. Sessile sponges rely on the ebb and flow of ambient water. By contrast, the jellyfish, which can be quite large, has a low oxygen need because its content of organic matter is less than 1 percent and its metabolizing cells are located just beneath the surface, so that diffusing distances are small.

Organisms too large to satisfy their oxygen needs from the environment by diffusion are equipped with special respiratory structures in the form of gills, lungs, specialized areas of the intestine or pharynx (in certain fishes), or tracheae (air tubes penetrating the body wall, as in insects). Respiratory structures typically have an attenuated shape and a semi-permeable surface that is large in relation to the volume of the structure. Within them there is usually a circulation of body fluids (blood through the lungs, for example). Two sorts of pumping mechanisms are frequently encountered: one to renew the external oxygen-containing medium, the other to ensure circulation of the body fluids through the respiratory structure. In air-breathing vertebrates, alternately contracting sets of muscles create the pressure differences needed to expand or deflate the lungs, while the heart pumps blood through the respiratory surfaces within the lungs. Oxygenated blood returning to the heart is then pumped through the vascular system to the various tissues where the oxygen is consumed.

NNAMDI AZIKIWE UNIVERSITY, AWKA

BIO102 COURSE

TOPIC: ENVIRONMENTAL HEALTH AND DISEASES

KINGNAS

FOR

100 LEVEL STUDENTS

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ENVIRONMENTAL HEALTH AND DISEASES

Definition of Terms

Environment

Environment is defined as the complex of physical, chemical and biotic factors that act upon an organism or an ecological community and ultimately determines its form and survival. It is categorized into air environment, water environment, socio-economic environment and cultural environment.

Health

Health is a state of complete physical, mental, and social well being and not merely absence of disease or infirmity.

Environmental Health

Environmental health involves those aspects of the human health and diseases that are determined by the factors in the environment .It includes both the direct pathological effects of chemicals, radiation and some biological agents, and the effects on health and well-being of the broad physical, psychological, social and cultural environment.

Environmental health is also defined as the control of all factors in man's environment which exercise, or may exercise a deleterious effect on his physical development, health or survival. It comprised the aspect of human health, including quality of life, which is determined by physical, biological, chemical social and psychological factors in the environment.

It is an organized community efforts which monitor the relationship between man and his environment in other to achieve a better healthy living. The components of environmental health as specified by world health organization include: waste management, food hygiene and safety, pest and vector control, housing sanitation, air quality monitoring, occupational health and safety, water resources management and sanitation, environmental health impact and assessment, health education and promotion, collection and analysis of environmental data, epidemiological study and control of communicable diseases, noise pollution monitoring and prevention, control of frontiers , air and seaport and border crossing, protection of public amenities and recreational environment, promotion and enforcement of environmental health quality standards.

Causes of environmental health problems

The failure of the numerous efforts to address the problem of environmental health has been attributed to various factors. Prominent among these are unhealthy socio-cultural practices, poor environmental sanitation education and awareness, low literacy level, bad governance over the years, disregard for the rule of law and other forms of indiscipline.

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Environmental Diseases

Environmental diseases are illnesses and conditions that result from man-made environmental problems. Illnesses and conditions caused by factors in the environment are collectively called environmental diseases. Pesticides, chemicals, radiation, air pollution, and water pollution, are some of the man-made hazards that are believed to contribute to human illnesses. Potential illness-causing agents are everywhere: at home, at work, and at play. However, the likelihood of an individual developing a specific disease depends on the hazards present in their particular environment and their genetic susceptibility to a specific hazard. For example, x-ray technicians are at risk for radiation-induced illnesses, whereas coal miners are prone to lung diseases caused by inhalation of dust. Proper use of safeguards can prevent these and other environmental diseases.

What Are Some of the Common Environmental Diseases?

1. Lung Diseases

Any substance other than air that is breathed into the lungs has the potential to cause damage to these organs. For example, air pollution, including smoke from other peoples' cigarettes (secondhand smoke), and workplace chemicals can lead to lung diseases. Examples of lung diseases include:

Asthma, a condition in which breathing is difficult, affects millions of Americans. Environmental triggers for asthma are everywhere and include naturally occurring triggers such as animal dander, plant pollen, dust, and mold, and manmade triggers such as chemicals. Not everyone is sensitive to these triggers, but many people are sensitive to some of them.

Black lung disease is an illness in which coal miners' lungs become coated with coal dust, causing a chronic condition in which breathing becomes difficult and painful.

Bronchitis, an inflammation of the airways of the lungs, can be caused by breathing in certain chemicals or smoke. Welders and fire fighters are some of the people at risk for this condition. Smokers are also at increased risk for the development of bronchitis and lung cancer.

Breathing asbestos (a natural mineral fiber) can lead to asbestosis (as-bes-TO-sis), a severe lung disorder, and lung cancer. Schools, homes, and businesses that have asbestos in them as a fire retardant put people at risk when the asbestos dust begins to leak into the air during repairs and renovations. These buildings used asbestos in the walls and ceilings as insulation before laws were passed to ban their use.

Silicosis is a lung disease caused by exposure to the silica dust in clay. Pottery workers are at risk of developing this disease.

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2. Cancers

In addition to lung cancer, other cancers have been linked to environmental toxins (poisons). For example, pesticides, herbicides, and radioactive substances have the potential to cause cancer. There are almost 2,000 chemicals that are suspected of causing cancer. Of these, only several hundred have had their use restricted by law in the United States.

Asbestos, chromium, and coal tar have been linked to lung cancer. Construction workers, welders, and steelworkers may be repeatedly exposed to these compounds. People working in plastics manufacturing are at risk for liver or bladder cancer. People who work with radioactive substances are at increased risk for cancer caused by radiation. Fortunately, legal restrictions and careful oversight of hazardous materials can reduce risks.

3. Gulf War Syndrome

Many United States veterans have complained about disabling symptoms that they attribute to their participation in the 1991 Persian Gulf War. Gulf War Syndrome encompasses symptoms such as chronic fatigue, aching muscles and joints, skin rashes, memory loss, miscarriages, and babies born with birth defects.

Gulf War Syndrome is a mysterious condition. While veterans are experiencing very real medical problems, physicians and scientists cannot agree on the source of these problems. Some people believe that the syndrome actually consists of multiple illnesses for which symptoms overlap. Veterans could be reacting to chemical weapons, biological weapons, pesticides, vaccines, oil fires, or infectious diseases that they were exposed to in the Gulf War. A presidential commission set up to study the controversy surrounding the Gulf War Syndrome proposed that stress was the major cause behind the symptoms the veterans were experiencing. A decade after it began, the controversy continues as conflicting reports emerge.

4. Birth Defects

Infertility, miscarriage, stillbirth (the baby is born dead), childhood cancer, and birth defects may have links to various environmental toxins. When a pregnant woman is exposed to lead, her child has a higher than usual risk of being born with behavior and nervous system problems. Exposure to radiation, chemical wastes, pesticides, solvents, paints, lead, and methyl mercury all can cause problems in a developing fetus.

5. Chemical Poisonings

Lead is a serious environmental hazard to children in many parts of the world, including the United States. It affects children's mental and physical development, and high doses can cause paralysis and death. People can be exposed to lead through lead paint, leaded gasoline, lead water pipes, and certain ceramics. Although lead is no longer used in most of these products in the United States, it can still be found in older homes and in some imported products.

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The metals mercury and cadmium can cause nerve damage, cancer, and liver and skin diseases. Mercury has been used since the beginning of the nineteenth-century in many industrial processes. In earlier times, people who worked with mercury were often unknowingly poisoned. Mercury can accumulate in the food chain and present a health risk. For example, some fish in the Great Lakes are contaminated with mercury they acquired through eating plants and other fish. Eating a lot of these contaminated fish can transfer unhealthy levels of mercury to a person. The effects are cumulative, (build up), because the body cannot rid itself of mercury. Mercury can be found in fluorescent lights, latex paint, batteries, dental fillings, and mercury thermometers.

Other sources of environmental poisoning can arise from the manufacture of refrigerants, plastics, and other industrial products and from the manufacture and misuse of pesticides.

6. **Neglected tropical diseases (NTDs),**

Neglected tropical diseases (NTDs), a group of communicable diseases found in predominantly tropical and subtropical conditions, affect the poorest populations who often live in remote, rural areas, in urban slums, or in conflict zones. The Centre for Disease Control (CDC) reports that one hundred percent of low-income countries are concurrently affected by at least five neglected tropical diseases. With little political voice, neglected tropical diseases typically have low status among public health priorities. In 2012, the WHO published a comprehensive plan that established targets for the reduction of NTDs, representing a shift in these priorities. In 2010, an estimated 2 billion individuals needed treatment for neglected tropical diseases. By 2016, the estimate decreased to 1.5 billion confirming the great gains have been through approaches in treatment and prevention.

Treatment and Prevention

Many measures can be taken almost immediately to reduce the environmental disease burden, such as the promotion of safe household water storage, improved hygiene, and the control of the use of toxic substances in the home and workplace. Though many of these diseases can be treated with simple, inexpensive medications, global strategies to control NTDs also include **preventive chemotherapy, intensified case management, vector control, veterinary public health measures and improved water quality and sanitation.**

In 2013, the WHO categorized seventeen diseases as neglected tropical diseases, namely **buruli ulcer, Chagas diseases, dengue fever, guinea-worm, liver flukes, trachoma, tapeworms, yaws, soil-transmitted helminthiases, river blindness, rabies, lymphatic filariasis, leishmaniasis, schistosomiasis, sleeping sickness and hydatid disease. Recent additions include mycetoma and snakebite envenoming.** This text provides a comprehensive review of a few of the major neglected tropical diseases:

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Trachoma

Trachoma is the leading cause of preventable blindness of infectious origins caused by the bacterium *Chlamydia trachomatis*. Trachoma is spread easily by contact with an infected person's hands or clothing, or by flies that come in contact with the eyes or nose of an infected person. Infected individuals do not instantly go blind; instead, the disease manifests gradually. Scarring from repeated infections causes the eyelashes to turn inward and scratch the cornea, a condition known as trichiasis, leading slowly and painfully to complete blindness. Since trachoma is transmitted through close personal contact, it tends to occur in clusters, often infecting entire families and communities. Estimates suggest that 1.9 million people are blind due to trachoma while another 21 million are actively infected with the disease.

Blindness from trachoma strikes adults in their prime years, hindering their ability to care for themselves and their families. Women, traditionally the caretakers of the home, are four times more likely than men to be affected by trichiasis due to increased contact with infected children and limited financial agency in health-related decisions necessary for treatment and prevention.

Treatment and Prevention

Trachoma is treatable and preventable with an approach known as the SAFE strategy. Recommended by the World Health Organization, the SAFE strategy is a comprehensive public health approach that combines treatment (Surgery and Antibiotics) with prevention (Facial cleanliness and Environmental improvement). The strategy works as follows:

S= Surgery to correct advances of the disease
A= Antibiotics, such as azithromycin, to treat infected individuals
F= Face washing to reduce transmission
E= Environmental improvement through increased access to clean water and improved sanitation

Through the work of national governments and various partners, the implementation of the SAFE Strategy has led to measurable results in the elimination of trachoma. As of 2016, over 260,000 individuals received surgery for trichiasis and over 85 million antibiotic treatments were given worldwide.

River blindness

Onchocerciasis, also known as river blindness, infects approximately 18 million people, of which the majority live in sub-Saharan Africa. Over 120 million globally are at risk for contracting river blindness. Approximately 6.5 million individuals experience severe itching due to onchocerciasis and another 270,000 have been left permanently blind from the disease. People living near the fast-moving streams of Africa are at high risk since these areas are breeding grounds for the *Simulium* black fly. Infected flies carry larvae of the filarial parasitic worm *Onchocerca volvulus* from person to person. Larvae, which enter the skin at the black fly

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bite site, move throughout the body and then die, resulting in a variety of health problems including skin rashes and de-pigmentation, itching, and skin nodules. Visual impairment and blindness occur if larvae enter the eye.

Treatment and Prevention

The severe socioeconomic impact of river blindness prompted the creation of the highly successful Onchocerciasis Control Program in 1974. This program focused on 11 countries in West Africa that aimed to make ivermectin treatment more accessible and included intensified spraying of black fly breeding sites with environmentally safe insecticides. The program ended in 2002 after the successful end to transmission of the disease in 10 of the 11 participating countries. In 2016, over 133 million treatments were provided globally. Current elimination barriers include political uncertainty and competing health priorities, though progress has been made and four countries within Central and South America have successfully eliminated river blindness since 2013.

Guinea Worm disease

Guinea worm disease, formally known as Dracunculiasis, is caused by the parasitic worm *Dracunculus medinensis*. It is transmitted through ingestion of stagnant water that is contaminated with guinea worm larvae. In the human body, the larvae are released and migrate into body tissues, and develop into adult worms. Female worms move through the person's tissue, causing intense pain, and eventually emerge through the skin, producing oedema, a blister and eventually an ulcer, accompanied by fever. Though rarely fatal, symptoms include joint pain, vomiting and severe burning. There are no drugs available for treatment of the disease and the worm is instead physically removed over a period of several weeks. Prevention is key in the eradication efforts of guinea worm, specifically by protecting water sources and filtering contaminated water.

Treatment and Prevention

The global transmission of guinea worm disease has significantly been reduced with only three reported cases as of April 2018 within the endemic countries of Chad, Ethiopia and Mali. The fight to end guinea worm was greatly influenced by The Carter Center which led the Guinea Worm Eradication Program along with partnerships with local ministries of health and global organizations beginning in the 1980s. The program's success was a direct result of community-based interventions that focused on water filtration and health education. Guinea worm disease is on track to be the first eradicated neglected tropical diseases due to these concerted efforts on a global and local level.

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Schistosomiasis

Schistosomiasis is acquired through contact with parasitic worms that are released by freshwater snails. Â Inside the body, the larvae develop into worms. Female worms release thousands of eggs which are passed out of the body and released into bodies of freshwater by urination or defecation. They migrate to snails and begin the cycle again. Some eggs trapped in the body migrate to specific organs and inflict major damage such as scarring and tearing of the bladder and kidneys. Estimates suggest that over 205 million individuals required treatment for schistosomiasis in 2015. Schistosomiasis is prevalent in 78 countries, with 92 % of those requiring treatment living in sub-Saharan Africa. The most susceptible populations include children under age 14, as well as individuals with responsibilities and occupations near infested water sources.

Treatment and Prevention

Improved sanitation and hygiene are critical to the elimination of schistosomiasis. Additional measures include efforts to eliminate the snail species required for transmission. Preventative chemotherapy, where individuals are treated without being diagnosed, has been shown to be successful in decreasing the disease burden of schistosomiasis globally. Praziquantel is the only recommended drug to treat schistosomiasis and has also been shown to reduce the severity of symptoms in those re-infected with the disease.

Soil Transmitted Helminth Infections

Hookworm, roundworm and whipworm make up the group of intestinal soil-transmitted infections and are among the most common infections in the world. These infections are associated with poverty and poor sanitation given their transmission through contaminated food, soil, and feces. There are an estimated 1.5 billion individuals infected by soil-transmitted helminths.

Ascariasis, also known as roundworm, is an intestinal infection caused by the soil transmitted parasitic worm Ascarislumbricoidesart. It is most prevalent in warm tropical and sub-tropical climates in Sub-Saharan Africa and Southeast Asia. The disease flourishes in areas with poor sanitation and poorly irrigated crops treated with wastewater. It is the most common human worm infection. Between 807 million to 1.2 billion people worldwide are infected with ascariasis. It is transmitted by accidental ingestion of ascaris eggs in contaminated food, water or soil. The larvae migrate into the bloodstream and cause severe coughing and wheezing.

Whipworms, caused by the soil transmitted parasitic worm *Trichuris trichiura* affect approximately 604-795 million individuals. Individuals infected by only a few worms usually do not experience symptoms while those with a higher parasitic load may experience weakness, anemia, rectal prolapse and even hinder growth in children.

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Hookworm, caused by the soil transmitted parasitic worms *Necator americanus* and *Ancylostoma duodenale* affect approximately 576-740 million individuals globally. Infected individuals typically experience gastrointestinal symptoms.

Prevention & Treatment

Albendazole and mebendazole are the recommended medications for intestinal parasitic worms. These medications are easily accessible through the ministries of health in endemic countries. The WHO adopted a resolution to eliminate the disease burden due to soil-transmitted helminth infections by 2020 which would require periodic deworming of 75% of all at-risk school children by 2020. The program has been able to achieve some success in reducing the severity of worm infections in some areas, particularly in Africa. In 2017, 485 million school age children were treated with deworming tablets. Education efforts aimed at prevention through proper sanitation, hand washing, and food preparation techniques are critical to reducing the incidence of the disease in regions of Africa.

Lymphatic Filariasis

Lymphatic Filariasis, also known as elephantiasis is common in tropical and sub-tropical climates, in both rural and urban areas where infected mosquitoes tend to breed. Over 850 million people are at risk for becoming infected with lymphatic filariasis with approximately 40 million currently affected by the disease. Lymphatic Filariasis is transmitted when infected mosquitoes bite humans and pass microfilariae into the skin. Larvae migrate through the body and settle in the lymphatic system, where they lay eggs that pass into the blood. The worms can live for up to six years here. When they die, they can cause severe disfiguring ailments. Initially, no symptoms may be present, but as the disease progresses, the following may occur: swelling or hardening of the skin, severe swelling/lymphedema of extremities or genitals, and "filarial fevers" which are episodes of severe swelling in chronic patients. The disease has been associated with social stigma leading to emotional and mental suffering.

Treatment and Prevention

Unfortunately, there is no cure or vaccine for the disease, but symptoms can be managed with a treatment plan. The treatment to reduce the symptoms of Lymphatic Filariasis includes a combination of albendazole, diethylcarbamazine (DEC), and ivermectin medicines. Evidence suggests that treatment with these medications concurrently can eliminate the microfilaria from the bloodstream in a few weeks times. Careful cleansing of the skin to remove bacteria can reduce or reverse skin or tissue damage. The 1997 WHO's Global Program to Eliminate Lymphatic Filariasis was implemented to stop the spread of transmission and to lessen the severity of the disease in those already infected. Hygiene education efforts aimed at reducing acute episodes of filarial fevers have been effective at lessening the social stigmas associated with the manifestations of the disease. Since 1997, 18 countries have begun the process to

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confirm elimination while mass treatment is currently ongoing in 45 countries demonstrating great strides in the elimination of lymphatic filariasis.

Sick Building Syndrome

Sick Building Syndrome describes an elusive health problem in which people attribute a variety of symptoms to the buildings where they work. Common complaints include headaches, dizziness, nausea, tiredness, concentration problems, sensitivity to odors, dry itchy skin, a dry cough, and irritated eyes, nose, and throat. Generally, as soon as affected people leave the building, their symptoms vanish and they feel well again.

Sick Building Syndrome is elusive in that neither one specific illness nor a common cause has been identified. No specific set of symptoms is common to all people complaining of Sick Building Syndrome. While nothing has been proven, theories as to the causes of Sick Building Syndrome abound. Many factors may contribute to it, including humidity, poor ventilation, and poor temperature control. Pollution from outdoor sources, (for example car exhaust, pollen, or smoke) or chemicals from inside a building, (cleaning supplies, glues, upholstery, or copy machine chemicals), may affect people who are sensitive to them. Bacteria, viruses, and molds also can invade buildings and make people sick.

ANTHROPOGENIC IMPACTS OF THE ENVIRONMENT

Human impact on the environment or anthropogenic impact on the environment includes changes to biophysical environments and ecosystems, biodiversity, and natural resources caused directly or indirectly by humans, including global warming environmental degradation (such as ocean acidification), mass extinction and biodiversity loss, ecological crisis, and ecological collapse. Modifying the environment to fit the needs of society is causing severe effects, which become worse as the problem of human overpopulation continues. Some human activities that cause damage (either directly or indirectly) to the environment on a global scale include human reproduction, overconsumption, overexploitation, pollution, and deforestation, to name but a few. Some of the problems, including global warming and biodiversity loss pose an existential risk to the human race, and overpopulation causes those problems.

The term anthropogenic designates an effect or object resulting from human activity. The term was first used in the technical sense by Russian geologist Alexey Pavlov, and it was first used in English by British ecologist Arthur Tansley in reference to human influences on climax plant communities. The atmospheric scientist Paul Crutzen introduced the term "Anthropocene" in the mid-1970s. The term is sometimes used in the context of pollution emissions that are produced from human activity but also applies broadly to all major human impacts on the environment.

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Human overpopulation and effects on the environment

David Attenborough described the level of human population on the planet as a multiplier of all other environmental problems. In 2013, he described humanity as "a plague on the Earth" that needs to be controlled by limiting population growth.

Some deep ecologists, such as the radical thinker and polemicist PenttiLinkola, see human overpopulation as a threat to the entire biosphere. In 2017, over 15,000 scientists around the world issued a second warning to humanity which asserted that rapid human population growth is the "primary driver behind many ecological and even societal threats."

Overconsumption

Overconsumption is a situation where resource use has outpaced the sustainable capacity of the ecosystem. It can be measured by the ecological footprint, a resource accounting approach which compares human demand on ecosystems with the amount of planet matter ecosystems can renew. Estimates indicate that humanity's current demand is 70% higher than the regeneration rate of all of the planet's ecosystems combined. A prolonged pattern of overconsumption leads to environmental degradation and the eventual loss of resource bases.

Humanity's overall impact on the planet is affected by many factors, not just the raw number of people. Their lifestyle (including overall affluence and resource utilization) and the pollution they generate (including carbon footprint) are equally important. In 2008, The New York Times stated that the inhabitants of the developed nations of the world consume resources like oil and metals at a rate almost 32 times greater than those of the developing world, who make up the majority of the human population.

Technology

The applications of technology often result in unavoidable and unexpected environmental impacts, which according to the I = PAT equation is measured as resource use or pollution generated per unit GDP. Environmental impacts caused by the application of technology are often perceived as unavoidable for several reasons. First, given that the purpose of many technologies is to exploit, control, or otherwise "improve" upon nature for the perceived benefit of humanity while at the same time the myriad of processes in nature have been optimized and are continually adjusted by evolution, any disturbance of these natural processes by technology is likely to result in negative environmental consequences. Second, the conservation of mass principle and the first law of thermodynamics (i.e., conservation of energy) dictate that whenever material resources or energy are moved around or manipulated by technology, environmental consequences are inescapable. Third, according to the second law of thermodynamics, order can be increased within a system (such as the human economy) only by increasing disorder or entropy outside the system (i.e., the environment). Thus, technologies can create "order" in the

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human economy (i.e., order as manifested in buildings, factories, transportation networks, communication systems, etc.) only at the expense of increasing "disorder" in the environment.

Environmental impact of agriculture

The environmental impact of agriculture varies based on the wide variety of agricultural practices employed around the world. Ultimately, the environmental impact depends on the production practices of the system used by farmers. The connection between emissions into the environment and the farming system is indirect, as it also depends on other climate variables such as rainfall and temperature.

The environmental impact of agriculture involves a variety of factors from the soil, to water, the air, animal and soil diversity, plants, and the food itself. Some of the environmental issues that are related to agriculture are climate change, deforestation, genetic engineering, irrigation problems, pollutants, soil degradation, and waste.

The environmental impact of fishing can be divided into issues that involve the availability of fish to be caught, such as overfishing, sustainable fisheries, and fisheries management; and issues that involve the impact of fishing on other elements of the environment, such as by-catch and destruction of habitat such as coral reefs.

Environmental impact of irrigation

The environmental impact of irrigation includes the changes in quantity and quality of soil and water as a result of irrigation and the ensuing effects on natural and social conditions at the tail-end and downstream of the irrigation scheme. The impacts stem from the changed hydrological conditions owing to the installation and operation of the scheme. An irrigation scheme often draws water from the river and distributes it over the irrigated area. As a hydrological result it is found that the:

- downstream river discharge is reduced
- evaporation in the scheme is increased
- groundwater recharge in the scheme is increased
- level of the water table rises
- drainage flow is increased.

These may be called direct effects.

Effects on soil and water quality are indirect and complex, and subsequent impacts on natural, ecological and socio-economic conditions are intricate. In some, but not all instances, water logging and soil salinization can result. However, irrigation can also be used, together with soil drainage, to overcome soil salinization by leaching excess salts from the vicinity of the root zone.

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Irrigation can also be done extracting groundwater by (tube) wells. As a hydrological result it is found that the level of the water descends. The effects may be water mining, land/soil subsidence, and, along the coast, saltwater intrusion.

Irrigation projects can have large benefits, but the negative side effects are often overlooked. Agricultural irrigation technologies such as high powered water pumps, dams, and pipelines are responsible for the large-scale depletion of fresh water resources such as aquifers, lakes, and rivers. As a result of this massive diversion of freshwater, lakes, rivers, and creeks are running dry, severely altering or stressing surrounding ecosystems, and contributing to the extinction of many aquatic species.

It increases the moisture contents of the soil and high humidity.

Environmental impact of coal mining and burning

The environmental impact of coal mining and -burning is diverse. Legislation passed by the US Congress in 1990 required the United States Environmental Protection Agency (EPA) to issue a plan to alleviate toxic air pollution from coal-fired power plants.

Environmental impact of electricity generation

The environmental impact of electricity generation is significant because modern society uses large amounts of electrical power. This power is normally generated at power plants that convert some other kind of energy into electricity. Each such system has advantages and disadvantages, but many of them pose environmental concerns.

Environmental impact of nuclear power

The environmental impact of nuclear power results from the nuclear fuel cycle processes including mining, processing, transporting and storing fuel and radioactive fuel waste. Released radioisotopes pose a health danger to human populations, animals and plants as radioactive particles enter organisms through various transmission routes.

Radiation is a carcinogen and causes numerous effects on living organisms and systems. The environmental impacts of nuclear power plant disasters such as the Chernobyl disaster, the Fukushima Daiichi nuclear disaster and the Three Mile Island accident, among others, persist indefinitely, though several other factors contributed to these events including improper management of failsafe systems and natural disasters putting uncommon stress on the generators. The radioactive decay rate of particles varies greatly, dependent upon the nuclear properties of a particular isotope. Radioactive Plutonium-244 has a half-life of 80.8 million years, which indicates the time duration required for half of a given sample to decay, though very little plutonium-244 is produced in the nuclear fuel cycle and lower half-life materials have lower activity thus giving off less dangerous radiation.

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Environmental impact of the oil shale industry

The environmental impact of the oil shale industry includes the consideration of issues such as land use, waste management, and water and air pollution caused by the extraction and processing of oil shale. Surface mining of oil shale deposits causes the usual environmental impacts of open-pit mining. In addition, the combustion and thermal processing generate waste material, which must be disposed of, and harmful atmospheric emissions, including carbon dioxide, a major greenhouse gas. Experimental in-situ conversion processes and carbon capture and storage technologies may reduce some of these concerns in future, but may raise others, such as the pollution of groundwater.

Environmental impact of petroleum

The environmental impact of petroleum is often negative because it is toxic to almost all forms of life. Petroleum, a common word for oil or natural gas, is closely linked to virtually all aspects of present society, especially for transportation and heating for both homes and for commercial activities.

Environmental impact of reservoirs

The environmental impact of reservoirs is coming under ever increasing scrutiny as the world demand for water and energy increases and the number and size of reservoirs increases.

Dams and the reservoirs can be used to supply drinking water, generate hydroelectric power, increasing the water supply for irrigation, provide recreational opportunities and flood control. However, adverse environmental and sociological impacts have also been identified during and after many reservoir constructions. Although the impact varies greatly between different dams and reservoirs, common criticisms include preventing sea-run fish from reaching their historical mating grounds, less access to water downstream, and a smaller catch for fishing communities in the area. Advances in technology have provided solutions to many negative impacts of dams but these advances are often not viewed as worth investing in if not required by law or under the threat of fines. Whether reservoir projects are ultimately beneficial or detrimental—to both the environment and surrounding human populations— has been debated since the 1960s and probably long before that. In 1960 the construction of LlynCelyn and the flooding of CapelCelyn provoked political uproar which continues to this day. More recently, the construction of Three Gorges Dam and other similar projects throughout Asia, Africa and Latin America have generated considerable environmental and political debate.

Manufactured products

Cleaning agents

Environmental impact of cleaning agents

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The environmental impact of cleaning agents is diverse. In recent years, measures have been taken to reduce these effects.

Environmental impact of nanotechnology

Nanotechnology's environmental impact can be split into two aspects: the potential for nanotechnological innovations to help improve the environment, and the possibly novel type of pollution that nanotechnological materials might cause if released into the environment. As nanotechnology is an emerging field, there is great debate regarding to what extent industrial and commercial use of nanomaterials will affect organisms and ecosystems.

Environmental impact of paint

The environmental impact of paint is diverse. Traditional painting materials and processes can have harmful effects on the environment, including those from the use of lead and other additives. Measures can be taken to reduce environmental impact, including accurately estimating paint quantities so that wastage is minimized, use of paints, coatings, painting accessories and techniques that are environmentally preferred. Some scientists suggest that by 2050 there could be more plastic than fish in the oceans.

Environmental impact of pesticides

The environmental impact of pesticides is often greater than what is intended by those who use them. Kills both target and non-target organisms. Over 98% of sprayed insecticides and 95% of herbicides reach a destination other than their target species, including non-target species, air, water, bottom sediments, and food. Pesticide contaminates land and water when it escapes from production sites and storage tanks, when it runs off from fields, when it is discarded, when it is sprayed aerially, and when it is sprayed into water to kill algae.

The amount of pesticide that migrates from the intended application area is influenced by the particular chemical's properties: its propensity for binding to soil, its vapor pressure, its water solubility, and its resistance to being broken down over time. Factors in the soil, such as its texture, its ability to retain water, and the amount of organic matter contained in it, also affect the amount of pesticide that will leave the area. Some pesticides contribute to global warming and the depletion of the ozone layer.

Can bring about resistance which warrants that a greater dosage is used.

Environmental impact of mining

The environmental impact of mining includes erosion, formation of sinkholes, loss of biodiversity, and contamination of soil, groundwater and surface water by chemicals from mining processes. In some cases, additional forest logging is done in the vicinity of mines to increase the available room for the storage of the created debris and soil. Besides creating

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environmental damage, the contamination resulting from leakage of chemicals also affect the health of the local population. Mining companies in some countries are required to follow environmental and rehabilitation codes, ensuring the area mined is returned to close to its original state. Some mining methods may have significant environmental and public health effects.

Environmental impact of transport

The environmental impact of transport is significant because it is a major user of energy, and burns most of the world's petroleum. This creates air pollution, including nitrous oxides and particulates, and is a significant contributor to global warming through emission of carbon dioxide, for which transport is the fastest-growing emission sector. By subsector, road transport is the largest contributor to global warming.

Environmental regulations in developed countries have reduced the individual vehicles emission; however, this has been offset by an increase in the number of vehicles, and more use of each vehicle. Some pathways to reduce the carbon emissions of road vehicles considerably have been studied. Energy use and emissions vary largely between modes, causing environmentalists to call for a transition from air and road to rail and human-powered transport, and increase transport electrification and energy efficiency.

Other environmental impacts of transport systems include traffic congestion and automobile-oriented urban sprawl, which can consume natural habitat and agricultural lands. By reducing transportation emissions globally, it is predicted that there will be significant positive effects on Earth's air quality, acid rain, smog and climate change.

The health impact of transport emissions is also of concern. A recent survey of the studies on the effect of traffic emissions on pregnancy outcomes has linked exposure to emissions to adverse effects on gestational duration and possibly also intrauterine growth.

Environmental impact of aviation

The environmental impact of aviation occurs because aircraft engines emit noise, particulates, and gases which contribute to climate change and global dimming. Despite emission reductions from automobiles and more fuel-efficient and less polluting turbofan and turboprop engines, the rapid growth of air travel in recent years contributes to an increase in total pollution attributable to aviation. In the EU, greenhouse gas emissions from aviation increased by 87% between 1990 and 2006. Among other factors leading to this phenomenon are the increasing number of hypermobile travellers and social factors that are making air travel commonplace, such as frequent flyer programs.

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Environmental impact of roads

The environmental impact of roads includes the local effects of highways (public roads) such as on noise, light pollution, water pollution, habitat destruction/disturbance and local air quality; and the wider effects including climate change from vehicle emissions. The design, construction and management of roads, parking and other related facilities as well as the design and regulation of vehicles can change the impacts to varying degrees.

Environmental impact of shipping

The environmental impact of shipping includes greenhouse gas emissions and oil pollution. In 2007, carbon dioxide emissions from shipping were estimated at 4 to 5% of the global total, and estimated by the International Maritime Organization (IMO) to rise by up to 72% by 2020 if no action is taken. There is also a potential for introducing invasive species into new areas through shipping, usually by attaching themselves to the ship's hull.

Environmental impact of bush burning

It reduces the organic content of the soil.

It increases the inorganic content of the soil.

It kills micro organisms in the soil.

Environmental impact of tillage

It aerates the soil

It leads to leaching.

It leads to loss of fertility.

Environmental impact of different farming methods

Shifting cultivation results to loss of vegetation

Monocropping leads to loss of soil fertility and rapid spread of pests and diseases in the farms.

Overcrowding

Lack of space leads to overcrowding. Causes of overcrowding are: barriers, immigration and increase in birthrate.

Environmental impact of overcrowding:

Epidemics, fighting, cannibalism, decrease in natality.

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General Impact of Environmental Degradation

Human activity is causing environmental degradation, which is the deterioration of the environment through depletion of resources such as air, water and soil; the destruction of ecosystems; habitat destruction; the extinction of wildlife; and pollution. It is defined as any change or disturbance to the environment perceived to be deleterious or undesirable.

Mass extinction, defaunation, decline in biodiversity, Holocene extinction, biodiversity loss and Ecological collapse

Biodiversity generally refers to the variety and variability of life on Earth, and is represented by the number of different species there are on the planet. Since its introduction, Homo sapiens (the human species) has been killing off entire species either directly (such as through hunting) or indirectly (such as by destroying habitats), causing the extinction of species at an alarming rate. Humans are the cause of the current mass extinction, called the Holocene extinction, driving extinctions to 100 to 1000 times the normal background rate. Though most experts agree that human beings have accelerated the rate of species extinction, some scholars have postulated without humans, the biodiversity of the Earth would grow at an exponential rate rather than decline. The Holocene extinction continues, with meat consumption, overfishing, ocean acidification and the amphibian crisis being a few broader examples of an almost universal, cosmopolitan decline in biodiversity. Human overpopulation (and continued population growth) along with profligate consumption are considered to be the primary drivers of this rapid decline. A 2017 statement by 15,364 scientists from 184 countries warned that, among other things, this sixth extinction event unleashed by humanity could annihilate many current life forms and consign them to extinction by the end of this century.

Defaunation is the loss of animals from ecological communities.

It is estimated that more than 50 percent of all wildlife has been lost in the last 40 years. It is estimated that by 2022, 70% of the world's wildlife will be lost. In South America, this is believed to be a 70 percent loss. A May 2018 study published in PNAS found that 83% of wild mammals, 80% of marine mammals, 50% of plants and 15% of fish have been lost since the dawn of human civilization. Currently, livestock make up 60% of the biomass of all mammals on earth, followed by humans (36%) and wild mammals (4%).

Death of coral reefs

Because of human overpopulation, coral reefs are dying around the world. In particular, coral mining, pollution (organic and non-organic), overfishing, blast fishing and the digging of canals and access into islands and bays are serious threats to these ecosystems. Coral reefs also face high dangers from pollution, diseases, destructive fishing practices and warming oceans. In order to find answers for these problems, researchers study the various factors that impact reefs. The list of factors is long, including the ocean's role as a carbon dioxide sink, atmospheric changes,

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ultraviolet light, ocean acidification, biological virus, impacts of dust storms carrying agents to far flung reefs, pollutants, algal blooms and others. Reefs are threatened well beyond coastal areas.

General estimates show approximately 10% world's coral reefs are already dead It is estimated that about 60% of the world's reefs are at risk due to destructive, human-related activities. The threat to the health of reefs is particularly strong in Southeast Asia, where 80% of reefs are endangered.

Global Warming

Global warming is the result of increasing atmospheric carbon dioxide concentrations which is caused primarily by the combustion of fossil energy sources such as petroleum, coal, and natural gas, and to an unknown extent by destruction of forests, increased methane, volcanic activity and cement production. Such massive alteration of the global carbon cycle has only been possible because of the availability and deployment of advanced technologies, ranging in application from fossil fuel exploration, extraction, distribution, refining, and combustion in power plants and automobile engines and advanced farming practices. Livestock contributes to climate change both through the production of greenhouse gases and through destruction of carbon sinks such as rain-forests. According to the 2006 United Nations/FAO report, 18% of all greenhouse gas emissions found in the atmosphere are due to livestock. The raising of livestock and the land needed to feed them has resulted in the destruction millions of acres of Rainforest and as global demand for meat rises, so too will the demand for land. Ninety-one percent of all rainforest land deforested since 1970 is now used for livestock Potential negative environmental impacts caused by increasing atmospheric carbon dioxide concentrations are rising global air temperatures, altered hydrogeological cycles resulting in more frequent and severe droughts, storms, and floods, as well as sea level rise and ecosystem disruption.

Ozone Layer Depletion

The ozone layer is basically found at a height of about 20-30km above sea level. The ozone layer provides a protective layer, which prevents the penetration of the sun's harmful ultra-violet rays. Dr Joe Farmer\s finding in 1985 has proved that the ozone layer depletion was even greater than predicted. The pollution of the atmosphere from the release of Chlorofluoro Carbon (CFC) gases causes the depletion of the ozone layer and this result in environmental degradation. Most of the skin diseases and low productivity in agriculture are caused by ultra-violet rays. Also an increase in ultra-violet rays affects water bodies, disturbs aquatic life which supports the food chain, and causes the death of fishes that feed us..It also causes the deterioration of synthetic materials such as paints and other products used in the building industry, invariably causing deterioration of housing delivery to our ever-increasing human population.

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Land degradation

Land degradation is a process in which the value of the biophysical environment is affected by a combination of human-induced processes acting upon the land. It is viewed as any change or disturbance to the land perceived to be deleterious or undesirable. Natural hazards are excluded as a cause; however human activities can indirectly affect phenomena such as floods and bush fires.

This is considered to be an important topic of the 21st century due to the implications land degradation has upon agronomic productivity, the environment, and its effects on food security. It is estimated that up to 40% of the world's agricultural land is seriously degraded. Others include:

Desertification

Ocean acidification

Ozone depletion

Water degradation

Disruption of the nitrogen cycle

Human impact on the nitrogen cycle

Of particular concern is N2O, which has an average atmospheric lifetime of 114–120 years, and is 300 times more effective than CO2 as a greenhouse gas. NOx produced by industrial processes, automobiles and agricultural fertilization and NH3 emitted from soils (i.e., as an additional byproduct of nitrification) and livestock operations are transported to downwind ecosystems, influencing N cycling and nutrient losses. Six major effects of NOx and NH3 emissions have been identified:

- decreased atmospheric visibility due to ammonium aerosols (fine particulate matter [PM])
- elevated ozone concentrations
- ozone and PM affects human health (e.g. respiratory diseases, cancer)
- increases in radiative forcing and global warming
- decreased agricultural productivity due to ozone deposition
- ecosystem acidification and eutrophication.

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POLLUTION

Pollution is the introduction of contaminants into the natural environment that causes adverse change. Pollution can take the form of chemical substances or energy, such as noise, heat or light. Pollutants, the components of pollution, can be either foreign substances/energies or naturally occurring contaminants. Pollution is often classed as point source or nonpoint source pollution. In 2015, pollution killed 9 million people in the world.

The major forms of pollution are listed below along with the particular contaminant relevant to each of them:

- Air pollution: the release of chemicals and particulates into the atmosphere. Common gaseous pollutants include carbon monoxide, sulfur dioxide, chlorofluorocarbons (CFCs) and nitrogen oxides produced by industry and motor vehicles. Photochemical ozone and smog are created as nitrogen oxides and hydrocarbons react to sunlight. Particulatmatter, or fine dust is characterized by their micrometre size PM_{10} to $PM_{2.5}$.
- Light pollution: includes light trespass, over-illumination and astronomical interference.
- Littering: the criminal throwing of inappropriate man-made objects, unremoved, onto public and private properties.
- Noise pollution: which encompasses roadway-noise, aircraft noise, industrial noise as well as high-intensity sonar.
- Plastic pollution: involves the accumulation of plastic products and microplastics in the environment that adversely affects wildlife, wildlife habitat, or humans.
- Soil contamination occurs when chemicals are released by spill or underground leakage. Among the most significant soil contaminants are hydrocarbons, heavy metals. MTBEherbicides, pesticides and chlorinated hydrocarbons.
- Radioactive contamination, resulting from 20th century activities in atomic physics, such as nuclear power generation and nuclear weapons research, manufacture and deployment. (See alpha emitters and actinides in the environment.)
- Thermal pollutio, is a temperature change in natural water bodies caused by human influence, such as use of water as coolant in a power plant.
- Visual pollution, which can refer to the presence of overhead power lines, motorway billboards, scarred landforms (as from strip mining), open storage of trash, municipal solid waste or space debris.
- Water pollution, by the discharge of wastewater from commercial and industrial waste (intentionally or through spills) into surface waters; discharges of untreated domestic sewage, and chemical contaminants, such as chlorine, from treated sewage; release of waste and contaminants into surface runoff flowing to surface waters (including urban runoff and agricultural runoff, which may contain chemical fertilizers and pesticides; also including human feces from open defecation - still a major problem in many developing countries); groundwater pollution from waste disposal and leaching into the ground, including from pit latrines and septic tanks; eutrophication and littering.

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Pollutants

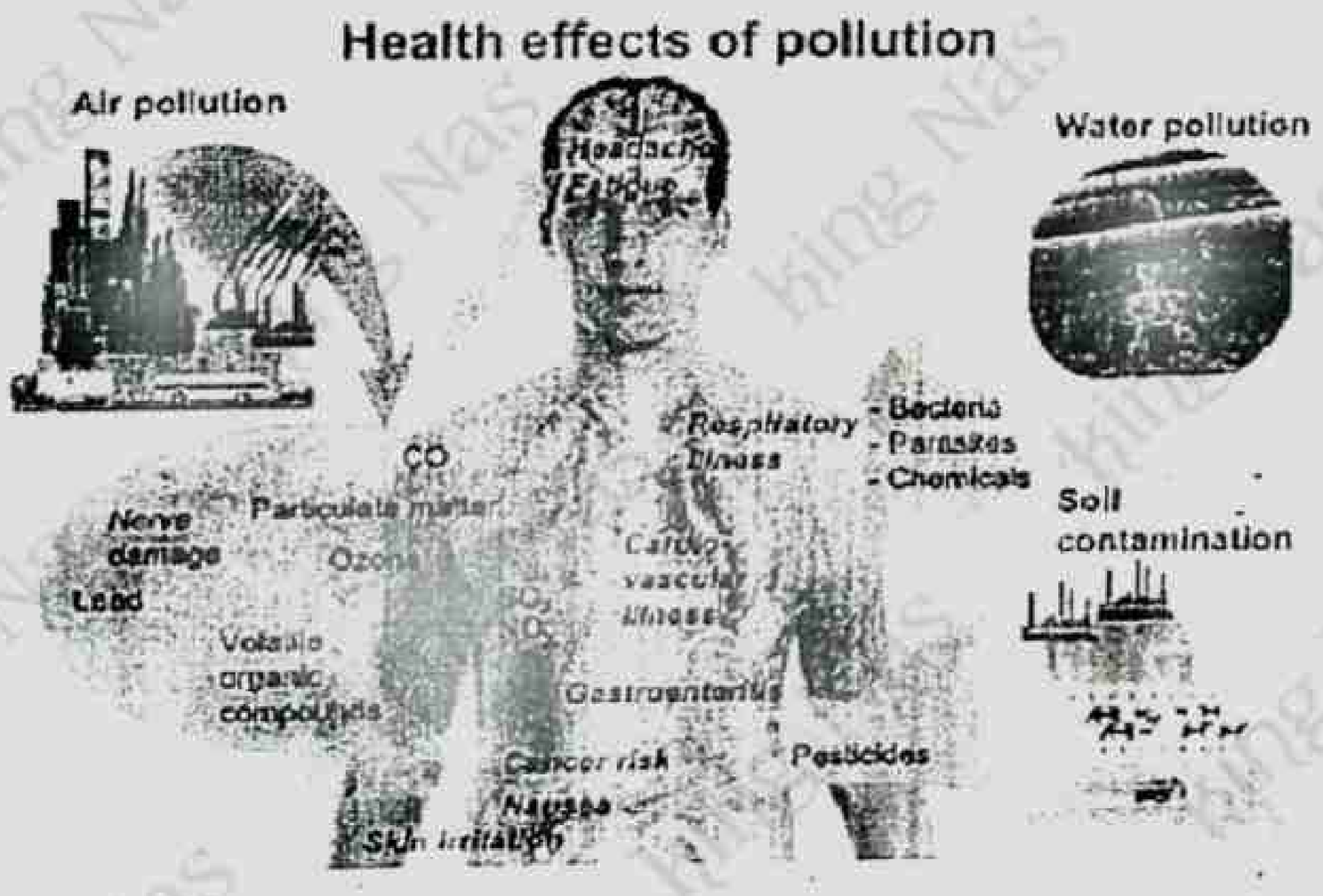
A pollutant is a waste material that pollutes air, water, or soil. Three factors determine the severity of a pollutant: its chemical nature, the concentration and the persistence.

Air pollution comes from both natural and manmade sources. Though globally man made pollutants from combustion, construction, mining, agriculture and warfare are increasingly significant in the air pollution equation.

Motor vehicle emissions are one of the leading causes of air pollution. China, United States, Russia, Mexico, and Japan are the world leaders in air pollution emissions. Principal stationary pollution sources include chemical plants, coal-fired power plants, oil refineries, petrochemical plants, nuclear waste disposal activity, incinerators, large livestock farms (dairy cows, pigs, poultry, etc.), PVC factories, metals production factories, plastics factories, and other heavy industry. Agricultural air pollution comes from contemporary practices which include clear felling and burning of natural vegetation as well as spraying of pesticides and herbicides

About 400 million metric tons of hazardous wastes are generated each year. The United States alone produces about 250 million metric tons. Americans constitute less than 5% of the world's population, but produce roughly 25% of the world's CO2, and generate approximately 30% of world's waste. In 2007, China has overtaken the United States as the world's biggest producer of CO2.

Human Health Effects:



Adverse air quality can kill many organisms including humans. Ozone pollution can cause respiratory disease, cardiovascular disease, throat inflammation, chest pain, and congestion. Water pollution causes approximately 14,000 deaths per day, mostly due to contamination of drinking water by untreated sewage in developing countries. An estimated 700 million Indians have no access to a proper toilet, and 1,000 Indian children die of diarrhoeal sickness every day. Nearly 500 million Chinese lack access to safe

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drinking water. 656,000 people die prematurely each year in China because of air pollution. In India, air pollution is believed to cause 527,700 fatalities a year. Studies have estimated that the number of people killed annually in the US could be over 50,000. Oil spills can cause skin irritations and rashes. Noise pollution induces hearing loss, high blood pressure, stress, and sleep disturbance. Mercury has been linked to developmental deficits in children and neurologic symptoms. Older people are majorly exposed to diseases induced by air pollution. Those with heart or lung disorders are under additional risk. Children and infants are also at serious risk. Lead and other heavy metals have been shown to cause neurological problems. Chemical and radioactive substances can cause cancer and as well as birth defects.

Effects in the Environment

Pollution has been found to be present widely in the environment. There are a number of effects of this:

- Biomagnification describes situations where toxins (such as heavy metals) may pass through trophic levels, becoming exponentially more concentrated in the process.
- Carbon dioxide emissions cause ocean acidification, the ongoing decrease in the pH of the Earth's oceans as CO_2 becomes dissolved.
- wrtThe emission of greenhouse gases leads to global warming which affects ecosystems in many ways.
- Invasive species can out-compete native species and reduce biodiversity. Invasive plants can contribute debris and biomolecules (allelopathy) that can alter soil and chemical compositions of an environment, often reducing native species competitiveness.
- Nitrogen oxides are removed from the air by rain and fertilise land which can change the species composition of ecosystems.
- Smog and haze can reduce the amount of sunlight received by plants to carry out photosynthesis and leads to the production of tropospheric ozone which damages plants.
- Soil can become infertile and unsuitable for plants. This will affect other organisms in the food web.
- Sulfur dioxide and nitrogen oxides can cause acid rain which lowers the pH value of soil.
- Organic pollution of watercourses can deplete oxygen levels and reduce species diversity.

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Regulation and monitoring of pollution

To protect the environment from the adverse effects of pollution, many nations worldwide have enacted legislation to regulate various types of pollution as well as to mitigate the adverse effects of pollution.

Pollution control

Pollution control is a term used in environmental management. It means the control of emissions and effluents into air, water or soil. Without pollution control, the waste products from overconsumption, heating, agriculture, mining, manufacturing, transportation and other human activities, whether they accumulate or disperse, will degrade the environment. In the hierarchy of controls, pollution prevention and waste minimization are more desirable than pollution control. In the field of land development, low impact development is a similar technique for the prevention of urban runoff.

Practices

- Recycling
- Reusing
- Waste minimisation
- Mitigating
- Preventing
- Compost

Pollution control devices for Air pollution control

Thermal oxidizer

Dust collection systems

Baghouses

Cyclones

Electrostatic precipitators

Scrubbers

Baffle spray scrubber

Cyclonic spray scrubber

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    - Ejector venturi scrubber
    - Mechanically aided scrubber
    - Spray tower
    - Wet scrubber
- Sewage treatment
    - Sedimentation (Primary treatment)
    - Activated sludge biotreaters (Secondary treatment; also used for industrial wastewater)
    - Aerated lagoons
    - Constructed wetlands (also used for urban runoff)
- Industrial wastewater treatment
    - API oil-water separators
    - Biofilters
    - Dissolved air flotation (DAF)
    - Powdered activated carbon treatment
    - Ultrafiltration
- Vapor recovery systems
- Phytoremediation

Health Effects of Toxic Chemicals

There are many materials used in the workplace that can be hazardous. However, in order for them to affect your health, they must come into contact with the body or be absorbed into the body. When assessing the potential health effects from working with a particular material, it is necessary to understand the difference between "toxicity" and "hazard".

1. **Toxicity** is the ability of a substance to produce an unwanted effect when the chemical has reached a sufficient concentration at a certain site in the body.

The more toxic a material is, the smaller the amount of it is necessary to be absorbed before harmful effects are caused. The lower the toxicity, the greater the quantity of it is necessary to be absorbed. Generally, the toxicity of a chemical is determined through experiments on animals (quite often rats), and measured in terms of the amount of material necessary to cause death in 50% of the test animals. These values are called LD50 (lethal dose) or LC50 (lethal concentration), and are usually given in weight of material per kg of body weight or airborne concentration of material per set time period respectively.

1. **Hazard** is the probability that this concentration in the body will occur.

Toxicity is an inherent property of the material. A material may be very toxic, but not hazardous, if it is handled properly and is not absorbed into the body. On the other hand, a material may have a very low toxicity, but be very hazardous.

Example:

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- An open container of an acid is much more hazardous than a closed container of the same material.
- Two liquids may possess the same degree of toxicity but present different degrees of hazard:
- One material may be non-irritating to the eyes and nose and odourless. The other may be irritating to the eyes or respiratory system and possess a pungent odour.
- The latter material, because of its warning properties presents a lesser degree of hazard.

In order for toxicants to affect the human system either they must cause damage to external tissues, such as the skin or eyes, or they must be able to enter the body by some mechanism.

Routes of Entry

There are three primary routes of entry into the body: ingestion, skin or eye absorption, inhalation and injection.

Ingestion- This means taking a material into the body by mouth (swallowing). Ingestion of toxic materials may occur as a result of eating in a contaminated work area.

Absorption- Substances that contact the eye and the skin may be either absorbed into the body or cause local effects. For the majority of organic compounds, the contribution from skin absorption to the total exposure should not be neglected.

Inhalation- This means taking a material into the body by breathing it in. In the lungs, very tiny blood vessels are in constant contact with the air we breathe in. As a result, airborne contaminants can be easily absorbed through this tissue. In the occupational environment, this is generally the most important route of entry.

Injection- This means taking in biological or chemical substances into the body when the skin is accidentally punctured with a sharp object (e.g. contaminated needle).

Health Effects – Chronic vs Acute

Once a toxic substance has contacted the body it may have either acute (immediate) or chronic (long term) effects.

Example: Spilling acid on your hand will cause **immediate harm,** i.e. a burn to the skin.

Exposure to asbestos or tobacco smoke may result in lung cancer after as much as twenty years (this is a long term effect).

Exposure – Chronic vs Acute

Exposure can be classified as chronic or acute. In chronic exposures, the dose is delivered at some frequency (daily or weekly usually) over a period of time. In acute exposures, the dose is

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delivered in a single event and absorption is rapid. Usually, a chronic exposure occurs at low concentration and acute exposure at high concentration.

Some materials may only cause harm if given acutely, not having any effect in the long term. Other materials may not exhibit an effect in the short term, but may cause problems after prolonged exposure.

Physiological Classification of Materials

This classification identifies toxic materials on the basis of biologic action.

Irritants – refers to some sort of aggravation of whatever tissue the material comes in contact with. e.g. ammonia, nitrogen dioxide.

Asphyxiants – exert their effects through a depletion of oxygen to the tissues e.g. simple asphyxiants – carbon dioxide, nitrogen, methane, hydrogen chemical asphyxiants – carbon monoxide, hydrogen cyanide, hydrogen sulphide.

Narcotics or Anaesthetics – the main toxic action is the depressant effect upon the Central Nervous System. e.g. – many organics, chloroform, xylene.

Systemic Poisons – the main toxic action includes the production of internal damage e.g. Hepatotoxic agents – toxic effects produce liver damage. eg. carbon tetrachloride.

e.g. Nephrotoxic agents – toxic effects produce kidney damage eg. some halogenated hydrocarbons

Carcinogens – agents/compounds that will induce cancer in humans. e.g. benzene, arsenic, inorganic salts of chromium, nickel, beryllium.

Mutagens – agents that affect the cells of the exposed people in such a way that it may cause cancer in the exposed individual or an undesirable mutation to occur in some later generation. e.g. radiation, variety of chemical agents that alter the genetic message.

Teratogens – Agents or compounds that a pregnant woman takes into her body that generate dy defects in the fetus e.g. Thalidomide, possibly steroids

Sensitizers – Agents that may cause allergic or allergic-like responses to occur. After an initial exposure to a substance an individual may become sensitized to that substance. Subsequent exposures to the same substance, often at a much lower concentration than before, produce an allergic response. This response may be a skin rash (dermatitis) or an asthmatic-like attack, depending on the route of exposure. e.g. cutting oils, isocyanates in polyurethane foam operations and paint spraying operations, some laboratory solvents.

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Toxic chemicals present in the environment: sources and toxic action

- There are a large number of chemicals in the environment. Some of these are toxic and the rest are non-toxic. Many metals which are listed as environmental hazards are essential dietry trace elements required for normal growth and development of living things. These elements include Al, Sb, As, Be, Bi, Cd, Co, Cu, In,Pb, Mo, Ag, Te, Sn, Ti, W, U, and Zc. In such cases, it is the amount or the concentration of the respective element which makes it toxic. Even the well known toxic element like As, Pb, and Cd are required in trace quantities for the growth of animals. A typical example is that of Al, which biologically inert, but it causes brain damage, bone disease and anaemia in patients subjected to haemodialysis using water containing 100-1000 part per billion of aluminum. In fact 100-1000 ppb is the normal concentration of aluminium in drinking water where aum is used for water treatment.

The fact is that everyone is exposed to toxic chemicals at home, at work, at school and in the outdoor environment. The workplace is an important source of exposure. Thus, roofers, construction workers and automobile engineers are exposed to particulate matter besides other pollutants. Farm workers are exposed to pesticides. Anyone who cleans anything may be exposed to chlorinated solvents. Office going people come in contact with photocopier, tonners, correction fluids and rubber, cement. Hospital workers surrounded by disinfectants. Everyone is exposed at home to cleaners' detergents, paints and others. Animals, plants and microorganisms encounter many of these substances as environmental pollutants. It is in fact because of the presence of toxic chemicals in the environment that environmental toxicology is important.

Toxic Chemicals in the Air

The usual major pollutants (like oxides of carbon, nitrogen and sulphur) particulate pollutants and metallic pollutants pose serious health hazards, it is necessary to control these pollutants. Besides the above, the US Environmental Protection Agency along with consumer products Safety Commission has identified extremely hazardous substances in the atmosphere. These are acrylonitrate, arsenic asbestors, berellium, Cadmium, chlorinated solvents, chlorofluorocarbon, chromates, coke-oven emission, diethylstilbestrol, dibromochloropropene, ethylbromide, ethylene oxide, lead, mercury, nitrosoamines, ozone, polybrominated biphenyls, polychlorinated biphenyls, radiation, sulphur dioxide, vinyl chloride, and toxic waste disposal emissions and leachates.

Toxic Elements in Water

A list of toxic elements found in natural and waste water is given in Table 1 below. Some of these are essential at low levels (necessary as nutrients for animal and plant life) but are toxic at higher levels. Besides the toxic elements the water bodies contain a large number of pesticides, which are primarily agricultural runoffs.

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Element	Sources	Effects and Significance
Arsenic	Mining by-products, pesticides, chemical waste	Toxic, possibly carcinogenic
Cadmium	Industrial discharge, mining waste, metal plating, water pipes	Replaces zinc biochemically, causes high blood pressure, kidney damage, destruction of testicular tissue and red blood cells.
Beryllium	Coal, nuclear power and space, industries.	Acute and chronic toxicity, possibly carcinogenic
Boron	Coal, detergent formulations industrial wastes.	Toxic to some plants
Chromium	Metal plaiting, cooling tower water additives, (chromate) normally found as Cr(vi) in polluted water.	Essential trace elements possibly carcinogen as Cr(vi).
Copper	Metal plating, industrial and domestic wastes, mining, mineral leaching	Essentially trace elements not very toxic to animals, toxic to plants and algae at moderate level.
Fluorine (fluoride ion)	Natural geological sources, industrial wastes water additives	Prevents both decay at about 1mg/l, causes mottled teeth and bone damage at about 5mg/l.
Lead	Industry, mining, plumbing, coal, gasoline	Toxic, (anaemia, kidney disease, nervous disorder), wildlife destroyed.
Manganese	Mining, industrial waste, acid mine drainage, microbial action on manganese at low pH	Relatively non-toxic to animals, toxic to plants at higher levels, stains materials (bathroom fixtures and clothing).
Mercury	Industrial waste, mining, natural sources.	Toxic to animals.
Selenium	Natural geological sources, sulphur, coal.	Essential at low level but toxic at higher level.
Zinc	Industrial waste, metal plating, plumbing.	Essential in metallo-enzymes, toxic to plants at higher levels.

Toxic Waste in Soils

Soil has been used as a dumping ground for most of the waste products including toxic wastes. The main sources of these wastes are industrial wastes, urban and domestic wastes, radioactive wastes, agrochemicals like fertilizers and pesticides, plastics and polymers, chemical and metallic pollutants, biological agents.

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CONSERVATION OF NATURAL RESOURCES

Conservation is the planned use or exploitation of natural resources to ensure their continuous use and preserve the original nature of the environment. Natural resources are divided into two they are renewable and non-renewable natural resources.

Renewable natural resources are the resources that can be replaced. They are water, air, food, forest, soil; animals etc. non-renewable are the ones that cannot be replaced in future. The quantity of non-renewable resources is fixed on earth. Examples of non-renewable natural resources are crude oil, cola, bauxite limestone, tin, oil, gas etc.

METHODS OF CONSERVATION

Methods of conserving water

Irrigation processes, construction of dams and reservoirs. control of the use of underground water effective control of water pollution, treatment of and recycling of used water and proper channeling of rain water to wells, tanks and collecting centers.

Benefits of water conservation

Source of food, source of water for irrigation, domestic use, hydro-electricity, transportation.

Water preserves natural resources

Methods of conserving forest

Establishment of a forest reserve, legislation, restricting the felling down of economic trees, proper recycling of paper and other forest products, proper treatment of the soil, reforestation, and efficient forest management measures.

Benefits of forest conservation

Supplies timber and other forest products

Checks erosion and desert encroachment

Preserves natural species

Encourages rainfall and helps to maintain natural succession

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Methods of conserving wildlife

By restricting hunting processes, legislation to maintain population of animals, artificial stocking and game farming

Benefits of conserving wildlife

Source of food/protein and raw material for industries example hides and skins, ivory

Boosts tourism education or recreation

Can be used for ecological or scientific studies.

Methods of conserving fisheries

Control of water pollution, artificial stocking, fishing regulations, construction of fish ponds, and seed multiplication, control of aquatic weeds.

Benefits of conserving fisheries

- Source of food/protein

Boosts tourism education or recreation

Can be used for ecological or scientific studies.

Methods of conserving soil

Through the control of soil erosion, improvement of soil fertility through the addition of manures, planting of cover crops, crop rotation, nitrogen cycle etc. By afforestation to prevent wind erosion and desertification; through the control of soil erosion.

Benefits of conserving soil

Source of plant nutrients for growth

Helps to preserve soil fertility.

It is a reservoir of water for soil organisms

A source of raw materials for some industries e.g use of clay in ceramics industries

Methods of conserving mineral resources

By substituting fossil fuel with solar energy

Recycling used mineral products

Preventing accidental waste of minerals example oil spillage

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Efficient use of metal scraps example iron and tin.

Reduction in the use of mineral resources generally.

Benefits of conserving mineral resources

Mineral resources are source of energy, raw materials for industries and source for common salt for food, and source of foreign exchange.

Benefits of petroleum and natural gas

Ensures the continuous supply of raw materials for petrochemical industries

Continuous use of fossil fuel

Ensures employment opportunities.

Problems encountered in conservation

Economic problems: Lack of money to carry out the conservation processes.

Limited land area for development farming: This has led to the destruction of natural forests.

Lmited source of energy: this also has led to destruction of forests for firewood

Maintenance of river for hydro-electricity and irrigation has resulted in the alteration of the ecosystem.

Overgrazing, overfishing or over cropping activities.

The nuclear power which is an alternative source of energy is dangerous and causes thermal pollution.

Bush burning bush fire.

Lack of public awareness and good management.

AGENCIES RESPONSIBLE FOR CONSERVATION

River Basin Authorities

PHCN ((Power Holding Company of Nigeria).

Ministry of Agriculture: 1. Department of Agriculture 2. Department of wildlife conservation 3. Department of Fisheries

Ministry of Energy Resources.

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National Environmental Sanitation Authority

Nigerian Conservation Foundation

Urbanization: Causes and Effects

More than half of the world's population lives in urban areas. Due to the ongoing urbanization and growth of the world's population, there will be about 2.5 billion more people added to the urban population by 2050, mainly in Africa and Asia. The world's urban areas are highly varied, but many cities and towns are facing problems such as a lack of jobs, homelessness and expanding squatter settlements, inadequate services and infrastructure, poor health and educational services and high levels of pollution.

Urbanization is an increase in the number of people living in towns and cities. Urbanization occurs mainly because people move from rural areas to urban areas and it results in growth in the size of the urban population and the extent of urban areas. These changes in population lead to other changes in land use, economic activity and culture. Historically, urbanization has been associated with significant economic and social transformations. For example, urban living is linked with higher levels of literacy and education, better health, lower fertility and a longer life expectancy, greater access to social services and enhanced opportunities for cultural and political participation. However, urbanization also has disadvantages caused by rapid and unplanned urban growth resulting in poor infrastructures such as inadequate housing, water and sanitation, transport and health care services.

The rapid increase in urban populations has meant that peri-urban areas are growing much more quickly than formal urban centers. **Peri-urban** areas are those areas immediately around a town or city. They are areas in transition from countryside to city (rural to urban), often with undeveloped infrastructure, where health and sanitation services are under pressure and where the natural environment is at risk of degradation.

Causes of urbanization

Urbanization in the developing world occurs for two main reasons: the natural increase of population and rural to urban migration.

Natural increase of population

As birth rates decline over time, according to the demographic transition model, the role of natural increase in determining the pace of urban population growth becomes less important in comparison to migration.

Rural to urban migration

In developing countries, urbanization usually occurs when people move from villages to settle in cities in hope of gaining a better standard of living. The movement of people from one place to another is called **migration**. Migration is influenced by economic growth and development and

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by technological change and possibly also by conflict and social disruption. It is driven by **pull factors** that attract people to urban areas and **push factors** that drive people away from the countryside.

Employment opportunities in cities are one of the main pull factors. Many industries are located in cities and offer opportunities of high urban wages. There are also more educational institutions providing courses and training in a wide range of subjects and skills. People are attracted to an urban lifestyle and the 'bright lights' of city life. All of these factors result in both temporary and permanent migration to urban areas.

Poor living conditions and the lack of opportunities for paid employment in rural areas are push factors. People are moving away from rural areas because of poor health care and limited educational and economic opportunities as well as environmental changes, droughts, floods, lack of availability of sufficiently productive land, and other pressures on rural livelihoods.

Rural to urban migration can be a selective process, as some types of people are more likely to move than others. One of the factors involved is gender, because employment opportunities vary greatly with different jobs for men and women. Another factor is age. Young people are more likely to move to towns, with more elderly people and children left in rural areas. Selectivity in migration affects the population in both the rural and the urban areas. If more men move to towns and cities than women, this leaves a predominantly female society in rural areas.

Impacts of urbanization

Although people are pulled towards the advantages of cities, the impacts of urbanization are mixed. First we will look at the many positive impacts of urbanization before going on to describe some of the challenges created by rapid unplanned urban growth.

Thriving towns and cities are an essential element of a prosperous national economy. The gathering of economic and human resources in one place stimulates innovation and development in business, science, technology and industry. Access to education, health, social services and cultural activities is more readily available to people in cities than in villages. In cities, child survival rates are better than in rural areas because of better access to health care. The density of urban populations makes it easier and less costly for the government and utilities to provide essential goods and services. For example, the supply of basic facilities such as fresh water and electricity can be achieved with less effort and less cost per person.

Schools, colleges and universities are established in cities to develop human resources. A variety of educational courses are available, offering students a wide choice for their future careers. People of many classes and religions live and work together in cities, which creates better understanding and harmony and helps break down social and cultural barriers. Cities also have advanced communication and transport networks.

However, these many benefits of urban life do not apply to all. Rapid population increases and unplanned growth create an urban sprawl with negative economic, social, and environmental consequences. In Nigeria, the rate of urban growth often strains the capacity of local and national

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government to provide urban residents with even the most basic services of housing, water supply, sewerage and solid waste disposal.

Housing

In developing countries, about a third of urban inhabitants live in impoverished slums and squatter settlements. **Slums** are urban areas that are heavily populated and have sub-standard housing with very poor living conditions, creating several problems.

In Nigeria, about 60% of the houses in the city were classed as slums due to the physical deterioration of its housing, overcrowding, high density, poor access and lack of infrastructure services.

Slum areas typically suffer from:

- poor housing with small, overcrowded houses built very close together using inadequate materials and with uncertain electricity supply
- restricted access to water supplies
- little or no sanitation/latrine facilities and no solid waste disposal, which leads to a polluted and degraded local environment
- inadequate health care facilities which, coupled with the poor living conditions, increases sickness and death rates
- insecure living conditions – slum dwellers may be forcibly removed by landowners or other authorities.

Water supply and sanitation

The provision of water and sanitation services to growing urban settlements, peri-urban and slum areas presents critical challenges. The increased demand for water from the growing population can place added stress on already stretched resources. In and around cities, water is commonly in short supply and subject to increasing competition by different users. Urban growth leads to increasing demand for water for industrial and domestic use, which conflicts with agricultural demands.

It is especially difficult to provide water and sanitation services to deprived areas and the poorest people. Many people in these areas live without access to safe drinking water and proper sanitation. Even where adequate water supplies are available, sanitation and wastewater disposal are often inadequate or missing. Pit latrines and septic tanks are the usual methods for human waste disposal but they have limited capacity and are not always adequate to cope with the quantity of waste produced by many people living close together. Overflowing latrines and septic tanks contaminate surface water and create a serious health risk.

The lack of these essential services threatens not only the health and the environment of people in slum areas, but also that of people living in formal urban areas. This is true of many cities with a million or more inhabitants, as well as smaller cities and towns.

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Wastes and pollution

Urbanisation affects land, water, air and wildlife because of the number of people, the amount of buildings and construction, and the increased demands on resources. It has impacts on the physical environment in several ways.

Water quality

In developing countries, including Nigeria, many rivers in urban areas are more like open sewers. The lack of sanitation and sewerage systems has a dramatic impact on urban watercourses. People use the rivers to dispose of all their wastes from homes, industries and commercial businesses. Wastewater from human settlements contains organic material and nutrients; industrial wastewater contains many different types of toxic pollutant. These make the water unsafe for humans to use for many purposes including drinking and irrigation, as well as harming the fish and other animals and plants living in the water. Any change to the quality of surface water also affects groundwater because they are linked by the processes of the water cycle so pollutants from the surface will infiltrate down and contaminate soil and groundwater as well.

Solid waste

In many towns and cities solid waste management is inefficient or non-existent. Solid waste management means the proper collection, transfer, recycling and disposal of all the solid material we throw away, including plastics, paper and cardboard, food wastes, electrical waste, etc. It also includes industrial, hospital and institutional wastes which often contain pathogens as well as hazardous and toxic chemicals, which need special care.

Urban waste often ends up in illegal dumps on streets, open spaces, wastelands, drains or rivers. This is frequently a problem in peri-urban areas, which are convenient for dumping wastes because of the availability of open space and ease of access from central urban areas. This can lead to the pollution of groundwater and surface waters which may be used as a source for drinking water. Sometimes the wastes are collected and taken to legalised waste disposal sites but these are not always properly managed to protect water bodies and groundwater.

The combustion of solid waste creates yet another environmental problem. People want to get rid of the wastes and they will burn them in their backyards if there is no collection system.

Air quality

Air quality in towns and cities is frequently very poor as a result of air pollution from many different sources. These include:

- vehicle exhausts
- smoke from domestic fires

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- outputs from factory chimneys
- diesel-powered generators
- dust from construction works and city streets.

Poor air quality has a significant impact on the health of many urban residents as well as leaving a damaging and unsightly layer of dust on plants, buildings and other surfaces.

Health

Urbanisation can have both positive and negative effects on health. The main benefits are associated with easier access to hospitals, clinics and health services in general. If you live close to these services you can reach a doctor in minutes rather than hours or days, so this improves emergency care and general health. There are also benefits from easier access to sources of information such as radio and television which may be used to communicate information about health to the general public. For example, women living in towns and cities are more likely to be informed about family planning, which results in reduction in family size and less frequent childbirth, with consequent benefits to general health.

However, urban life can also damage health. Poor environment, housing and living conditions are the main reasons for poor health in urban areas. Contamination of water sources can cause epidemics of waterborne disease. Close proximity to other people can make the spread of many types of infectious disease more likely. The polluted air can also cause respiratory disease and contribute to premature deaths among more vulnerable sections of the population such as older people and children.

Food

Population movements also put pressure on food supplies and on food distribution. As people migrate to the cities, they tend to use purchased food instead of their own crops and this makes them more vulnerable to changes in food prices. As the population grows and the demand for water and land increases, it becomes difficult to increase food production in a sustainable way. The increase in urban demand, combined with a loss of agricultural land, means more pressure on rural people to produce food for the growing number of urban people.

Furthermore, pollution from urban areas can disrupt food supply. For example, fisheries are often damaged by urban domestic wastes and liquid effluents from city-based industries. Effluent is another word for wastewater that flows out from a source.

Economic and social systems

The process of urbanization has positive as well as negative economic and social changes. The positive effects include economic development, and education. However, urbanisation places stresses on existing social services and infrastructure. Crime, prostitution, drug abuse and street children are all negative effects of urbanisation. Also there tends to be a lack of social support for children in school and home by their hard-working, usually poor, parents. Inadequate income, overcrowded housing and poor living conditions create a fertile ground for the development of violence. Violent crime is more visible in the cities than in rural areas and it affects people's

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everyday life, their movements and the use of public transport. Crime in the city can create a sense of insecurity in its inhabitants. This unsafe feeling in city streets separates residential areas into higher-income and lower-income groups, which reduces the sense of community and forms areas with dissimilar incomes, costs and security levels.

Health Effects of Environmental Degradation

The urban environment is an important factor in determining the quality of life in urban areas. Some urban environmental problems include inadequate water and sanitation, lack of rubbish disposal, and industrial pollution. Unfortunately, reducing the problems and ameliorating their effects on the urban population are expensive.

The health implications of these environmental problems include respiratory infections and other infectious and parasitic diseases.

Capital costs for building improved environmental infrastructure — for example, investments in a cleaner public transportation system such as a subway — and for building more hospitals and clinics are higher in cities, where wages exceed those paid in rural areas.

And urban land prices are much higher because of the competition for space. But not all urban areas have the same kinds of environmental conditions or health problems. Some research suggests that indicators of health problems, such as rates of infant mortality, are higher in cities that are growing rapidly than in those where growth is slower.

Environmental Effects of Urbanization

Urban populations interact with their environment. Urban people change their environment through their consumption of food, energy, water, and land. And in turn, the polluted urban environment affects the health and quality of life of the urban population.

People who live in urban areas have very different consumption patterns than residents in rural areas. For example, urban populations consume much more food, energy, and durable goods than rural populations. In China during the 1970s, the urban populations consumed more than twice as much pork as the rural populations who were raising the pigs.[11] Witheconomic development, the difference in consumption declined as the rural populations ate better diets. But even a decade later, urban populations had 60 percent more pork in their diets than rural populations. The increasing consumption of meat is a sign of growing affluence in Beijing; in India where many urban residents are vegetarians, greater prosperity is seen in higher consumption of milk.

Urban populations not only consume more food, but they also consume more durable goods. In the early 1990s, Chinese households in urban areas were two times more likely to have a TV, eight times more likely to have a washing machine, and 25 times more likely to have a refrigerator than rural households. This increased consumption is a function of urban labor markets, wages, and household structure.

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areas than in rural villages. For example, urban populations have many more cars than rural Energy consumption for electricity, transportation, cooking, and heating is much higher in urban populations per capita. Almost all of the cars in the world in the 1930s were in the United States. Today we have a car for every two people in the United States. If that became the norm, in 2050 there would be 5.3 billion cars in the world, all using energy.

In China the per capita consumption of coal in towns and cities is over three times the consumption in rural areas.[14] Comparisons of changes in world energy consumption per capita and GNP show that the two are positively correlated but may not change at the same rate.[15] As countries move from using noncommercial forms of energy to commercial forms, the relative price of energy increases. Economies, therefore, often become more efficient as they develop because of advances in technology and changes in consumption behavior. The urbanization of the world's populations, however, will increase aggregate energy use, despite efficiencies and new technologies. And the increased consumption of energy is likely to have deleterious environmental effects.

Urban consumption of energy helps create heat islands that can change local weather patterns and weather downwind from the heat islands. The heat island phenomenon is created because cities radiate heat back into the atmosphere at a rate 15 percent to 30 percent less than rural area. The combination of the increased energy consumption and difference in albedo (radiation) means that cities are warmer than rural areas (0.6 to 1.3 C). And these heat islands become traps for atmospheric pollutants. Cloudiness and fog occur with greater frequency. Precipitation is 5 percent to 10 percent higher in cities; thunderstorms and hailstorms are much more frequent, but snow days in cities are less common.

Urbanization also affects the broader regional environments. Regions downwind from large industrial complexes also see increases in the amount of precipitation, air pollution, and the number of days with thunderstorms. Urban areas affect not only the weather patterns, but also the runoff patterns for water. Urban areas generally generate more rain, but they reduce the infiltration of water and lower the water tables. This means that runoff occurs more rapidly with greater peak flows. Flood volumes increase, as do floods and water pollution downstream.

Many of the effects of urban areas on the environment are not necessarily linear. Bigger urban areas do not always create more environmental problems. And small urban areas can cause large problems. Much of what determines the extent of the environmental impacts is how the urban populations behave — their consumption and living patterns — not just how large they are.

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BIO 102 GENERAL BIOLOGY 2

BASIC CHARACTERISTICS, IDENTIFICATION AND CLASSIFICATION OF BACTERIA WITH EXAMPLES

Introduction

Bacteria are single-celled microorganisms that exist in diverse shapes, sizes, and environments. We have studied their basic characteristics in the previous module (BIO 101). These include their cell wall structure, reproduction, metabolism, genetic diversity, and ecology. In this module, we shall look into how to identify and classify bacteria with examples.

IDENTIFICATION OF BACTERIA

Sample Collection and Preparation

- Collect samples from various sources (e.g., soil, water, clinical specimens) using sterile techniques.
- Sample is prepared by diluting, if necessary, and plating it on appropriate growth medium

Common sample types include:

1. **Clinical Samples**: These are collected from patients and can include:
    - Blood
    - Urine
    - Sputum
    - Cerebrospinal fluid
    - Tissue biopsies
    - Swabs from infected sites (e.g., throat, wound, nasal)
2. **Environmental Samples**: These are collected from natural or artificial environments and can include:
    - Soil
    - Water (rivers, lakes, oceans, drinking water)
    - Air
    - Surfaces in healthcare settings (e.g., hospital rooms, equipment)
3. **Food Samples**: These are used in the food industry to ensure safety and quality, and can include:
    - Raw meats
    - Dairy products
    - Fruits and vegetables
    - Processed foods
4. **Industrial Samples**: These are collected from industrial environments to monitor contamination and can include:
    - Biofilms in pipes and reactors
    - Fermentation products
    - Industrial waste

5. **Research Samples**: These can come from various sources depending on the focus of the research and can include:
    - Model organisms (e.g., mice, plants)
    - Synthetic constructs (e.g., genetically modified bacteria)
    - Environmental DNA (eDNA) samples for metagenomic studies

Methods of examination

1. **Culture-Based Methods**

- **Selective and Differential Media**: Culturing bacteria on specialized media allows for the selective growth of certain bacterial species based on specific nutritional requirements or metabolic characteristics.

2. **Macroscopic Examination**

- **Colony Morphology**: Observing the appearance of bacterial colonies on agar plates can provide initial clues about the identity of bacteria. Characteristics such as size, shape, color, texture, and margin are noted.

3. **Microscopic Examination**

- **Gram Staining**: One of the most widely used staining techniques to differentiate bacteria based on their cell wall structure. Gram-positive bacteria retain the crystal violet stain, appearing purple under the microscope, while gram-negative bacteria appear pink after counterstaining with safranin.
- **Cell Morphology**: Microscopic examination allows the observation of bacterial cell shape, arrangement, and presence of specific structures such as flagella or capsules.

4. **Biochemical Tests**

Various biochemical tests are performed to assess the metabolic capabilities of bacteria. Common tests include:

- **Catalase Test**: Differentiates bacteria based on their ability to produce the enzyme catalase. Positive: *Staphylococcus*, Negative: *Streptococcus*
- **Oxidase Test**: Detects the presence of cytochrome c oxidase in bacteria. Positive: *Pseudomonas aeruginosa*, Negative: *Escherichia coli*
- **Indole Production Test**: Determines the ability of bacteria to produce indole from tryptophan. Positive: *Escherichia coli*, Negative: *Enterobacter cloacae*
- **Citrate Utilization Test**: Determines the ability to use citrate as the sole carbon source. Positive: *Klebsiella pneumoniae*, Negative: *Escherichia coli*
- **Coagulase Test:** To differentiate *Staphylococcus aureus* (coagulase-positive) from other *Staphylococcus* species (coagulase-negative).
- **Methyl Red (MR) Test**: Detects the production of stable acids from glucose fermentation. Positive: *Escherichia coli*, Negative: *Enterobacter aerogenes*
- **Sugar Fermentation Tests**: Assess the ability of bacteria to ferment specific sugars, producing characteristic metabolic byproducts.

5. **Molecular Techniques**

- **Polymerase Chain Reaction (PCR)**: Amplifies specific regions of bacterial DNA for identification purposes. Techniques such as:
    - **16S rRNA Sequencing**: Targets the highly conserved 16S ribosomal RNA gene for phylogenetic analysis and bacterial identification.
    - **Whole Genome Sequencing (WGS)**: Provides comprehensive genetic information for accurate identification and characterization of bacteria.

6. **Serological Methods**

- **Immunological Assays**: Serological techniques involve the use of antibodies to detect specific antigens or antibodies produced by bacteria. Examples include:
    - **Enzyme-Linked Immunosorbent Assay (ELISA)**: Detects bacterial antigens or antibodies in patient samples.
    - **Western Blotting**: Identifies specific proteins within bacterial extracts using antibodies.

7. **Anaerobic Culture Techniques**: Some bacteria require anaerobic conditions for growth, necessitating specialized culture techniques such as anaerobic chambers or specific

Types of Bacterial Isolation Media

1. **General Purpose Media**

    Nutrient Agar (NA)

    - Composition: Peptone, beef extract, sodium chloride, agar.
    - Use: Supports the growth of a wide range of non-fastidious bacteria.
    - Example: Used for routine culturing and sub-culturing of bacteria.

2. **Enriched Media**
    - **Blood Agar (BA)**
        - Composition: Nutrient agar base supplemented with 5-10% sheep or horse blood.
        - Use: Supports the growth of fastidious bacteria and differentiates bacteria based on hemolytic properties.
        - Example: *Streptococcus pyogenes* (beta-hemolysis), *Streptococcus pneumoniae* (alpha-hemolysis).
    - **Chocolate Agar (CHOC)**
        - Composition: Nutrient agar with lysed red blood cells.
        - Use: Supports the growth of fastidious respiratory bacteria.
        - Example: *Haemophilus influenzae*, *Neisseria gonorrhoeae*.
3. **Selective Media**
    - **MacConkey Agar (MAC)**
        - Composition: Peptone, lactose, bile salts, crystal violet, neutral red, agar.

        - Use: Selective for Gram-negative bacteria and differentiates lactose fermenters from non-fermenters.
        - Example: *Escherichia coli* (lactose fermenter, pink colonies), *Salmonella* spp. (non-lactose fermenter, colorless colonies).
    - **Mannitol Salt Agar (MSA)**
        - Composition: Peptone, beef extract, mannitol, sodium chloride, phenol red, agar.
        - Use: Selective for staphylococci due to high salt concentration; differentiates mannitol fermenters.
        - Example: *Staphylococcus aureus* (mannitol fermenter, yellow colonies), *Staphylococcus epidermidis* (non-mannitol fermenter, red colonies).
4. **Differential Media**
    - **Eosin Methylene Blue Agar (EMB)**
        - Composition: Peptone, lactose, eosin Y, methylene blue, agar.
        - Use: Differentiates lactose fermenters, which produce dark colonies with metallic sheen.
        - Example: *Escherichia coli* (dark colonies with green metallic sheen), *Enterobacter aerogenes* (pink colonies).
    - **Xylose Lysine Deoxycholate Agar (XLD)**
        - Composition: Yeast extract, xylose, lysine, lactose, sucrose, deoxycholate, phenol red, agar.
        - Use: Selective and differential for enteric pathogens, especially *Salmonella* and *Shigella*.
        - Example: *Salmonella* spp. (red colonies with black centers), *Shigella* spp. (red colonies).
5. **Specialized Media**
    - **Thiosulfate-Citrate-Bile Salts-Sucrose Agar (TCBS)**
        - Composition: Peptone, yeast extract, bile salts, citrate, thiosulfate, sucrose, bromothymol blue, thymol blue, agar.
        - Use: Selective for *Vibrio* species.
        - Example: *Vibrio cholerae* (yellow colonies due to sucrose fermentation), *Vibrio parahaemolyticus* (green colonies).
    - **Lowenstein-Jensen Medium (LJ)**
        - Composition: Eggs, glycerol, malachite green, potato flour, mineral salts.
        - Use: Selective for *Mycobacterium* species.
        - Example: *Mycobacterium tuberculosis* (growth of rough, buff-colored colonies).

CLASSIFICATION OF BACTERIA

Classifification based on Cell Wall Composition

- **Gram-positive Bacteria**
    - Example: *Staphylococcus aureus* - A spherical, Gram-positive bacterium known for causing skin infections, pneumonia, and food poisoning.
- **Gram-negative Bacteria**

- Example: *Escherichia coli* - A rod-shaped, Gram-negative bacterium commonly found in the intestines of humans and other warm-blooded animals. While most strains are harmless, some can cause foodborne illness and urinary tract infections.

Classification based on Morphology

- **Cocci (Spherical Bacteria)**
  - **Monococcus**: Single spherical cell.
    - Example: *Micrococcus luteus*
  - **Diplococci**: Pairs of spherical cells.
    - Example: *Neisseria gonorrhoeae* (causes gonorrhea)
  - **Streptococci**: Chains of spherical cells.
    - Example: *Streptococcus pyogenes* (causes strep throat)
  - **Staphylococci**: Clusters of spherical cells, resembling grapes.
    - Example: *Staphylococcus aureus* (can cause skin infections)
  - **Tetrads**: Groups of four spherical cells.
    - Example: *Micrococcus tetragenus*
  - **Sarcinae**: Cubic configuration of eight spherical cells.
    - Example: *Sarcina ventriculi*
- **Bacilli (Rod-shaped Bacteria)**
  - **Single Bacillus**: Single rod-shaped cell.
    - Example: *Escherichia coli* (commonly found in the intestines)
  - **Diplobacilli**: Pairs of rod-shaped cells.
    - Example: *Coxiella burnetii* (causes Q fever)
  - **Streptobacilli**: Chains of rod-shaped cells.
    - Example: *Streptobacillus moniliformis* (causes rat-bite fever)
  - **Coccobacilli**: Short and plump rod-shaped cells.
    - Example: *Haemophilus influenzae* (can cause respiratory infections)
- **Spirilla (Spiral-shaped Bacteria)**
  - **Spirillum**: Rigid spiral-shaped cells.
    - Example: *Spirillum minus* (causes rat-bite fever)
  - **Spirochetes**: Flexible, tightly coiled spiral-shaped cells.
    - Example: *Treponema pallidum* (causes syphilis)
    - Example: *Borrelia burgdorferi* (causes Lyme disease)
- **Vibrios (Comma-shaped Bacteria)**
  - **Vibrio**: Curved rod-shaped cells, resembling a comma.
    - Example: *Vibrio cholerae* (causes cholera)
- **Pleomorphic Bacteria (Variable Shape)**
  - Bacteria that can alter their shape or size in response to environmental conditions.
    - Example: *Mycoplasma pneumoniae* (lacks a cell wall and causes atypical pneumonia)

4. Based on Oxygen Requirement

Bacteria can also be classified based on their oxygen requirements. Here are the main types, along with examples for each:

1. **Obligate Aerobes**
    - Require oxygen for growth as they use it for aerobic respiration.
    - Example: *Mycobacterium tuberculosis* (causes tuberculosis)
    - Example: *Bacillus subtilis* (common soil bacterium)
2. **Obligate Anaerobes**
    - Cannot survive in the presence of oxygen; they obtain energy through anaerobic respiration or fermentation.
    - Example: *Clostridium botulinum* (causes botulism)
    - Example: *Bacteroides fragilis* (part of the normal gut flora, but can cause infections)
3. **Facultative Anaerobes**
    - Can grow in the presence or absence of oxygen; they use aerobic respiration when oxygen is available and switch to anaerobic respiration or fermentation when it is not.
    - Example: *Escherichia coli* (commonly found in the intestines)
    - Example: *Staphylococcus aureus* (can cause skin infections)
4. **Microaerophiles**
    - Require oxygen at lower concentrations than atmospheric levels (typically 2-10%); higher concentrations of oxygen can be harmful to them.
    - Example: *Helicobacter pylori* (causes stomach ulcers)
    - Example: *Campylobacter jejuni* (causes food poisoning)
5. **Aerotolerant Anaerobes**
    - Do not require oxygen for growth but can tolerate its presence; they obtain energy through fermentation.
    - Example: *Lactobacillus* species (important in food fermentation processes, such as yogurt and sauerkraut)
    - Example: *Streptococcus pyogenes* (causes strep throat)

5. Based on Metabolic Characteristics

- **Autotrophic Bacteria**
    - Example: *Nitrosomonas europaea* - An autotrophic bacterium that oxidizes ammonia to nitrite as part of the nitrogen cycle.
- **Heterotrophic Bacteria**
    - Example: *Pseudomonas aeruginosa* - A versatile heterotrophic bacterium found in soil, water, and on plants, known for its opportunistic pathogenicity in humans, particularly in immunocompromised individuals.

6. Based on Ecological Roles

- **Symbiotic Bacteria**
  - Example: *Rhizobium leguminosarum* - A symbiotic bacterium that forms nitrogen-fixing nodules on the roots of leguminous plants, aiding in nitrogen fixation and enhancing plant growth.
- **Pathogenic Bacteria**
  - Example: *Salmonella enterica* - A pathogenic bacterium responsible for causing salmonellosis, a common foodborne illness characterized by symptoms such as diarrhea, fever, and abdominal cramps.

Systematic Classification of Bacteria

Systematic classification, also known as taxonomy, is the science of organizing living organisms into hierarchical categories based on their evolutionary relationships, genetic similarities, and shared characteristics. Bacteria, being one of the most diverse groups of organisms, are classified into several levels of hierarchy. Here's an overview of the systematic classification of bacteria with examples:

Domain: Bacteria

- Bacteria belong to the domain Bacteria, one of the three domains of life, along with Archaea and Eukarya.

Kingdom: Eubacteria

Phylum: Firmicutes

- **Class: Bacilli**
  - *Example*: *Bacillus subtilis* - A Gram-positive, rod-shaped bacterium commonly found in soil and the gastrointestinal tract of humans and animals. It's widely studied for its ability to produce antibiotics and other bioactive compounds.
- **Class: Clostridia**
  - *Example*: *Clostridium botulinum* - A spore-forming, anaerobic bacterium known for producing botulinum toxin, the most potent toxin known, causing botulism, a severe and potentially fatal paralytic illness.

Phylum: Proteobacteria

- **Class: Alphaproteobacteria**
  - *Example*: *Rhizobium leguminosarum* - A nitrogen-fixing bacterium that forms symbiotic relationships with leguminous plants, helping them convert atmospheric nitrogen into a form usable by the plants.
- **Class: Betaproteobacteria**
  - *Example*: *Neisseria gonorrhoeae* - A gram-negative diplococcus bacterium responsible for the sexually transmitted infection gonorrhea.
- **Class: Gammaproteobacteria**

    - *Example*: *Escherichia coli* - A facultatively anaerobic, gram-negative bacterium commonly found in the lower intestine of warm-blooded organisms. While most strains are harmless, some can cause severe foodborne illness.
- **Class: Deltaproteobacteria**
    - *Example*: *Desulfovibrio vulgaris* - A sulfate-reducing bacterium found in various environments, including soil, sediment, and the human gastrointestinal tract.

Phylum: Actinobacteria

- **Class: Actinobacteria**
    - *Example*: *Mycobacterium tuberculosis* - A rod-shaped, acid-fast bacterium responsible for causing tuberculosis in humans and other animals. It's one of the most significant bacterial pathogens worldwide.

Phylum: Cyanobacteria

- **Class: Cyanobacteria**
    - *Example*: *Anabaena sp.* - A filamentous, photosynthetic bacterium capable of nitrogen fixation. It forms symbiotic relationships with certain plants and is a prominent member of aquatic ecosystems.

Phylum: Spirochaetes

- **Class: Spirochaetes**
    - *Example*: *Borrelia burgdorferi* - A spiral-shaped, gram-negative bacterium responsible for causing Lyme disease, a multisystemic illness transmitted through the bite of infected ticks

LECTURE NOTE ON BIO 102 (GENERAL BIOLOGY 2) SECOND SEMESTER 2024

BY Dr N.V. Elosiuba

FUNGI

Up until the mid-20th century many Scientists classified fungi as plants. Until recently the biology of fungi was poorly understood. Based on fossil evidence, fungi appeared in the Precambrian era, about 450 million years ago. Fungi, like most plants are sessile and seem to be rooted to a spot. They possess a stem like structure similar to plants, as well as having a root-like fungal mycelium in the soil. The field of Mycology, which is the scientific study of fungi, brought a lot of progress in field of fungal biology.

The word fungus comes from the Latin word for mushrooms. The Kingdom Fungi includes a large group of living organisms collectively referred to as Eumycota (True fungi). About 100, 000 species of fungi have been identified by scientists which are only a fraction of about 1 million species probably present on Earth. Organisms like mushrooms, yeasts, mould of all types are all fungi. *Penicillium notatum* used in the production of Penicillin drug is also a member of the Kingdom Fungi.

Fungi were once considered as plants but, molecular studies have thrown more light and it was discovered that fungi have more of animal characteristics than plants hence they were carved into their own separate kingdom. Fungi play very important roles in the ecosystem. They are decomposers and play great roles in the cycling of nutrients, by breaking down organic materials into simple molecules. They often interact with other organisms to form mutualistic associations. They could also be harmful, causing diseases in plants, animals and humans

BASIC CHARACTERISTICS OF Fungi

Fungi are eukaryotic organisms with a complex cellular organization. Their nucleus is membrane bound. The DNA in the nucleus is wrapped around with histone proteins (which package DNA into structural units called nucleosomes). Few fungal cells have structures that can be compared to bacterial plasmids

(loop of DNA) however there is no horizontal transfer of genetic information as is often the case in plasmids.

Fungi contain mitochondria, endoplasmic reticulum and golgi apparatus. They do not have chloroplasts and chlorophyll; however they have other cellular pigments giving them multiple colours ranging from red, green to black. The pigments in fungi are associated with their cell wall and function protectively against U.V. radiations. Some of these pigments are poisonous. Fungal cells have thick cell walls made of chitin and glucans which give structural strength to the cell walls. Chitin is similarly found in the exoskeleton of insects.

Fungi have plasma membrane similar to other eukaryotes except that the structure is stabilized by ergosterol a steroid molecule replacing the cholesterol found in animals cell membranes. Fungi are non-motile, it is only the gametes of the primitive phylum Chytridiomycota that produces flagella. Most fungi are multicellular, while some are unicellular example the yeast cells. They display two distinctive morphological stages;

- The vegetative stage
- The reproductive stage

They reproduce by means of spores. They exhibit alternation of generation (they alternate between two different life stages or generations in their life cycle) a haploid stage (gametophyte) a diploid stage (sporophyte). They develop from the spores, they have no embryonic stage. Reproduction can be by sexual or asexual means.

Fungi have a filamentous structure except in yeast cells (Yeast cells are single celled). These filamentous/thread-like structures are known as hyphae, which come together to form a mesh-like structure called mycelium. Mycelium is a network of fungal threads or hyphae. It often grows underground. The fruiting bodies of fungi such as mushrooms sprout from mycelium.

CLASSIFICATION OF FUNGI:

Based on mode of nutrition, fungi can be classed into 3.

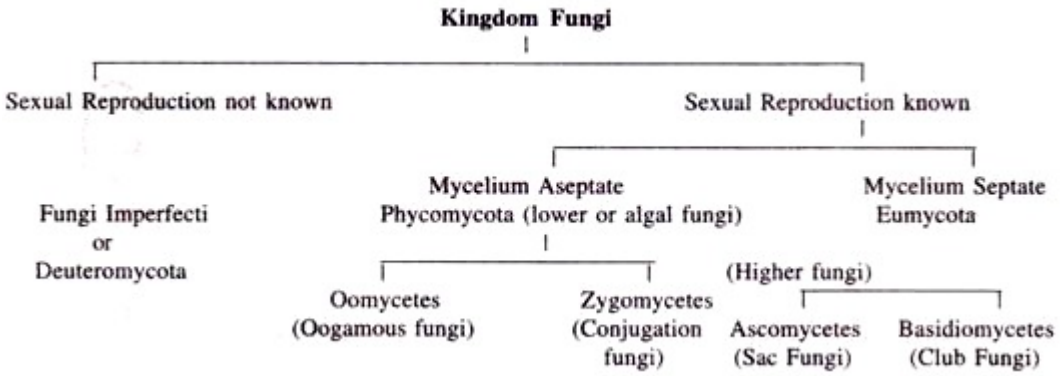
Saprophytic mode of nutrition: These saprotrophs feed on dead and decaying organic matter. Examples include *Rhizopus stolonifer* – the black mould of

bread. *Penicillium chrysogenum* – used in the production of penicillin antibiotic. Aspergillus – a spore forming mould fungus.

Parasitic mode of nutrition: these live on animals and plants causing harm to them. Examples include *Taphrina deformans* – causing peach leaf curl disease (phyto-pathogenic fungi). Those that live on animals are called zoo-pathogenic fungi. Examples include ringworm infections, epidermophytosis and *Tinea vescicolor, Candida spp*. Those that live on insects are called entomopathogenic fungi.

Symbiotic mode of nutrition: Interdependent relationship whereby both organisms are mutually beneficial e.g. Lichens (relationship between algae and fungi) and Mycorrhiza (relationship between fungi and plants).

Fungi can also be classified based on their spore formation and reproductive strucures.



1. Deuteromycota – The fungi imperfecti

This is an artificial class of fungi created to include all those fungi whose sexual stage is either absent or not known. Some are unicellular like yeast cells, but most are multicellular with septate mycelium. They reproduce asexually by spore formation (Conidia formation). They live mainly on land. Most food spoilages are caused by this group of fungi and are also responsible for most human fungal infections. Examples include red-rot of

sugar cane caused by *Colletotrichum falcatum. Helminthosporium oryzae*, the leaf spot of rice, early blight of potato and tomatoes etc. They occur as saprophytes on a wide range of substrates but a large number of them are parasites of plants and animals.

2. Oomycota - (Oogamous fungi)

Mycelium is coenocytic (multinucleate and aseptate). Hyphal wall contain cellulose and in some cases chitin. Asexual reproduction involves spore formation contained in sacs or sporangia. In aquatic conditions sporangia produce zoospores, but in terrestrial condition sporangia behave as spores known as conidia hence the name conidiosporangia. Sexual reproduction is by gametangial contact in which the male sex organ or antheridium passes its product into the female sex organ oogonium or archegonium through a fertilization tube. The product of sexual reproduction here is **oospore. Examples** include blight diseases, white rust disease, damping off, downy mildew, etc.

3. Zygomycota – (The conjugation fungi)

Their mycelium is coenocytic. Hyphal wall contain chitin or fungus cellulose. There is absence of motile cells (zoospores and plasmogametes). Mitospores are non-motile and are called sporangiospores. These are terrestrial fungi which are mostly saprotrophic and rarely parasitic. Sexual reproduction occurs through gametangial copulation or conjugation, hence they are called conjugation fungi. Sexual reproduction gives rise to to **zygospore** (diploid spore). Examples include squirting fungus (*Pilobolus crystallins*), *Rhizopus stolonifer* – black bread mould, *Mucor caninus* – dung mould, both are saprotrophic.

4. Ascomycota – (The sac fungi)

This is a class of diverse fungi having over 30000 species. They are saprotrophs, decomposers, coprophilous (living on dung) and parasitic. They have mycelium with septate hyphae, with the exception in yeasts. Cell wall contain chitin and fungus cellulose. No motile structures in their life cycle. In most ascomycetes asexual reproduction are by formation of conidia borne on conidiophores, which maybe branched or unbranched.

We find them growing on bread, citrus fruit, old leather etc. Examples are *Penicillium* and *Aspergillus*. They also include pigmented moulds (brown, green, blue, pink), powdery mildew, yeasts, morels and truffles. Sexual reproduction can take place in this group also.

5. Basidiomycota – (the club fungi).

These are the most advanced and most commonly seen fungi as their fructifications are large and conspicuous. The class contain about 25,000 species. They are the best wood decomposers. They can decompose both lignin and cellulose with the help of secreted enzymes. They have no motile structures or cells. They have two types of mycelia, 1^0 and 2^0 mycelium. Sexual reproduction is by Plasmogamy (fusion of protoplasts without fusion of nuclei. Sex organs are not involved. Examples include: Rusts – *Puccinia graminic tritici* (black rust of wheat), *Puccinia glumarun* (yellow rust of wheat). Smuts: *Ustilago maydis* – (smut of corn). *Ustilago triciti* – (loose smut of wheat). Mushrooms: with umbrella-like basidiocarp – *Agaricus campestris, Voluariella volvacea* – (paddy straw mushroom). Toadstool: these are non-edible, often poisonous with white spores *Amanita acesarea* (Caesaer' mushroom). *Amanita phalloides* (death cap), Bracket fungi or shelf fungi also known as **polypores.**